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NEIGHBORHOOD EFFECTS ON SOCIAL ISOLATION AND LONELINESS
AMONG COMMUNITY-DWELLING OLDER ADULTS

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ABSTRACT OF THE DISSERTATION

Neighborhood Effects on Social Isolation and Loneliness among Community-Dwelling

Older Adults

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Researchers have determined that social isolation and loneliness are major threats to the wellbeing of older adults. To better understand these mechanisms and to design effective interventions for reducing social isolation and loneliness among older adults, growing interests have focused on exploring risk factors. Neighborhoods are important social settings for older adults, as well as locations where social programs aiming at enhancing social engagement are implemented. However, although individual-level risk factors have been widely examined (e.g., age, race/ethnicity, income, and health), relatively fewer studies have empirically tested neighborhood-level risk factors for, and protective factors against, social isolation and loneliness among older adults.

This study, therefore, aims at exploring the neighborhood-level risk and protective factors for social isolation and loneliness among community-dwelling older adults, focusing specifically on neighborhood socioeconomic structure and racial/ethnic composition. These structures were measured using a series of proxy variables, including neighborhood deprivation, racial/ethnic diversity, income inequality, and dynamics of

neighborhood income and racial/ethnic composition. Two-level data were constructed. Individual survey data were collected in 2015 from 3,701 older adults, defined as 60 and older, who resided in New Jersey. The individual data were then combined with census tract-level sociodemographic and economic indicators from the 2015 American Community Survey (ACS). Subsamples of older White and African American adults were subsequently created and compared. Hotspot analyses were conducted to explore the concentrations of socially isolated and lonely older adults across the state. Multilevel linear regressions were conducted to examine the effects of selected neighborhood indicators on social isolation (distal and proximal) and loneliness among older adults.

Four major findings were identified from the regression models: 1) the association between neighborhood deprivation and social isolation/loneliness is primarily due to the concentration of older adults with low socioeconomic status (SES) living in relatively deprived neighborhoods; 2) greater racial/ethnic diversity is associated with more social isolation among older White adults, with only those living in neighborhoods with intermediate levels of diversity reporting statistically positive effects of diversity on loneliness; 3) there is an overall trend toward the increase in diversity and the decrease in percentage of Whites in NJ neighborhoods over time; the change in the percentage of Whites, but not neighborhood income, affects distal social isolation among older African American adults; 4) older adults who live alone and/or without a partner tend to relocate to apartments and senior living communities. In addition, no significant association was identified for income inequality, suggesting that income inequality is more likely to function at a larger geographic scope, such as at the state and country level, or may not be a spatial predictor of social isolation/loneliness among older adults.

The findings highlight the importance of neighborhood socioeconomic structure and racial/ethnic composition for the social wellbeing of older adults. Neighborhood-level risk factors are likely to reduce older adults' motivation and willingness to engage in social activities and likely to impose barriers to accessing social activities. Conversely, however, some findings also imply the possibility of older adults expanding/enhancing their social connections or adjusting their social needs as coping and adaptation strategies when facing individual or neighborhood impediments to social engagement. Practitioners and policy makers should be aware of the importance of neighborhood contexts for older adults to effectively engage in social activities and to obtain support, information, and resources through this process. Community-based programs that aim at reducing social isolation and loneliness should assess the overall neighborhood socioeconomic and political structures (e.g., interracial relationships and the power structure) as well as the more direct neighborhood and individual-level factors (e.g., the transportation system and individual financial barriers) to identify the most vulnerable populations and to better design and distribute services and resources. In addition to directly targeting social isolation, practitioners should respect the choices of older adults in forming their own social support networks and provide direct resources and services to supplement the inadequacies of their social connections.

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Introduction

Many older adults experience life transitions such as the loss of partners and peers, changing social roles, and diminished mobility capacity that make them more at risk for social isolation and loneliness (Cornwell et al., 2008), which are major risk factors that can affect wellbeing (Cacioppo & Cacioppo, 2014; Cacioppo et al., 2010; Nicholson, 2012; Steptoe et al., 2013). Older adults are also more physiologically vulnerable to the detrimental effects of loneliness (Hawkley & Cacioppo, 2007).

This study explores neighborhood-level risk factors for social isolation and loneliness among older adults. Older adults are defined as those over 60 years of age in this study. Scholars have examined social isolation and loneliness among older adults from various perspectives, including health effects and possible mechanisms, comparisons of the effects between social isolation and loneliness, intervention evaluations, and individual risk factors (Courtin & Knapp, 2017). However, relatively few studies have explored the influence of neighborhood characteristics on social isolation and loneliness among older adults, even though neighborhoods are important settings for social participation and community-based social services targeting older adults.

Although limited studies have directly addressed the correlation of neighborhood characteristics and social isolation/loneliness, rich empirical evidence and theoretical frameworks have examined neighborhood effects on a broader range of social indicators among older adults and the general population. For example, a high degree of neighborhood deprivation has been associated with lower levels of social trust and reduced social contact (Bowling & Stafford, 2007; Franzini et al., 2005; Verhaeghe &

Tampubolon, 2012), and neighborhood racial/ethnic diversity has been associated with the social withdrawal of Whites (Putnam, 2007; Portes & Vickstrom, 2011). There are also emerging studies on the loss of social connections and a declining sense of belonging among older African American adults residing in neighborhoods undergoing gentrification (Buffel & Phillipson, 2019; Burns et al., 2011; Versey, 2018). This evidence sheds light on the possible impacts of neighborhood economic and social structures on the social wellbeing of older adults.

Background and Significance: Social Isolation and Loneliness Are Public Health Concerns

Social isolation and loneliness as related but distinctive concepts

Although social isolation and loneliness are closely related concepts, there are important distinctions. Social isolation is the state of minimal contact with and connection to other people, and loneliness is the negative feeling associated with a lack of desired contact (Wenger et al., 1996). Social isolation refers to the objective determination of social disconnectedness while loneliness refers to the subjective response to social isolation (Victor et al., 2000). Therefore, scholars sometimes name loneliness as perceived social isolation (Cacioppo & Cacioppo, 2014; Cacioppo et al., 2010).

In fact, people who are socially isolated may not be lonely and those who are lonely may be socially connected. Perlman and Peplau (1981) defined loneliness as “the unpleasant experience that occurs when a person's network of social relations is deficient in some important way, either quantitatively or qualitatively” (p. 31). Indeed, the quality of connections is more important in regards to whether people feel loneliness (Cacioppo & Hawkley, 2009). Empirical studies have also found that the self-reported quality of social connections with spouses and broader social network are significant predictors of loneliness among middle-aged and older adults (Hawkley et al., 2008). In addition, respondents’ personalities, operative cultural norms, childhood environment, and social needs all influence how people perceive their degree of social isolation (Cacioppo & Hawkley, 2009). In short, loneliness is a subjective response to a person’s unmet needs for expected social connections, both as regards quality and quantity (De Jong Gierveld

& Kamphuls, 1985), which depends on the gap between their objective social connectedness and their perceived need or expectations regarding social connections.

The two concepts also differ in how they are measured by researchers. Loneliness as a psychological indicator is often measured via a scale, and the most widely used is the abbreviated version of the UCLA loneliness scale created by Hughes and colleagues (2004). The original version contains 20 items, but the abbreviated version asks just three questions regarding whether participants felt a “lack of companionship,” “left out,” and “isolated from others” (Hughes et al., 2004). Another loneliness scale and its shortened version, developed by De Jong Gierveld and colleagues (1985; 2006), targeted the older adult population with two subscales corresponding to Weiss’s (1973) typology of emotional loneliness and social loneliness. Emotional loneliness is related to the lack of close and intimate attachment, while social loneliness refers to the absence of social connections and social integration (Russell et al., 1984; Weiss, 1973).

Unlike the measurement of loneliness, no consistent measure for social isolation has yet been developed to facilitate comparison across empirical studies, due in part to the fact that the measure of social isolation is often constructed as an index instead of a scale. That is, social isolation has usually been calculated as the composite score of a broad range of items measuring the social network, social contacts, living arrangement, marital status, and social engagement (Coyle & Dugan, 2012; Shankar et al., 2011; Wenger & Burholt, 2004). Because survey questions vary from study to study, the items used to construct a social isolation index will also vary, sometimes substantially. In addition, as living arrangements and the absence of partners are not strongly correlated

with other social isolation indicators, sometimes these factors are not considered as parts of the social isolation construct (Cornwell & Waite, 2009a; Victor et al., 2000).

As discussed in the following sections, social isolation and loneliness represent overlapping and distinctive risk factors among older adults. Distinguishing the differences as well as the similarities between these two concepts and their measures is also important in this study as neighborhood effects on social isolation can be very different from neighborhood effects on loneliness.

Social isolation and loneliness are threats to health and wellbeing

Social isolation and loneliness can negatively affect both the physical and mental wellbeing of older adults. That is, higher social isolation and loneliness correlate with declining neurobiological functioning and declining physical and mental health as well as increased risks of all-cause mortality, dementia, depression, cognitive decline, falls, and re-hospitalization among older adults (Cacioppo & Cacioppo, 2014; Cacioppo et al., 2010; Nicholson, 2012). Detrimental health effects also lead to higher health care utilization and costs (Gerst-Emerson & Jayawardhana, 2015), low life satisfaction (Swami et al., 2007), and low quality of life (Ekwall et al., 2005; Hawton et al., 2011).

In other words, not only do isolated older adults have a higher risk of health problems partly because they feel lonely (Chen & Feeley, 2014; Cornwell & Waite, 2009b), but Hawkley and Cacioppo (2010) proposed what they call the loneliness model, which suggested that loneliness is related to feeling unsafe and perceiving threats from social life. These negative social expectations cause additional negative social interactions with others, which result in older adults thus afflicted further distancing

themselves from society. The researchers described this process as a self-reinforcing loop that reinforces negative emotions like depression, stress, pessimism, anxiety, and low self-esteem (Hawkley & Cacioppo, 2010) and causes a series of physiological dysfunctions (Cacioppo et al., 2002; Cacioppo & Cacioppo, 2014; Hawkley et al., 2010), including cardiovascular activation, sleep dysfunction, metabolic syndrome, increased hypothalamic pituitary adrenocortical activity, decreased inflammatory control and increased glucocorticoid insensitivity, diminished immunity, and diminished impulse control (Cacioppo et al., 2002; Cacioppo & Cacioppo, 2014; Hawkley et al., 2010).

In addition to the detrimental health outcomes of loneliness or perceived social isolation, the objective status of social isolation carries additional threats to the health and wellbeing of older adults. Social isolation means suffering from the lack of social connections and contacts, which have a wide range of functions for protecting health of older adults through social support, social influence, social engagement and attachment, and access to resources and material goods (Berkman et al., 2000). Social support can help an older adult deal with stressful situations while social influence encourages adopting normative health behaviors and avoiding health risk behaviors (House et al., 1988). Socially isolated older adults are more likely to be inactive and to smoke due to a lack of social control (Shankar et al., 2011). The observed higher risk of all-cause mortality of isolated older adults may be partially due to their lack of prompt medical attention because of the absence of contacts and connections (Stephoe et al., 2013).

Social isolation and loneliness are prevalent among older adults

The prevalence of social isolation and loneliness among older adults makes both problems public health concerns. However, the estimate of the overall rate of social

isolation among older adults varies by the criteria used. In a study by the National Council on Aging, researchers identified risky older adults as those who lived alone and faced one or more barriers for social connection, including language, geographic location, and disability. They calculated the prevalence of social isolation based on the above criteria and concluded that 17% of older Americans over the age of 65 are at high risk for social isolation (Ortiz, 2011). Cudjoe et al. (2020) estimated that 24% of community-dwelling older adults in the US who are 65 and older were socially isolated, while 4% were severely socially isolated in 2011. Walker and Herbitter (2005) identified the percentage of older adults at risk of social isolation by using primary factors associated with isolation, such as marital status, disability, living alone, income, and English fluency. Following this methodology, based on the US census data the Administration for Community Living calculated that 28% of non-institutionalized adults age 65 and over lived alone in 2017 and 33% of women age 65 and over were widows (ACL, 2017), making them high risk groups for social isolation.

Regarding rates of loneliness, studies using a national representative sample from the US reported a prevalence of loneliness in 19.3% of older adults over age 65 (Theeke, 2009) and 16.9% among those over the age of 50 (Theeke, 2010). Similarly, based on a national survey conducted by AARP among US older adults between 62 and 91, researchers reported that 19% of respondents identified as frequently feeling lonely while 29% of the total sample reported feeling occasionally lonely (Hawkley et al., 2017).

In short, social isolation and loneliness are not temporary issues among older adults, but rather, an ongoing public health concern. With the baby boom generation progressing into later life, the population of older adults is growing dramatically.

Although loneliness peaks in both young adulthood and in old age (Victor & Yang, 2012), among older people, the effects of loneliness on their physiological functioning are more detrimental (Hawkley & Cacioppo, 2007). And the current and future generations of older adults are more likely to live in isolation than their predecessors. That is, the modern lifestyle that values independence and individualism has led to smaller household sizes, decreased marriage rates, increased childlessness, less social engagement, and less religious affiliation (Holt-Lunstad, 2017).

Efforts for reducing social isolation and loneliness

Various programs have been designed to reduce social isolation and loneliness for older adults. The Campaign to End Loneliness was initiated in the UK to advocate for inputs from the government and private sector and to develop a research base for a better understanding of loneliness. Likewise, the AARP launched the Connect2Affect program to link people to resources for reducing social isolation among older adults in the US. Local efforts have been widespread, and various types of programs have been developed and distributed.

Mainstream intervention strategies have focused on enhancing social connections (befriending). Befriending strategies carry various forms, including telephone-based intervention, community activities and involvement, group support, home visits, social media and internet communications, and transportation services (Cattan et al., 2005; Findlay, 2003; MacLeod et al., 2018). The purpose of befriending programs is to expand the social networks of and increase social contacts for isolated older adults. Another approach to enable social connections is social skill training (Masi et al., 2011). Resilient training and cognitive behavioral therapies have been carried out to reduce loneliness

from the cognitive aspect (Cacioppo et al., 2015; MacLeod et al., 2018; Masi et al., 2011; Winningham & Pike, 2007).

Reviews of these programs have suggested mixed results as regards effectiveness. Community-based interventions, which take advantage of existing community resources and adopt a community development approach that engages older adults in program design and implementation, have proven more feasible and effective than one-to-one or small group interventions in reducing social isolation and loneliness among older adults (Cattan et al., 2005; Findlay, 2003; Gardiner et al., 2018). The new trend of adopting the use of computers and the internet, like the information and communication technology (ICT), have advantages of overcoming mobility barriers and have been reported to be effective means to reduce loneliness for older adults (Chen & Schulz, 2016; Choi et al., 2012; Czaja et al., 2018). The cognitive approach has also shown strong effects (Masi et al., 2011). Although some social support approaches failed to directly address loneliness, they have reduced isolation-related health risks and improved life quality by providing support for daily difficulties as well as assessments and monitoring for potential risks.

In spite of the increasing effort to connect older adults to their community, social isolation and loneliness among older adults remains a public health concern due to the continued prevalence of the problem among the older population and the relatively understudied mechanisms that both cause and ameliorate the problem. All of these initiatives call for a better understanding of the risk factors for social isolation and loneliness and related psychosocial mechanisms in order to develop effective and efficient prevention and intervention strategies.

Multilevel Risk Factors for Social Isolation and Loneliness

Studying the risk factors for social isolation and loneliness is of great importance for identifying at-risk groups and designing social programs to reduce social isolation and loneliness. The literature suggests multilevel factors that influence the social isolation and loneliness of older adults, including individual sociodemographic factors, physical and mental health, and life events, such as the loss of partners (Nicholson, 2012; Wenger et al., 1996). In addition, the structure of social networks and interpersonal connections, the personality of the individual, community and local contextual factors, and the broader social and cultural structure have been discussed as related to loneliness (Cacioppo & Hawkley, 2009).

Individual sociodemographics and health

In general, age has been generally associated with increased loneliness (Iliffe et al., 2007; Pinquart & Sörensen, 2003; Yang & Victor, 2011) and reduced social connections (Cornwell et al., 2008). However, the results are not consistent. Studies using US samples have reported a U-shape relationship between age with loneliness and frequency of contact (Utz et al., 2002; Cornwell et al., 2008). In addition, advanced age was found to be associated with high quality social connections (Heylen, 2010) and broader social engagement, including interactions with neighbors, religious participation, and volunteering (Cornwell et al., 2008). Scholars have argued that some life events, such as losing a partner and retirement, can trigger the expansion of social networks and social participation (Utz et al., 2002).

Depending on the context, being female can be a risk factor (Pinquart & Sörensen, 2003) or have no significant association with loneliness (Theeke, 2009; Victor

et al., 2006). The gender-related effect is culture-bound. The relatively disadvantageous socioeconomic position of women and women's caregiving responsibilities may be reasons for the increased incidence of loneliness among older women (Ballantyne, 1999; Theeke, 2009). However, as regards objective social isolation, women generally report more social connections and less isolation than men (Cornwell et al., 2008; Cudjoe et al., 2020; Iliffe et al., 2007).

Economic status and education are among the most consistent predictors of social isolation and loneliness. Rich empirical evidence indicates that limited financial resources and low educational achievement are risk factors for social isolation and loneliness among older adults (Cudjoe et al., 2018; Iliffe et al., 2007; Nicholson, 2009; Pinquart & Sörensen, 2003), which suggests a disparity between the haves and have-nots in regards to social isolation and loneliness.

Although the majority of studies have examined diminished health as an outcome of social isolation and loneliness, the opposite relationship was also discussed in some of the literature. Physical health, especially chronic illness and loss of functionality and mobility, can result in reduced social communications, hence leading to social isolation (Nicholson, 2009; 2012). Poor health and functionality, declining vision and hearing, the number of chronic illnesses, and motor impairment were all found to be associated with higher risks of loneliness (Aartsen & Jylhä, 2011; Savikko et al., 2005; Theeke, 2009).

Dimensions of social connectedness

Living alone is a common indicator for social isolation since it often includes the absence of daily support and contact within an older adult's household. In the US, older

adults live alone largely due to the loss of an intimate partner, and the absence of an intimate relationship is the primary reason for emotional loneliness. Having a partner also implies possessing social connections from both sides of the relationship, but because, in most situations, women play the role of social connector in a family, older men who lose their wives are at greater risk of isolation from a broader social world in addition to their loss of the emotionally intimate attachment they had to their spouse (Dykstra & De Jong Gierveld, 2004). Widowed older adults might intend to expand their informal social participation to cope with spousal loss, but limited economic resources and a lack of existing social connections can restrict their efforts (Utz et al., 2002).

In addition to the living arrangement and marital status, other dimensions of social connectedness also affect social isolation and loneliness. Limited social connections and contacts are the definition of social isolation, but the quality of social connections is a more important predictor for loneliness than the quantity of social connections (Cacioppo & Hawkley, 2009). The construct of the social network is also important. Older adults isolated from a broader non-kin social network are also vulnerable to loneliness, even though they are strongly connected within family networks (Litwin & Shiovitz-Ezra, 2011). However, although the dynamic has not been empirically examined, loneliness may also motivate an individual to build social connections. Like other physiological feelings of hunger and thirst, loneliness is an indication of a lack that one seeks to fill. However, conversely loneliness may also include the perception that social interactions are threatening, resulting in further isolation from society (Hawkley & Cacioppo, 2010). The outcome in this regard depends on multiple factors, such as demographics, personality, and the availability of coping resources.

Personalities and cultural norms

An individual's personality can influence how people perceive social isolation. Among older adults, neuroticism and anxiousness can lead to unrealistically high expectations for social connections to feel safe (Long & Martin, 2000). Extraversion, agreeableness, and openness are associated with a lower risk of loneliness regardless of the actual degree of social connectedness (Long & Martin, 2000; Schutter et al., 2020).

Broader social contexts and cultural norms can also affect the social status of older adults as well as how they perceive social isolation. The prevalence and experience of loneliness can be culture-bound and vary across regions (Perlman, 2004). For example, feeling lonely is more prevalent in regions where collectivism is prevalent, which implies that loneliness is a function of interactions between the environment and individuals (Jylhä & Jokela, 1990). Studies comparing the living arrangement of older adults between the Netherlands and Italy found that older adults prefer to live alone after losing their partners in the Netherlands, while Italian older adults prefer to live with adult children. Living alone, in contrast to living with adult children, can protect older adults from loneliness who live in the Netherlands while it can cause greater loneliness for those in Italy. The ideology of individualism in the Netherlands and the relatively traditional family-centric ideology in Italy may be the reason behind this disparity (De Jong Gierveld & Van Tilburg, 1999).

Neighborhood Effects on Social Isolation and Loneliness Among Older Adults

Neighborhood as an important social setting for older adults

Neighborhood-level characteristics are also important predictors of social isolation and loneliness among older adults. Neighborhoods, usually regarded as spatial units where residents are connected due to geographical proximity and shared environments (Chaskin, 1997), are important social settings by definition. Although a defined neighborhood does not assume a collection of inter-member social bonds as do related concepts (e.g., community and local community), a neighborhood is a social setting that can potentially facilitate interpersonal social networks and the development of sense of community, social identity, and social action (Chaskin, 1997). In many empirical studies regarding community-related themes, like social capital and social cohesion, neighborhoods were used as the proxy for local communities. Some predefined geographic units, like census tracts and block groups, were used as the units of analysis in quantitative studies (Dietz, 2002; Sampson et al., 2002).

For older adults, neighborhood-level risk and protective factors can be very important for the residents' social lives. Due to their declining mobility and retirement, the neighborhood becomes an important source of social support for older adults, especially since physical proximity is an important determinant of developing and maintaining friendships (Bromell & Cagney, 2014; Buffel et al., 2012). Although personal networks shrink, on average, as people get older, age is positively related to the increased frequency of socializing with neighbors, increased religious participation, and volunteering (Cornwell et al., 2008).

The neighborhood is also an important setting for developing and delivering social programs and initiatives to older adults. Many programs for older adults are arranged in community settings and dedicated to improving social infrastructure and thereby the wellbeing of older adults. For example, many social activities that have typically provided opportunities for social connections use senior centers and other community facilities and resources. Also, arguments about the benefits of aging in place for older adults have suggested that communities can provide feelings of attachment and security, existing social networks, and familiar environments for older adults (Gardner, 2014; Wiles et al., 2012). Reviews of intervention programs have also suggested that community-based interventions, which utilize existing community resources and adopt a community development approach that engages older adults in program design and implementation, are more feasible and effective than one-to-one or small group interventions in reducing social isolation and loneliness among older adults (Cattan et al., 2005; Findlay, 2003; Gardiner et al., 2018).

Previous studies on neighborhood effects on social isolation, loneliness, and general social wellbeing of older adults

Studies that explore relationships between neighborhood-level characteristics with individuals' behaviors and health and socioeconomic outcomes are part of the "neighborhood effects" research. Wilson's (2012) book *The Truly Disadvantaged*, which focuses on the effects of concentrated poverty on the wellbeing of inner-city African Americans, popularized the studies of neighborhood effects and suggested that the social context and structure of neighborhoods can influence individuals' behaviors and wellbeing. Neighborhood effects have been widely studied as an interdisciplinary topic.

Earlier works stemmed from Wilson's efforts regarding segregation and disadvantaged neighborhoods, while later studies expanded to a wide range of topics such as crime, delinquency, public planning, and the mental and health wellbeing of residents at different life stages (Dietz, 2002; Sampson et al., 2002).

Studies of neighborhood effects on the general wellbeing of older adults have proliferated in recent years. These studies have tested a wide range of neighborhood factors—including socioeconomic composition (poverty rate and unemployment rate), racial composition, demographics (age composition and residential mobility), perceived resources and/or problems, physical environment, and social environment—for their association with the wellbeing of older adults (Yen et al., 2009). Some of the outcome measures that were most studied include general physical health, walking activity, depression, and cognitive functioning (Barnett et al., 2018; Besser et al., 2017; Li et al., 2005; Mendes de Leon et al., 2009).

Limited studies have directly addressed neighborhood effects on social isolation and loneliness among older adults, even though both are important indicators of older adults' wellbeing (Courtin & Knapp, 2017; Grenade & Boldy, 2008). Qualitative studies have suggested a higher risk of isolation for older adults living in high-crime neighborhoods (Portacolone, 2018; Portacolone et al., 2018). The fear of high crime, weak norms of reciprocity, and toxic relations in neighborhoods act as social-environmental risk factors for social isolation, while a dilapidated physical environment, problems with accessibility, and limited availability of social and health resources act as physical-environmental risk factors (Portacolone et al., 2018). Suen et al. (2017) tested the effects of a built environment on social satisfaction for older adults and found minor

but significant impacts. They used data collected by the local No Wrong Door initiative, which contained 2,049 cases measuring the social satisfaction of older adults residing in the Richmond, Virginia area. Compactness of development was negatively associated with social satisfaction, while accessibility to amenities and services was positively associated with social satisfaction (Suen et al., 2017). However, this study did not provide detailed information regarding the definitions and measures of the built environment variables. In addition, scholars have also suggested from a theoretical perspective that income inequality may impact social isolation and loneliness through collapsed social cohesion; however, no empirical studies have tested the effect (De Jong Gierveld et al., 2006).

Although studies directly addressing neighborhood effects on social isolation and loneliness are scarce, a broader body of literature has suggested possible relationships between neighborhood factors with broader measures of social wellbeing among older adults, including measures of social-related behaviors, attitudes, and feelings. Disadvantageous housing and neighborhoods reduce the possibility of social participation (Tomaszewski, 2013), while affluent areas with good facilities encourage social activities among older adults (Bowling & Stafford, 2007); and proximity to neighborhood resources has been widely identified as a positive predictor of social participation of older adults (Levasseur et al., 2015). Residential stability is related to residents' experience with their environment (Schieman, 2009; Schulz et al., 2008) and can modify the effects of neighborhood deprivation on individual social support (Schieman, 2005). Cornwell and Cagney (2014) compared self-reported social cohesion and social ties of a representative sample of older adults in the US and found that those living in moderately

dense neighborhoods reported the lowest social cohesion and social ties compared to those living in regions of high density and low density. Therrien and Desrosiers (2010) randomly selected three groups of older adults of comparable age and gender from three typical metropolitan, urban, and rural regions in Canada, and compared their social wellbeing. They reported that older adults living in metropolitan areas reported lower satisfaction with their social support than those living in rural and urban areas, but there was no difference regarding interpersonal relationships across the three regions (Therrien & Desrosiers, 2010). Age composition was also related to social connections and engagement of older adults. Older adults living in a community dominated by younger adults may have lower social cohesion and intergenerational connections compared to those in age-mixed neighborhoods and neighborhoods with an overrepresentation of older adults (Moorman et al., 2017). Buys (2001), through surveys and interviews with older adults residing in Australian retirement villages, reported that retirement villages, where older adults live closer to each other, can also facilitate informal social contacts among older adults.

Research gaps

As noted, although neighborhood effects studies have examined various social indicators as outcome measures, limited studies have directly addressed social isolation and loneliness. Social isolation is usually measured as a composite score of multiple dimensions of social disconnectedness and social disengagement (Coyle & Dugan, 2012; Shankar et al., 2011; Wenger & Burholt, 2004). Adopting a composite measurement can provide a more comprehensive assessment of the social wellbeing of older adults, as all sources of social disconnectedness and disengagement contribute collectively to social

isolation. The measure of social isolation also functions as the objective counterpart of loneliness. Loneliness, as a well-constructed psychological indicator for perceived social isolation, is considered a clinically meaningful indicator for mental health (Heinrich & Gullone, 2006), and constructing the composite measure of social isolation enables parsimonious comparisons between loneliness and the broad range of social network and engagement measures. Findings regarding the paired outcome variables can also be easily compared to identify the gaps between objective and subjective dimensions of social wellbeing.

Prior studies have adopted a broad range of neighborhood indicators, but limited studies have focused on the effects of neighborhood socioeconomic structure and racial/ethnic composition on the social wellbeing of older adults. Neighborhood socioeconomic structure and racial/ethnic composition have been widely examined as important predictors for important social wellbeing measures, like social ties, social capital, and social cohesion (Sampson et al., 2002). Accordingly, rich theoretical frameworks have been expanded or developed to explain possible mechanisms, like Putnam's "hunker down" theory, Wilson's discussions on concentrated poverty (2012), and Wilkinson's psychosocial effects of income inequality (Putnam, 2007; Wilkinson, 2002; 2005; Wilson, 2012). However, relatively fewer studies have integrated the rich evidence and theoretical frameworks to examine the effects of neighborhood socioeconomic structure and racial/ethnic composition on social isolation and loneliness among the older adult population.

Part of the reason is that health becomes a more prominent concern among older adults and thus studies on social isolation and participation among older adults have

conceptualized both primarily as health-related (Levasseur et al., 2015; Nicholson, 2012), while the socioeconomic and political dimensions of social isolation related to social exclusion, social segregation, and the lack of social capital are generally overlooked. In his study on concentrated poverty, Wilson (2012) discussed the social isolation of inner-city minority groups by emphasizing the importance of social networks on employment opportunities. In contrast, within the gerontology discourse, the focus is on the health and mental health consequences of social isolation (Nicholson, 2012). In addition, studies on social participation/engagement among older adults focus on different types of activities, ranging from daily activities, like dressing and preparing meals, to productive and meaningful social engagement; but limited studies have conceptualized social participation within the broader socioeconomic and political context (Levasseur et al., 2010).

Although it is reasonable to put emphasis on the health dimension of social wellbeing for older adults, examining the socioeconomic and political dimensions of social isolation and the related neighborhood socioeconomic and racial/ethnic structures can be important additions to the current literature; and not only because we can integrate the rich multidisciplinary evidence and theoretical frameworks into gerontology studies and identify the uniqueness of the older adult population, but because social relationships of older adults and the related environmental risk and protective factors are also embedded within the socioeconomic context. Recent studies on gentrification exemplify the importance of neighborhood socioeconomic and racial/ethnic structures on older adults' social wellbeing. Scholars have described how the influx of White middle-class residents in minority neighborhoods harms the social cohesion and existing social

connections among the long-term residents, especially the most vulnerable older adult population (Buffel & Phillipson, 2019; Burns et al., 2011; Versey, 2018). In addition, through the unequal distribution of resources, neighborhood socioeconomic structures can also contribute to major environmental risk and protective factors for social isolation, including the availability of services and social activities, neighborhood safety and walkability, the built environment, and the transportation system (Bailey et al., 2015; Farmer, 2011; Sallis et al., 2011). Therefore, this study aims to fill this gap by integrating the rich multidisciplinary theories and evidence to explore the effects of a wide range of neighborhood socioeconomic indicators on the social wellbeing of older adults.

Focus of the Current Study: The Economic and Racial/Ethnic Dimensions of Neighborhood Effects on Social Isolation and Loneliness

Neighborhood effects research has examined two dimensions of neighborhood structures: economic status and racial/ethnic composition. These are not only fundamental socioeconomic and demographic neighborhood indicators but also key indicators for studying segregation and social justice. Classic studies on concentrated poverty and social structure used key indicators that measured the economic and racial/ethnic dimensions of neighborhood characteristics. This study adopted this tradition and integrates the rich theoretical frameworks from previous theories and evidence. The two dimensions are presented in Table 1. Related indicators include percentage of Whites/minorities, neighborhood deprivation, diversity, and income inequality.

Table 1 The dimensions of neighborhood indicators

	Racial/ethnic dimension	Economic dimension
Between-neighborhood segregation	(Percentage of Whites/minorities)	Neighborhood deprivation
Within-neighborhood division (heterogeneity)	Percentage of Whites/minorities & diversity	Income inequality
Dynamics	Change in racial/ethnic composition	Change in economic conditions

Previous studies of concentrated poverty have focused on between-neighborhood difference, while social structure studies, like studies on racial/ethnic diversity and income inequality, have focused on within-neighborhood difference. Because the two approaches actually focus on the same topics, spatial-based intergroup differences and

inequality, but within different geographic scopes, the two approaches can be united under one framework. In Table 1, the neighborhood indicators were further categorized into measuring between-neighborhood segregation and measuring within-neighborhood division. Both concepts imply the idea of environmental justice and spatial-based social determinants of health disparity/inequality, which are major concerns of this study as well as of the social work profession and practice.

Between-neighborhood segregation, for example some neighborhoods are affluent but other neighborhoods are less-developed, is a measure of disparity across neighborhoods. The concentrated poverty and concentration of African Americans in inner cities are examples of between-neighborhood segregation (Lichter et al., 2012), and indeed there is frequent overlap. That is, because of the historical economic disadvantage and structural racism that minorities in the US experienced; neighborhoods with a concentration of minorities also indicate wealth segregation.

Within-neighborhood division indicates the differences among residents in the same neighborhoods, especially if there are internal conflicts among different groups based on their racial/ethnic identity and economic status. Residents living in the same neighborhood are able to share some public services and resources; however, the division within neighborhoods can result in lack of trust and competition (Buttrick & Oishi, 2017). For example, housing policies have advocated for mixed-income neighborhoods, but for disadvantaged groups, sometimes diversity may bring inequality and social exclusion (Berrey, 2005). This division within neighborhoods can also harm social cohesion among residents.

In addition to between-neighborhood segregation and within-neighborhood division, racial/ethnic and economic changes are also considered in this study.

Gentrification, usually indicated by the increase in middle-class residents and the percentage of Whites, is an example of how neighborhood dynamics matter as regards social cohesion and social contacts among older adults.

The following sections introduce each key indicator and draw on interdisciplinary theories and evidence to explore how neighborhood economic and racial/ethnic indicators can affect the social isolation and loneliness of older adults.

Between-neighborhood segregation: Neighborhood deprivation

Neighborhood deprivation describes the concentration of socioeconomic disadvantage in small areas and is correlated with a wide range of social and health outcomes of older adults (Fone et al., 2007; Lang et al., 2009; Lang, Llewellyn, Langa, Wallace, Huppert, & Melzer, 2008). Residential segregation based on racial/ethnic and economic status, as well as the uneven distribution of resources, is regarded as the source of concentrated poverty and associated social issues in deprived neighborhoods. Studies of neighborhood deprivation and residential segregation, which stemmed from studies on concentrated poverty in inner-city in the US (Wilson, 2012), revealed that residents living in deprived neighborhoods with higher proportions of African Americans were isolated from the outside world. Consequently, there are limited opportunities for residents to find decent jobs and to be self-sufficient (Wilson, 2012). Because these neighborhoods are segregated, residents are isolated from the mainstream society and their access to resource and opportunities are restricted. Scholars have also examined how concentrated

poverty can result in withdrawal from social interactions with neighbors and from participation in community activities (Massey & Denton, 1993; Wilson, 2012).

Neighborhood deprivation has been found to be associated with lower levels of social trust, fewer contacts with friends and relatives, and fewer social activities (Annear et al., 2009; Bowling & Stafford, 2007; Franzini et al., 2005; Verhaeghe & Tampubolon, 2012). Deprivation can also impede residents with average to low household income from receiving support from non-kin networks (Phan et al., 2009). The effect on broader social participation is more complicated. For example, empirical studies have suggested a positive relationship between concentrated poverty and neighborhood disadvantage with participation. For African Americans living in neighborhoods with high concentrations of African Americans, Rankin and Quane (2000) found that those living in poor communities participate more in solving neighborhood problems compared to those living in moderately wealthy communities. Swaroop and Morenoff (2006), using a representative sample collected from various types of neighborhoods in Chicago, found that participation is generally higher in more disadvantaged neighborhoods. These researchers adopted the social needs perspective to explain that residents living in advantaged neighborhoods are more likely to be motivated to advocate for their rights and needs. However, they also found that the development of social organizations is restricted in highly disadvantaged neighborhoods (Swaroop & Morenoff, 2006).

Deprived neighborhoods are characterized by many specific issues, including a deteriorated physical environment, low cohesion among residents, a high crime rate, low perceived safety, and less accessibility to services and resources. These factors can also be risk factors for social isolation and loneliness, especially among older adults. Social

disorganization theory focuses on neighborhood deterioration, which primarily describes the physical environment, and perceived neighborhood physical deterioration is related to decreased social support and community participation among older adults (Brown et al., 2009; Krause, 1993, 2006; Thompson & Krause, 1998). The social environment is also widely discussed in neighborhood studies. Neighborhood social cohesion and sense of community have been found to be related to social support and social interactions for the general population (Brown et al., 2009; Weijs-Perrée et al., 2015) and to the feeling of companionship for older adults (Bromell & Cagney, 2014). Fear of crime and perceived danger is also related to low social cohesion and having few social ties (Cornwell & Cagney, 2014; Ross & Jang, 2000), and perceived danger leads to reduced walkability in a neighborhood, which limits the formation and maintenance of social connections (Foster & Giles-Corti, 2008). Other environmental factors that can improve social activities among older adults include the availability of common spaces, transportation, and the user-friendliness of the walking environment (Kweon et al., 1998; Levasseur et al., 2015). King's study (2008) suggests that perceived safety and cohesion are more important predictors of the social engagement of older adults compared to the physical aspect of walkability.

In summary, neighborhood deprivation is considered a risk factor for reduced social cohesion, decreased interpersonal social ties, and increased barriers for social activities for older adults. At the same time, however, residents may be more likely to engage in community activities in deprived neighborhoods to initiate social changes.

Within-neighborhood division: Neighborhood heterogeneity

Neighborhood heterogeneity, which can be another cause of social isolation and loneliness (Putnam, 2007; Wilkinson, 2002; 2005), describes the social mix of residents within small areas and can be defined relative to many factors, including age composition and religious affiliation. This study focuses on economic and racial/ethnic composition measured in terms of racial/ethnic diversity and income inequality. Unlike neighborhood deprivation, which implies residential segregation and isolation from the broader society and limited resources, heterogeneous neighborhoods can be materially sufficient in general but internally divided due to the presence of partitions or isolated subgroups.

Racial/ethnic diversity

Racial/ethnic diversity has been a key topic in the discussion of social segregation, but it is a relatively new topic regarding neighborhood effects. Although racial/ethnic diversity has been widely studied across multiple disciplines, including sociology, political science, social epidemiology, public health, and community psychology, empirical evidence is mixed and results are sensitive to research design.

McPherson et al. (2001) laid the foundation for discussing the relationship between heterogeneity and social networks. They suggested that the formation and maintenance of social networks is based on a homophily principle, which results in homogenous personal networks. The homophily principle implies that people also tend to build connections with people of their own racial/ethnic group and hesitate to develop connections with people of other racial/ethnic groups.

Putnam (2007) turned the focus to neighborhood racial/ethnic diversity and its effects on social isolation, focusing on immigration into the US and its implications for

social cohesion. He compared two competing social psychology theories that explain interracial relationships: contact theory and conflict theory. Contact theory suggests that diversity fosters more interracial contacts, thereby enhancing out-group trust and solidarity (Allport, 1954; Pettigrew & Tropp, 2006). Conflict theory suggests that diversity fosters competition across racial/ethnic groups for limited resources, which brings interracial conflicts, distrust, and perceived threats (Blumer 1958; Blalock 1967). Putnam (2007) identified a third trend, which he called the “hunker down” effect, which suggests that, for all racial/ethnic groups, diversity results in social isolation and withdrawal from both in-group and out-group connections.

Rich empirical studies have been conducted, following Putnam’s framework, to test the relationship between racial/ethnic diversity and social capital/trust in different contexts. Most of the studies’ findings were consistent with Putnam’s model, especially studies using US samples (Portes & Vickstrom, 2011; van der Meer & Tolsma, 2014). Studies in the UK and the Netherlands found that, although ethnic diversity has negative effects on trust and social capital among Whites, it can also enhance positive inter-race contacts and reduce the perceived threat between ethnic groups (Laurence, 2011; Lancee & Dronkers, 2011; Schmid et al., 2014). Sturgis and colleagues (2011), using a large British sample, found that, although there is a negative relationship, the effect is only applied to trust in neighbors and not generalized to trust in strangers. They also argued that the effect is trivial and can be largely explained by neighborhood deprivation (Sturgis et al., 2011). Twigg et al. (2010) expanded the analysis and found the effect is also applicable to informal social control, another important indicator for collective efficacy. A study using both Canadian and US samples reported that, although

racial/ethnic diversity negatively impacts social trust in both countries, residents who interact frequently with neighbors are less affected compared to those who do not talk to neighbors (Stolle et al., 2008).

A similar discussion about the dilemma between diversity and sense of community was recently initiated by community psychologists (Townley et al., 2011). Neal and Neal (2014) simulated the formation of social networks based on homophily and proximity principles using agent-based models. The model suggested the incompatibility of diversity and social network density (Neal & Neal, 2014). A following study by Stivala et al. (2016) extended Neal and Neal's agent-based model by including both mutable and immutable features for homophily. Immutable features, such as race/ethnicity and income, are resistant to change. Mutable features, like music and fashion preferences, can be changed through cultural dissemination. The researchers found that sufficient diversity regarding mutable features can overcome the effects of diversity on immutable features. However, these studies used the agent-based model with a simulated group of people, and as of yet no study has used real survey data to examine the relationship between diversity and sense of community.

Studies of the negative effects of diversity on social relationships or cohesion or sense of community have faced the same dilemma: tradeoffs between diversity and cohesion. Scholars have proposed solutions that primarily focus on diluting racial/ethnic identity and enhancing other shared social identities. For example, Putnam (2007) has proposed that diminishing the importance of racial/ethnic identity and reconstructing a new identity can help to solve the dilemma regarding diversity and social solidarity. Empirical studies have revealed that living in a diverse community can enhance

intergroup contacts and reduce perceived threats (Laurence, 2011; Lancee & Dronkers, 2011; Schmid et al., 2014), which can be one approach to facilitate the reconstruction of identity. Stivala's model (2016) also suggests that diverse cultural-based mutable features can override the effects of racial/ethnic diversity, and Brodsky (2017) argued that we should value the diversity as the proxy of inclusion, rather than the diversity that implies inequality.

In general, most studies have supported a negative relationship between racial/ethnic diversity and social trust and sense of community for the general population. However, researchers have also suggested there are different stages of change. That is, when a society starts to change from homogenous to diverse, diversity harms social cohesion; but after a certain stage, a new identity and adaptation are developed that can override the negative effect of diversity. These forecasts imply a nonlinear relationship between diversity and social isolation/loneliness and a threshold when reconciliation is achieved.

Percentage of racial/ethnic groups

There are debates on whether a single diversity index is adequate to represent complicated racial/ethnic composition in a neighborhood (Abascal & Baldassarri, 2015; Laurence, 2011). The effects of minority concentration can be very different from the effects of diversity as measured by the diversity index, and consequently, Hou and Wu (2009) included both diversity and minority concentration (percentage of minorities) in their model. Although neighborhoods with a higher percentage of minority groups are more likely to also be diverse, their study found negative effects of minority concentration on social trust among Whites but positive effects of diversity.

Ethnic density theory focuses on explaining the effects of racial/ethnic composition on mental health among minority groups. Scholars have proposed a psychosocial process whereby the higher density of the same ethnicity group is related to reduced discrimination and higher social support. Rich empirical studies have supported the association between a concentration of neighbors from the same ethnic groups and fewer mental problems among minorities (Bécares et al., 2009; Halpern & Nazroo, 2000; Pickett & Wilkinson, 2008; Schofield et al., 2011). However, no empirical study has confirmed the pathway through social support for minorities. On the contrary, Phan et al. (2009) reported a non-significant effect of racial/ethnic density of same group on mutual social support from non-kin networks.

The ethnic density theory also follows from the fundamental principle of homophily for social networks, but it also adds complexities in the discussion of neighborhood racial/ethnic heterogeneity. In the US, homogenous neighborhoods are more likely to be White-dominated while diverse neighborhoods are more likely to be a mix of Whites and minorities. The social cohesion of Whites living in a diverse neighborhood can be either affected by diversity itself or by the lack of Whites as neighbors. For minorities, social cohesion of those living in a diverse neighborhood can be affected negatively by diversity and positively by a concentration of neighbors from the same racial/ethnic group. Therefore, when discussing neighborhood racial/ethnic composition, it is important to consider both the overall diversity and the composition of each racial/ethnic groups.

Racial/ethnic differences in the effects of racial/ethnic composition

Although racial/ethnic heterogeneity and its association with individual-level social isolation may harm social cohesion at the contextual level for both Whites and African Americans, the effects between Whites and African Americans can be different (Abascal & Baldassarri, 2015; Schmid et al., 2014).

For Whites, in most cases, diversity implies more minorities in the neighborhood and more out-group contacts. In addition, diversity may erode their sense of belonging and cohesion if the demographic composition of their neighborhood has changed due to minority immigrants. For African Americans, a high percentage of African Americans can form a subculture and social identity, which helps to build in-group social cohesion. More diversity, however, can have mixed effects. More minorities from other racial/ethnic groups, like Hispanics and Asians, may also harm cohesion. However, more minority groups can bring complicated intergroup interactions and change the power structure across different racial/ethnic groups. It may also bring a more open-minded and non-prejudiced environment for minority groups in the long term (Schmid et al., 2013). The influence can be quite complicated and varies depending on specific contexts, which can be difficult to determine without rich in-depth case studies.

Income inequality

Income inequality has been widely discussed as a risk factor for health via a psychosocial process as proposed by Wilkinson (2002, 2005). He suggested that income inequality affects health through reducing social cohesion and harming the quality of relations. Empirical studies have found supportive evidence that income inequality is

related to health status across countries, states, regions, and small areas (Wilkinson & Pickett, 2006).

However, current empirical studies, since most supportive empirical studies were based on analysis at aggregated levels to predict population-level health indicators (Kawachi et al., 1997), have some limitations. First, there are a limited number of studies that identify the effects of income inequality on individual-level social cohesion and trust, either as outcomes or as mediators between income inequality and health. Second, most studies were conducted based on large geographic settings, either at the international level or at the cross-regional level; fewer studies of smaller areas have yielded the same results (Wilkinson & Pickett, 2006). Only a multilevel analysis of older adults living in Japan supported the negative effects of community-level Gini coefficient on the general trust at the individual level (Ichida et al., 2009). However, most studies using multilevel models to test the effects of economic inequality on trust at the individual level have found no significant effect (Lancee & Dronkers, 2011; Portes & Vickstrom, 2011; Putnam, 2007). One reason is that the effects of income inequality might be sensitive to its measurement (Daly et al., 1998). Therefore, the validity of Wilkson's framework is in question.

Scholars seeking alternative explanations of the relationship between income inequality and health have questioned the existence of the psychosocial process and argued that political doctrine and structure, a declining welfare system, and the nature of social class are the fundamental causes for both higher economic inequality and lower social cohesion (Coburn, 2000; Muntaner & Lynch, 1999). That explains why the effects

are more salient at the scope of large geographic areas, like country and state, which are also political or cultural units.

In general, income inequality is another risk factor for health; however, evidence supporting the psychosocial process, born of reduced social cohesion and trust, as a cause is not solid. The relationship is more likely to function at a higher aggregate level of contexts. Based on Wilkinson's framework, scholars in the area of social isolation and loneliness have suggested including income inequality as a risk factor (De Jong Gierveld et al., 2006), but no empirical study has yet addressed this hypothesis.

Neighborhood dynamics in economic and racial/ethnic dimensions

Neighborhood dynamics, meaning the changes in the sociodemographic composition of neighborhoods, can also affect social cohesion and social connections among residents. Discussion of neighborhood dynamic traces back to classic models like the invasion-succession and neighborhood life models (Schwirian, 1983), and recent literature on neighborhood dynamics has focused on one specific type of neighborhood change: gentrification. Gentrification refers to the migration of middle- and upper-middle-class residents to originally inner-city neighborhoods. The process is always accompanied by the demolition of old buildings and displacement of long-term residents.

Scholars have described how vulnerable older adults can become "stuck" in gentrified neighborhoods and how gentrification processes can harm their social network and attachment to their community (Buffel & Phillipson, 2019; Burns et al., 2011; Versey, 2018). Discussions of gentrification include a primary focus on economic change and on racial/ethnic change. The wealthier newcomers bring dramatic increases in rents and real estate values. Older adults' younger family members who don't own a house and

cannot afford living there relocate to other neighborhoods, and the construction of new stores and institutions that are also unlikely to fit the needs of older adults (Burns et al., 2011; Versey, 2018). Versey (2018) argues that the visual race/ethnicity distinction also matters and the change in racial/ethnic composition can lead to the loss of racial/ethnic identity and neighborhood culture. However, positive changes brought by gentrification are also reported. With renovation efforts, some public facilities, like public libraries and community centers, are redeveloped and serve as social settings for older adults (Buffel & Phillipson, 2019; Burns et al., 2011). The pressure from gentrification leads some older adults to engage in social contacts in local social groups as a coping strategy, and small groups of older adults are able to engage in social actions to defend their rights in gentrified neighborhoods (Buffel & Phillipson, 2019). Positive interactions and development of new relationships with the newcomers also happens (Buffel & Phillipson, 2019). Although both positive and negative impacts have been reported, the majority of arguments hold a more negative view that gentrification can lead to the social isolation of older adults.

Although gentrification is a specific example of neighborhood demographic change that brings dramatic economic and demographic change to deprived urban neighborhoods, its association with the change of social life of older adults sheds some light on the relationship between general demographic changes and aging. Fast changes in sociodemographic compositions can upset the preexisting balance among different groups, resulting in the loss of social connections and reduced social cohesion. Residents may choose different adaptation strategies, including developing a new identity and social connections, or simply move from the now unsuitable environment. During the

process, residents may experience the destruction of old social connections and identity, and they also may experience the development of a new identity and new connections until a new equilibrium is reached. The fundamental mechanism for change in neighborhoods shares some tenets with static neighborhood effects, which highlights the importance of intergroup conflict and within-group collaboration.

Research Hypotheses

Prior studies have examined a wide range of risk factors for social isolation and loneliness; however, studies directly addressing neighborhood effects on social isolation and loneliness among older adults are still few in number. Neighborhoods have been found to be important for the social wellbeing of residents. Multidisciplinary studies have suggested that neighborhood deprivation, heterogeneity, and dynamics affect social relationships and social integration. This study integrated multidisciplinary theories and applied them to an exploration of neighborhood effects, with an emphasis on the effects of neighborhood economic and racial/ethnic indicators on social isolation and loneliness among older adults. This study can contribute to the current literature regarding testing neighborhood effects on the health and wellbeing of older adults, with a focus on their social functionality.

Due to the complicated and inconsistent relationships discussed in prior empirical studies, the nature of this study is exploratory, the results linked to the abovementioned theoretical frameworks for discussion and implication. Four preliminary hypotheses guide the analysis that focuses on each major independent variable:

- Higher neighborhood deprivation is associated with higher loneliness and social isolation.
- Higher racial/ethnic diversity is associated with higher loneliness and social isolation among White older adults.
- Higher percentage of African Americans is associated with lower loneliness and social isolation for African American older adults.
- Higher income inequality is associated with higher loneliness and social isolation.

Regarding the influence of neighborhood dynamics, because recent studies have focused on gentrification and because researchers have come to mixed conclusions, it is hard to state a set of comprehensive hypotheses at this point. Moreover, the effects of racial/ethnic change and income change were not clearly distinguished. However, some preliminary hypotheses born of existing evidence are:

- Changes in income/diversity/percentage of Whites are associated with higher loneliness among African American older adults.
- Changes in income/diversity/percentage of Whites are associated with higher social isolation among African American older adults.

Methods

Data

Individual survey data

The individual-level data are from the cross-sectional data of the 2015 New Jersey Older Adult Survey (NJOAS) collected from October 2015 to December 2015. The questionnaire was designed and the survey conducted by Rutgers University School of Social Work and Avalon Health Economics, and the study was overseen by New Jersey Division of Mental Health and Addiction Services (NJ DMHAS). The goals of the NJOAS include: to assess mental health and substance abuse among New Jersey older adults and to provide policy suggestions to better allocate resources for older adults. The survey collected data on social interactions, mental health, substance abuse, and demographics. The research team used random digit dialing (RDD) to collect a representative sample of older adults aged 60-and-over in the state of New Jersey. The sampling pool was adjusted to collect equally representative samples of geographic regions across the state.

The response rate was 3.7% and the cooperation rate was 37.5%, calculated based on the American Association for Public Opinion Research (AAPOR) response rate calculator 4.1 for RDD (AAPOR, n.d.). The denominator of the response rate includes all phone calls attempted, while the denominator of the cooperation rate excludes non-contacts and those with unknown eligibility. The sample size is 3,701, including 55.0% women, 87.6% Whites, 6.1% African Americans, and 2.2% Hispanic or Latino. Respondents aged 85 and over comprise 11.9% of the total sample, and 57.6% of all

respondents were married or had a partner and 32.9% were living alone in their households. Complete data for key variables was available for 92% of the sample.

The state dataset was selected because it comprises a large number of older adults living in a densely populated and diverse state. Compared to a national representative sample, this strategy eliminates variations across states and regions that were not due to neighborhood effects but differences in policies and cultures. However, the one-state study strategy may introduce limitations on generalization. Future studies can extend and replicate this study using data from other states and regions. The dataset also includes questions that measure a wide range of social isolation indicators, which will be discussed in the measurement section.

Neighborhood-level data

Neighborhood-level demographics and socioeconomic indicators are from the 2015 five-year estimates of the American Community Survey and are measured at census tract level. Data from the 2000 Decennial Census were combined with 2015 ACS data to calculate changes in neighborhood socioeconomic status and demographics. Census tracts were constructed by the US Census Bureau to represent a relatively homogeneous subdivision of counties (US Census Bureau, 2012), and were used as a proxy for neighborhoods in the current study. Previous studies have suggested that the census tract was the most commonly used proxy for a neighborhood to examine neighborhood effects on health and psychosocial indicators across populations (Arcaya et al., 2016). Even though this study uses the census tract as a proxy for a neighborhood, it should be acknowledged that no human-drawn geographic unit can represent the perfect

homogenous neighborhood. Over time, the census tract has been rezoned to capture demographic change but represents less homogenous units.

Regarding the neighborhood-level sample of this study, New Jersey has diverse neighborhoods, urban and rural, with a mixture of racial/ethnic groups. These varied types of neighborhoods make New Jersey an ideal place to study neighborhood effects that relate to income and race/ethnicity.

Combined two-level data

A total of 100% of the 3,701 cases were successfully geocoded by their residential address. This study constructs a two-level dataset with the original individual-level survey data and selected neighborhood-level indicators. Among the total 2,004 census tracts in New Jersey, the 3,701 older adults in the survey resided in 1,431 census tracts.

Measures

Dependent variables

Social isolation

Prior studies have adopted different approaches to measure social isolation. Most studies adopted a series of items for dimensions of social disconnections and disengagement to construct the measurement. Dimensions most commonly used were marital status, living arrangement, social network, and interpersonal contacts (with family members, friends, and neighbors), different types of social activities, and housebound status (Coyle & Dugan, 2012; Shankar et al., 2011; Wenger & Burholt, 2004; Wister et al., 2019).

In this study, items were selected in accordance with prior studies, including marital status, living arrangement, interpersonal networks and contacts with friends and family members, social participation, and housebound status. Marital status was recoded into a dichotomous variable with 0 representing “not in relationship” (single, divorced, separated, and widowed) and 1 representing “in a relationship” (married, civil union, and domestic partnership). Living arrangement was measured as a continuous variable as the number of people living in the household. Network and contacts with friends and family members were measured using the six-item abbreviated Lubben Social Network Scale (LSNS-6). Respondents were asked to rate from 1 (none) to 6 (nine or more) for three questions: people you “see or hear from at least once a month,” “feel at ease with that you can talk about private matters,” and “feel close to such that you could call on them for help.” The same questions were asked separately regarding family and friendships. The Cronbach’s alpha was .85 for the friend subscale and .84 for the family subscale. The advantage of using the LSNS-6 is that it collects broad information for frequency of contacts, network size, and quality of network. Composite scores for the network with friends and family members were created separately by calculating the mean of the three indicators. Social participation was measured using two items asking the frequency respondents “attend meetings of an organized group” and “volunteer.” The answers ranged from 1 (never) to 7 (several times a week). The social participation measure was adopted from the National Social Life, Health, and Aging Project (NSHAP)¹.

¹ Waite, Linda J., Edward O. Laumann, Wendy Levinson, Stacy Tessler Lindau, and Colm A. O’Muircheartaigh. National Social Life, Health, and Aging Project (NSHAP): Wave 1. ICPSR20541-v6. Ann Arbor, MI: Inter-university Consortium for Political and Social Research [distributor], 2014-04-30. doi:10.3886/ICPSR20541.v6

Housebound status was measured as the self-rated frequency of going out in a typical week. The answers ranged from 0 (rarely or never) to 4 (more than five times per week).

The composite social isolation score was generated using principle component analysis. Since the two items measure three different aspects of social participation, each was used as a unique indicator in the principle component analysis. Results suggested three factors with eigenvalues higher than 1 (factor 1: 2.05, factor 2: 1.47, and factor 3: 1.01). However, the inclusion of the third factor resulted in cross loading and the third factor did not carry specific meaning. In addition, the third factor carried a very low eigenvalue. Therefore, the two-factor solution was executed and resulted in two theoretically meaningful factors. The factor loadings are reported in Table 2.

Table 2 Factor loadings based on a principle component analysis for social isolation (n=3701)

	Factor 1 Distal social isolation (reverse coded)	Factor 2 Proximal social isolation (reverse coded)
In a relationship (marital status)	.33	.75
Living arrangement	.24	.77
Network and contacts with friends	.70	-.07
Network and contacts with family members	.62	.18
Participation in organizations	.62	-.40
Volunteering	.63	-.31
Frequency of going out	.48	-.12

All social isolation indicators have high factor loadings on the first factor except marital status and living arrangement, which formed the second factor. Conceptually, the

first factor represents a general concept of social connectedness within the broader society. It was then reverse coded as distal social isolation. The first factor explains 29% of the variance of all indicators. The second factor, in its reverse coded form, measures proximal social isolation, which are the sources of emotional loneliness as described in Weiss's classification of loneliness (Weiss, 1973). This explains an additional 21% of the total variance. In this study, the two dimensions were treated as two aspects of social isolation. Older adults can be actively engaged in social life but still have an absence of intimate relationships and stay at home alone. In contrast, they can be engaged in active communications within their household but isolated from the outside world. The first factor was labeled as distal social isolation, while the second was labeled as proximal social isolation. Both factors were calculated as a weighted sum of all indicators. The factor loadings were used as weights and the composite scores for both were standardized with a mean of 0 and standard deviation of 1.

Although the social isolation indicators form two unique factors regarding the definition of social isolation, it is conceptually acceptable to compute a composite score that combines all indicators. However, when examining the associations with neighborhood indicators, clear distinctions have to be made between the two types of social isolation. It only makes sense that neighborhood characteristics affect the distal social isolation among older adults, and it is unlikely that neighborhood characteristics affect changes in proximal relationships. Instead, it is reasonable to argue that those who lack proximal relationships prefer to reside in neighborhoods with certain characteristics. For example, older adults who are single or widowed are more likely to have lower household income that makes them to live in lower-SES neighborhoods. The

characteristics can include those that facilitate the development of distal social relationships. Consequently, aggregating both types of social isolation into one composite score may bias the results. The models predicting proximal social isolation were retained in this study, because examining the associations can help to locate older adults who lack proximal relationships and to explore their residential choices,.

Loneliness

Loneliness was measured via the widely used abbreviated UCLA scale (Hughes et al, 2004). Respondents were asked to rate “how often do you feel that you lack companionship,” “feel left out,” and “feel isolated from others?” The answers range from 0 (hardly ever) to 2 (often). The three-item loneliness scale yielded a consistent internal reliability score of .72 from two population-based samples in Hughes et al.’s study (2004), indicating acceptable reliability, and its correlation with the original R-UCLA loneliness scale is high ($r=.82$). The sample of this study yielded an identical Cronbach’s alpha of .72, suggesting acceptable internal reliability. The composite score of loneliness was created by adding all values of the three items, and it ranges from 0 to 6.

Predictor variables

Neighborhood deprivation index

The principle component analysis (PCA) has been widely used to construct the neighborhood deprivation index, and this study reviewed various studies that construct their unique deprivation index for various dimensions and indicators (Butler et al., 2013; Knighton et al., 2016; Messer et al., 2006; Singh, 2003). As racial/ethnic composition and income inequality will be included as distinctive measures, items for racial/ethnic

composition and income inequality were excluded from the index. Twenty-two items from the 2015 American Community Survey five-year estimates were initially included in the PCA, including seven dimensions: poverty (% of individuals below the poverty line, % of families below the poverty line, % of families under 150% of the poverty threshold, median per capita income, and median household income), education (% of 25+ adults with less than nine years of education, with less than a high school diploma, and with less than a college degree), employment (% of unemployed among the 16+ labor population and % of white-collar workers), housing (% of owner-occupied units, median house value, vacancy rate, % of units with >1 person/room, median gross rent, % of units without telephone service, and % of units without complete plumbing), vulnerable households (% of households on public assistance, % of female-headed households, % of single parent households with dependents, and % of households without a motor vehicle) and residential stability (% living in the same household last year).

When the first round of PCA was performed, three indicators yielded lower than .40 factor loadings, including % living in the same household last year (-.38), vacancy rate (.30), and units without complete plumbing (.29), and were excluded from the construction of the index. The final 19 items form five dimensions: poverty, education, employment, housing, and vulnerable households. Another round of PCA identified the first factor that is able to explain 61.5% of the total variation. All factor loadings are higher than .55 (see Table 3). In addition, the second and third factors, although with eigenvalues higher than 1, do not carry clear meanings. The first exacted factor was estimated as the neighborhood deprivation index. The deprivation score was standardized with the lowest value of -2.23 and the highest value of 3.25.

Table 3 Indicators forming the neighborhood deprivation index

Dimensions	Indicator	Mean	SD	Factor loadings
Poverty	% Individuals below poverty line	11.6	11.1	.91
	% Families below poverty line	9.4	1.4	.89
	% Families under 150% poverty threshold	18.7	15.4	.94
	Median per capita income	35929.6	16908.1	-.83
	Median household income	76205.9	36579.1	-.87
Education	% 25+ adults with less than 9 years of education	5.6	6.6	.75
	% 25+ adults with less than high school degree	12.1	1.2	.88
	% 25% adults with less than college degree	64.6	2.2	.81
Employment	% Unemployment	9.4	5.5	.66
	% White collar occupations	66.5	15.0	-.83
Housing	% Owner-occupied units	57.2	26.4	-.78
	Median house value	329119.1	16835.6	-.61
	% Units with >1 person/room (house crowd)	3.5	5.0	.66
	Median gross rent	1162.0	391.6	-.60
	% Units without telephone services	2.0	2.4	.56
Vulnerable households	% households on public assistance	3.1	3.8	.75
	% Female-headed households	31.3	12.2	.77
	% Single parent households	3.6	21.6	.84

% households without motor vehicle	12.8	14.2	.81
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Racial/ethnic diversity

Index of Diversity Racial/ethnic diversity has been measured in different ways in previous studies. The Simpson Index of Diversity (also called the Herfindahl index) (Simpson, 1949) has been widely adopted to measure neighborhood racial/ethnic diversity (Hipp, 2010; Laurence, 2011; Putnam, 2007). The formula is shown below.

$$D = 1 - \sum (n / N)^2$$

N represents the total population in a geographic unit, and n represents the population for each racial/ethnic group. We used five groups to calculate neighborhood diversity: non-Hispanic Whites, non-Hispanic Blacks, Hispanics and Latinos, Asians, and others. The value of neighborhood diversity ranges from 0 to 0.8, with the higher value indicating a more diverse neighborhood.

Percentage of African Americans

The percentage of racial/ethnic groups carried different meanings compared to diversity. A homogenous neighborhood can be homogenously White or a concentration of a given minority. Since the two types of homogenous neighborhoods were the most common based on the study sample, the percentage of African Americans was included in the models to distinguish them. The selection of a percentage of minority groups over using a percentage of Whites was due to a concern regarding multi-collinearity. The percentage of Whites was strongly correlated with the diversity index in the sample of White older adults ($r = -.88$) and strongly correlated with the deprivation index ($r = -.72$)

and percentage of African Americans ($r=-.77$) in the sample of African American older adults. The inclusion of the percentage of African Americans also reflected the framework of ethnic density theory in models predicting loneliness and social isolation among older adult African Americans.

Income inequality

The Gini coefficient was used to measure income inequality in this study, which has been calculated at the census tract level by the US Census Bureau. The Gini coefficient ranges from 0 to 1, with 0 representing perfect equality and 1 representing perfect inequality. It is calculated as the proportion of the area between the perfect equality line and the Lorenz curve divided by the total area below the perfect equality line. The Lorenz curve describes the cumulative distribution of income with households sorted by income in ascending order.

Neighborhood sociodemographic changes

Two aspects of change were measured: economic change and racial/ethnic change. Economic change, which reflects the process of gentrification, was measured as the change of median household income from 2000 to 2015. Racial/ethnic change was measured as two indicators: change in the diversity index and change in the percentage of Whites from 2000 to 2015. The two measures were contrasted in the same way as the static measures of racial/ethnic composition. In gentrification literature, it has also been argued that the increase in White population in gentrified neighborhoods did not equate to higher social mixing (Lees, 2008). Therefore, the inclusion of both measures can better capture the racial/ethnic dynamic of neighborhoods.

Racial differences between Whites and African Americans

Additional separated models were constructed for non-Hispanic Whites and non-Hispanic Blacks. There are 3,294 Whites and 229 African Americans in the sample, making both groups of sufficient sample size. For Hispanics and Asians, however, there was not a sufficient sample size to detect possible statistical effects. In addition, the Hispanic and Asian groups are very diverse groups; therefore, an accurate approach would be to look at each sub-ethnicity group, for example, Mexicans, Puerto Ricans, Chinese, and Indians. For each group, there are unique characteristics regarding acculturation and preferences for intergroup contacts. Hispanics and Asians are also relatively new immigrants to the US, and compared to other groups, a higher proportion of Hispanics and Asians use another language at home and cannot speak English fluently (Ryan, 2013). Therefore, there may be similarities across these groups as well.

Control variables

Neighborhood level control variables

Three additional neighborhood-level indicators were included as control variables. Population density was measured as people per square mile. Residential stability was measured as the percentage of residents who remained in the same residence in the past year. Age composition was calculated as the percentage of older adults aged 60 and above in the census tract. All the data were measured at census tract level.

Neighborhood control variables measured at the individual level

Three neighborhood indicators were measured at the individual level: housing type, unit type, and urban/rural. Housing type consisted of three categories: affordable

and subsidized housing, active adult retirement communities, and non-age or income-restricted communities. The addresses of residents were linked to administrative records from the New Jersey Department of Community Affairs to identify if they lived in affordable and subsidized housing (NJDCA, 2019). The locations of the active adult retirement communities were from commercial search websites for active adult communities and were matched with individual addresses. Housing type measures the direct and more proximate living environment for older adults, and the available resources, facilities, and opportunities for social engagement vary across the three living environments.

Unit type was a measure of whether participants lived in an apartment/townhouse or independent house. If the address of the participant contained an apartment number, the unit was considered an apartment or a townhouse. If the full address did not contain an apartment number, the unit was categorized as an independent house.

Urban/rural was categorized into three categories: urbanized area, urban cluster, and rural area. The New Jersey Department of Transportation (NJDOT) adopted the names of the three types of areas from the US Census Bureau's urban-rural classification but followed the criteria created by the Federal Highway Administration (FHWA) (NJGIN, 2017). Urbanized areas are the most populated, with 50,000 or more people within the identified urban area boundaries. Urban clusters contain 5,000 to 50,000 people. Rural areas are those not included within urbanized areas and urban clusters. The FHWA classification was developed for the purpose of transportation planning (FHWA, 2017). The addresses of participants were geographically linked to the three types of

regions using ArcMap. The urban/rural data were obtained from the New Jersey Geographic Information Network (NJGIN, 2017).

Individual-level control variables

Various individual-level factors were predictors for social isolation and loneliness as noted previously. This study included gender, age, self-rated health, income, and education as control variables. Gender was measured as a dichotomous variable with male=1 and female=0. Age, self-rated health, household income in the past year, and education were measured as continuous variables. Self-rated health asked respondents to rate their physical health from 1=poor to 5=excellent. Household income was self-rated and measured as eight groups, from low to high as 1=less than \$10,000 to 8=\$100,000 or more. Education data consisted of six groups, from 1=no education to 6=post graduate degree. Other important individual risk factors, living arrangement and marital status, were used as dimensions to construct social isolation and hence were not be duplicated as individual control variables.

Analytic strategy

Descriptive analysis

Descriptive statistics for key indicators at both the individual and census tract level were examined. In addition, a hotspot analysis using ArcMap 10.7 (Esri, 2019) was performed to examine the concentrations of older adults who reported loneliness and those who were isolated from either proximal or distal relationships. The Optimized Hot Spot Analysis was performed to automatically select appropriate scales of analysis and correct for multiple testing and spatial dependence. Loneliness and the two social

isolation measures were used to identify the clusters of older adults who reported high loneliness/social isolation and those who reported low loneliness/social isolation. The Getis-Ord G_i^* statistics, including z-value and p-value, were calculated for each participant to represent the intensity of hot spots and cold spots. Then the Inverse Distance Weighting (IDW) was used to create interpolation maps to reflect the distributions of loneliness and social isolation among older adults in NJ.

Neighborhood indicators, aggregated at the census tract level, were also presented in maps and compared with the distribution of loneliness and social isolation. Pearson correlations were then performed across dependent variables and neighborhood indicators to explore possible relationships. Correlation analyses based on the full sample, Whites subsample, and African Americans subsample were conducted and the results compared.

Regression analyses

A series of multilevel linear regression models with individual data nested under census tracts were conducted. The models were estimated using Stata 15 command `xtreg, re`. The random effect general least squares (GLS) estimation was used. The intraclass correlation coefficients (ICCs) were calculated to assess if neighborhood effects exist at the census tract level and if the multilevel specifications are necessary for each model. ICCs represent the proportion of variance attributable to the groups. If the multilevel structure was not supported, the ordinal least squares (OLS) models were used instead. In this study, the distribution of the African American sample is very sparse across census tracts with an average group size of only 1.1. Previous studies have suggested the random effects can be overestimated using unbalanced data with very small group size (<2)

(Clarke, 2008). Therefore, the ICCs for the African American samples were interpreted with caution.

The multilevel models were constructed as two-level random intercept linear models, as specified below:

$$Y_{ij} = \beta_0 + \beta_1 X_{1ij} + \beta_2 X_{2ij} + \beta_3 X_{3ij} + \dots + u_j + e_{ij}$$

$$u_j \sim N(0, \sigma_u^2)$$

$$e_{ij} \sim N(0, \sigma_e^2)$$

Where i represents the first-level unit, each older adult is indicated. The second-level unit, the census tract, is represented by j . Y_{ij} is the dependent variable for the j -th older adults living in the i -th census tract. β_0 is the intercept, and $\beta_1, \beta_2, \beta_3, \dots$ are the coefficients for independent variables or covariates $X_1, X_2, X_3, \dots$. There are two residuals, u_j and e_{ij} . The former, u_j , represents the census tract-level random effect, and e_{ij} represents the individual effect.

For each dependent variable, nested models were examined. The basic model (Model 1) with no predictor was first examined, as $Y_{ij} = \beta_0 + u_j + e_{ij}$. Model 2 was specified as independent variables only, including deprivation, diversity, percentage of African Americans, Gini, and the three dynamic measures of sociodemographic changes of neighborhoods (change in income, diversity, and percentage of Whites). Model 3 included additional neighborhood-level controls, such as: population density, stability, and percentage of older adults aged 60 and above. Model 4 included additional individual-level controls, such as: socioeconomic and demographics measures as well as the three living environment measures (housing type, unit type, and urban/rural). Model 5

was specified as including additional social isolation as control variables. For models predicting loneliness, Model 4 included the two social isolation measures as covariates. For models predicting social isolation, the proximal social isolation was controlled.

Including individual-level control variables in Model 4 and Model 5 is meant to control for selection bias when examining neighborhood effects, which means that disadvantaged populations are more likely to live in disadvantaged neighborhoods and the neighborhood effect may be only due to the aggregation of population (compositional effect vs. contextual effect) (Diez, 2002; Pickett & Pearl, 2001). However, that gives rise to over-control issues, which means that neighborhood characteristics may have indirect effects on the examined individual characteristics through individual control variables (Sampson et al., 2002). In this study, neighborhood effects on health, income, and education throughout an individual's lifespan may be the reason for social isolation or loneliness in their later life. In addition, the relationship between individual sociodemographic status and social isolation/loneliness can be bidirectional. In Model 4 and Model 5, the conservative approach was adopted to include all important individual-level control variables, which may lead to the underestimation of neighborhood effects.

Several robust tests were performed to address possible concerns with the regression models. Supplementary OLS models were estimated to compare with the estimation of the multilevel models. As the measure of diversity was computed based on percentages of racial/ethnic groups, separated models that only include either diversity or the percentages of racial/ethnic groups were performed to examine if the significance or coefficient for racial/ethnic measures were robust. The same strategies were also applied to examine the change in percentage of Whites and the change in diversity.

Considering the small sample of African American older adults, marginally significant results were reported and interpreted for African American models. However, the marginally significant results were not considered robust and were interpreted with caution.

Multiple imputation for missing data

Overall, the dataset has a relatively low proportion of missing cases, but the two variables with the largest proportion of missing cases were income, with 22% missing, and education, with 7% missing. Missing data mechanisms can be categorized into three types: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) (Graham, 2012). MCAR means that missing data are a random sample from the total sample and missing independent of any variables. However, based on the sample used in this study, older adults who did not report their income tended to fall into the following demographic categories: younger, more likely to be women with lower education attainment, less isolated, less likely to have a partner, and more likely to be minorities than those who reported their income. Basic demographic indicators were less important regarding those who did not report education, although they tended to be less likely to have a partner than those who reported their education. These differences suggest that the missing data are not an instance of MCAR and therefore imputation is necessary in order to reduce bias caused by missing data.

MAR describes the case when missing data is fully explainable by other observed variables in the dataset, and in contrast, MNAR describes the situation wherein missing data can be due to both observed variables and unobserved variables. A tricky case can be that missingness is dependent on the missing variable itself. For example, income can be

MNAR since people with higher income tend not to report their income in surveys, which means that whether income is reported depends on income level (Graham, 2012). The same explanation can apply to education, as older adults with lower educational attainment may be less likely to report their education due to the social desirability bias. However, technically it is very difficult to identify MNAR and to apply appropriate techniques to reduce bias. In addition, studies have also suggested that MNAR instances of missing data can be treated as MAR when proper proxy variables are included and the assumption of MAR will not result in substantial bias (Collins et al., 2001). Therefore, income and education were imputed as MAR in this study.

Since there are at least two variables to impute, the multiple imputation with chained equation (MICE) was used to impute for all included variables with missing values. The dependent variable, loneliness, was also included in the imputed model. However, in models predicting loneliness, the imputed cases of loneliness were excluded. Marital status was estimated using a logistic model. Education and self-rated physical health were estimated using ordered logistic models. Loneliness and income were treated as continuous variables and estimated using linear regression models. A total of 30 imputations were executed.

Results

Descriptives statistics

Descriptive statistics at the individual level are presented in Table 4. The statistics are based on all valid cases for each variable. The average level of loneliness is .65, with a standardized deviation of 1.19. The distal social isolation was calculated as a standardized composite score, with the range from -2.75 standardized deviation to 2.81 standardized deviation. Proximal social isolation was also standardized, with the range from -4.89 to 2.65. Social isolation indicators used for estimating distal social isolation and proximal social isolation were summarized as well. The percentage of participants in a relationship, either in marriage, a civil union, or a domestic partnership, was 57%. An average of 1.98 people lived in the same household, with the lowest number of people living in a household being 1 and the highest 10. The average network and contacts with friends score was 3.92 and with family members was 4.26, equating to “three to four” friends or family members. The network with family members was a little larger than the network with friends. Frequency of attending meetings of an organized group was rated as 2.33 and volunteering as 2.18 on a 1-to-7 scale. Frequency of going out was rated as 3.33, representing an average of between “three to five times per week” and “more than five times per week.”

Table 4 Descriptive statistics of individual level indicators

	N	Mean or %	SD	Min	Max
		(Unless otherwise specified)			
Dependent variables					
Loneliness (0-6)	3,623	.65	1.19	0	6
Distal social isolation*	3,701	.00	1.00	-2.75	2.81

Proximal social isolation*	3,701	.00	1.00	-4.89	2.65
Social isolation indicators					
<i>In relationship (Marital status)</i>	3,665	57.6			
<i>Counts of people living in household (Living arrangement)</i>	3,701	1.98	1.04	1	10
<i>Network and contacts with friends</i>	3,688	3.92	1.33	1	6
<i>Network and contacts with family members</i>	3,698	4.26	1.25	1	6
<i>Social participation—organized groups</i>	3,642	2.33	1.98	1	7
<i>Social participation—volunteering</i>	3,647	2.18	1.93	1	7
<i>Frequency of going out</i>	3,701	3.33	.94	1	4
Covariates					
Racial/ethnic					
<i>Non-Hispanic Whites</i>		87.6			
<i>Non-Hispanic African Americans</i>	3,576	6.1			
<i>Hispanics</i>		2.2			
<i>Others</i>		4.2			
Income (Median and IQR)	2,859	6	3	1	8
Male	3,701	45.0			
Age	3,701	73.37	8.38	60	98
Self-rated health (Median and IQR)	3,690	4	1	1	5
Education (Median and IQR)	3,420	4	2	1	6
Housing type					
<i>Active adult retirement housing</i>		8.5			
<i>Affordable and subsidized senior housing</i>	3,701	2.6			
<i>Other residential communities</i>		89.0			
Unit type					
<i>Apartment or townhouse</i>	3,701	1.8			
<i>Independent House</i>		89.2			

Urban/rural		
<i>Urbanized areas</i>	3,701	92.3
<i>Urban clusters</i>		2.9
<i>Rural areas</i>		4.9

Notes: The two social isolation measures were calculated after multiple imputation. Medians and IQRs (interquartile range) are reported instead of means and standard deviations for income, self-rated health, and education.

Socioeconomic statistics and the three living environment indicators are listed under covariates in Table 4. Among the sample, 87.6% were non-Hispanic Whites, 6.1% were non-Hispanic African Americans, 2.2% were Hispanics, and 4.2% were others. The median household income was rated as 6 on a 1-to-8 scale, equating to “\$50,000 to \$75,000.” Men comprised 45% of the sample and the average age was 73.37. The median self-rated physical health was 4, or “very good.” The median education level was 4, “some college.” Regarding housing type, 8.5% lived in active adult retirement housing, 2.6% lived in affordable or subsidized housing, and the majority (89%) lived in non-age or income-restricted residential communities. Those living in an apartment or townhouse were 10.8% of the sample, and 89.2% lived in independent housing. The majority (92.3%) of the NJ sample lived in urbanized areas, 2.9% lived in urban clusters, and only 4.9% lived in rural areas.

Neighborhood indicators are summarized in Table 5. Statistics of our sampled census tracts were contrasted with all census tracts in New Jersey. The sampled older adults lived in 1,431 census tracts out of 1,999 census tracts. In general, the economic and living conditions of the sampled census tracts were better than the average. The deprivation index was calculated as a standardized composite score based on all census tracts. The mean deprivation index of the sampled census tracts was -.24, lower than the

average of all census tracts. The median household income for sampled census tracts was 82910.7, higher than the statewide average of 76591.3. The sampled census tracts also had a lower diversity rating of .40 compared to the statewide diversity rating of .42. The sampled census tracts had an average of 11.3% African American population compared to the statewide average of 14.4%. Even though the census tracts in the sample had a higher economic rating overall, they had a slightly lower Gini coefficient of .41 (the statewide average is .42). Regarding changes in economic conditions and demographics in the past 15 years, the sampled census tracts had a greater income increase of 22264.3 compared to 20806.3 statewide. The sampled tracts also experienced a greater increase of diversity, .09 compared to .07 statewide. In general, the percentage of Whites in the state decreased from 2000 to 2015, but the sampled census tracts experienced a greater decrease of -9.0 compared to the average of -8.5. The sampled census tracts were also less densely populated, with a population density of 5719.6 people per square mile, than the statewide average of 8457.7 people per square mile. Residents in the sample also lived in slightly more stable census tracts, with 90.6% having remained at the same address for at least a year. The level of statewide stability was also high, with 89.9% living at the same address for at least a year. The percentage of older adults, aged 60 and above, was also higher in sampled census tracts, with the average being 22.3 compared to 20.7 statewide. One-sample t-tests were performed to compare the sampled census tracts with the statewide average. Significant differences were identified for all neighborhood indicators, suggesting that older adults generally tend to live in neighborhoods that are less deprived, less diverse, more equal, less populated, and that have a higher proportion of older adults.

Table 5 Descriptive statistics of neighborhood indicators

Neighborhood indicators	All census tracts n=1999 ^c		Sampled census tracts n=1431		One-sample t test ^d
	Mean	SD	Mean	SD	
Neighborhood deprivation ^a	0	1	-.24	.83	t = -11.1, p=.000
Median household income	76591.3	36294.0	82910.7	34792.0	t = 6.87, p=.000
Diversity	.42	.19	.40	.19	t = -3.16, p=.002
Percentage of Whites	56.6	30.6	64.5	26.6	t = 11.36, p=.000
Percentage of African Americans	14.4	21.8	11.3	18.4	t = -6.30, p=.000
Gini ^a	.42	.06	.41	.06	t = -2.75 p=.006
Δ Income ^b	20806.3	22747.7	22264.3	21244.2	t = 2.60, p=.010
Δ Diversity	.07	.11	.09	.10	t = 6.60, p=.000
Δ Percentage of Whites	-8.5	9.4	-9.0	8.8	t = -2.25, p=.025
Population density	8457.7	11951.6	5719.6	8659.8	t = -11.96, p=.000
Stability	89.8	6.3	90.6	5.6	t = 5.68, p=.000
Percentage over 60	20.6	9.9	22.3	10.2	t = 6.20, p=.000

Notes: a. Deprivation and Gini scores were estimated based on 1992 census tracts due to missing data. No participant from the survey resided in any of the seven census tracts with missing data. b. Income change was calculated using 2015 household income adjusted for 2015 inflation and 2000 household income adjusted for 1999 inflation. The income change does not reflect the real change or increase in dollars. However, the change is valid to compare across cases. c. Census tracts with no population or fewer than 100 residents were removed from the sample. d. one-sample t-tests were conducted to examine if the sampled census tracts are different from all census tracts. Due to the large number of indicators and the large sample size, the cutoff point of $p < .05$ tends to be easier to achieve and overestimates the group differences. Therefore, the p values are reported in the table instead of the significance levels.

Hot Spot Analysis: The clustering of loneliness and social isolation

Results of the hot spot analysis were mapped in Figure 1 to Figure 3. Generally, a higher z-score (red color) represents that the spatial clustering of those who reported greater loneliness or social isolation. A lower z-score (blue color) represents spatial clustering of those who reported less loneliness or social isolation. In this section, the spatial clustering and distribution of loneliness and social isolation are discussed first, and then the different patterns of clustering across loneliness, distal social isolation, and proximal social isolation are compared. The thematic maps regarding neighborhood static and dynamic indicators are shown in Appendices to explore possible relationships between neighborhood indicators with the concentration of lonely and/or isolated older adults.

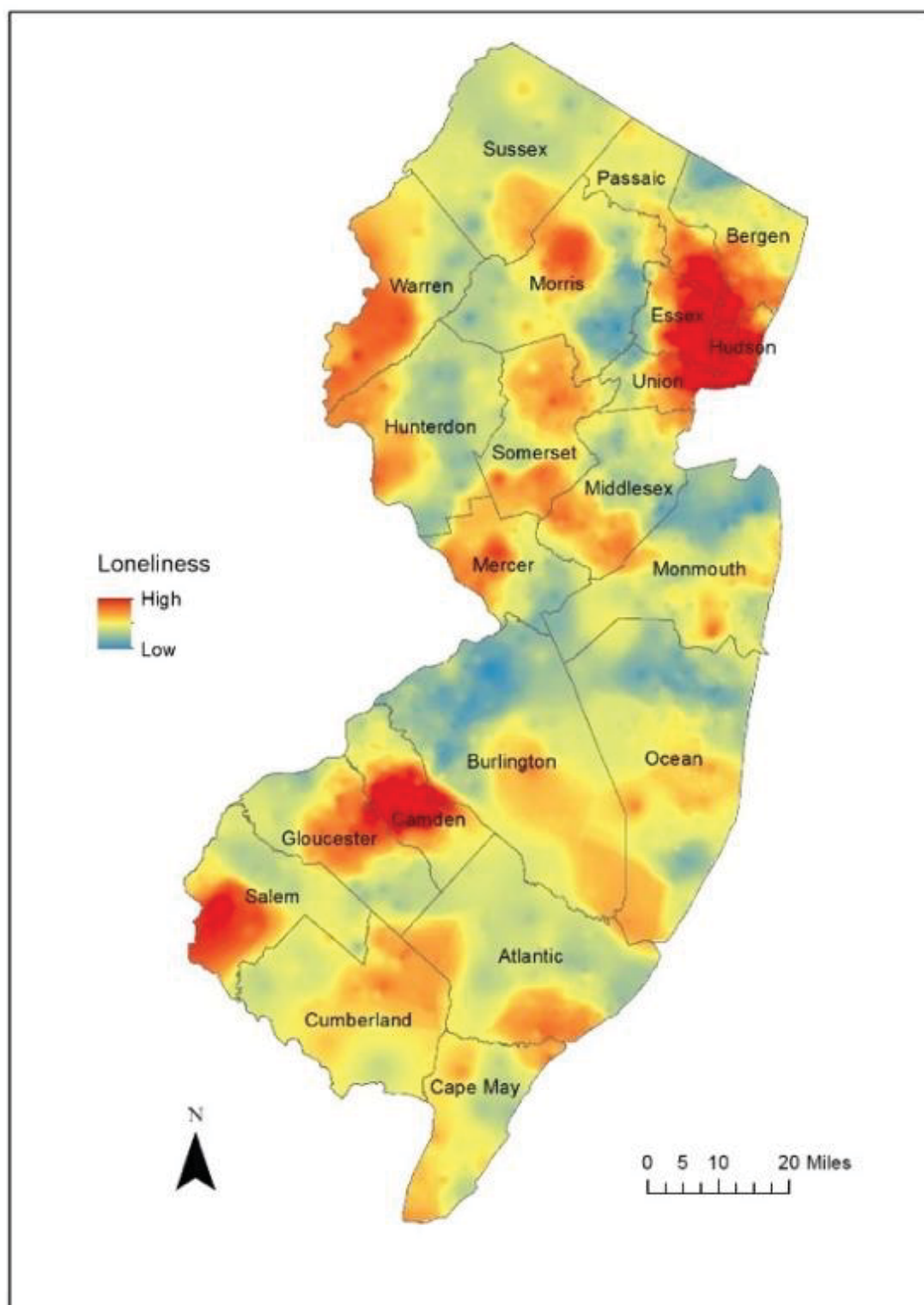


Figure 1 Hotspots of loneliness among older adults in NJ

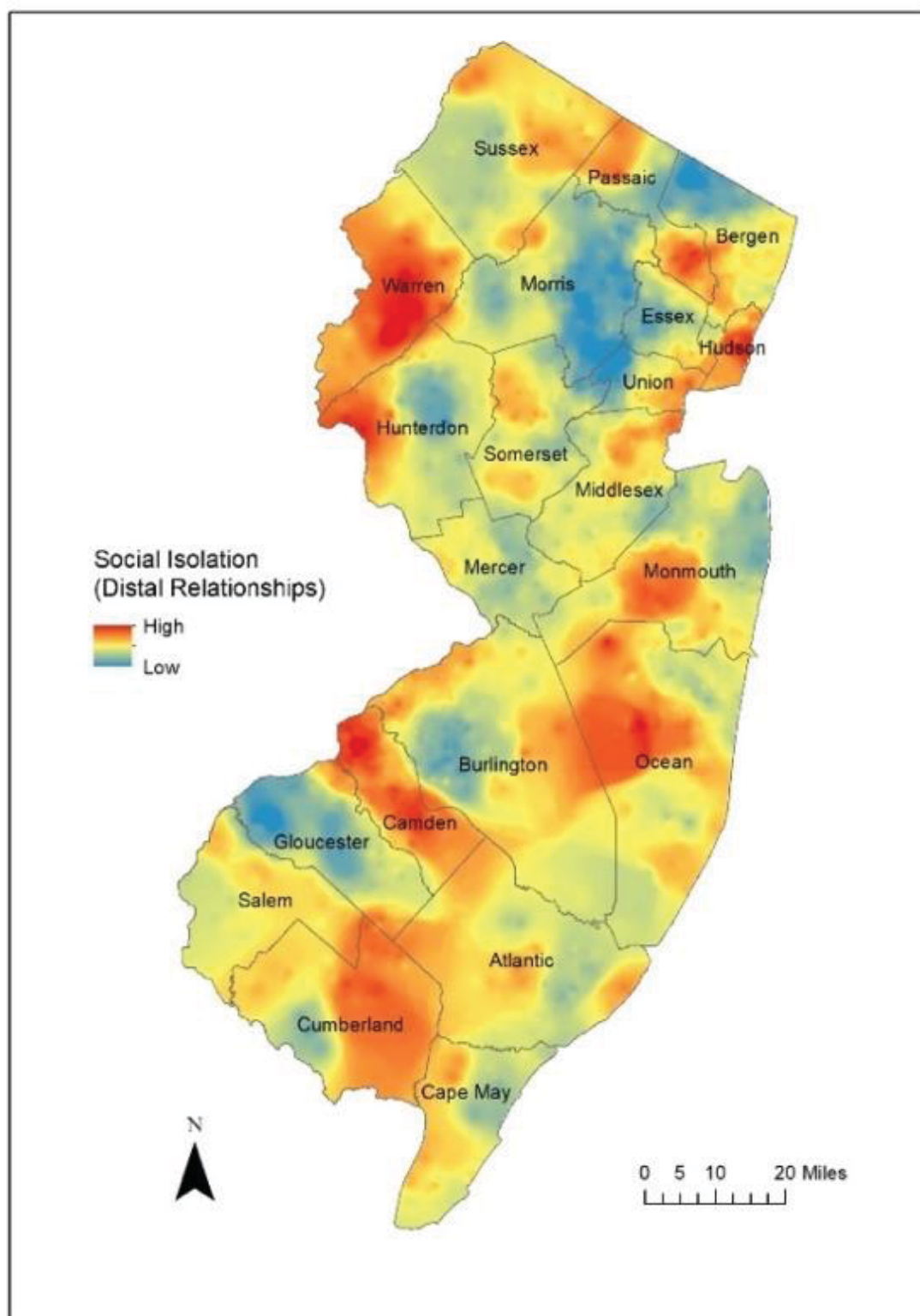


Figure 2 Hotspots of distal social isolation among older adults in NJ

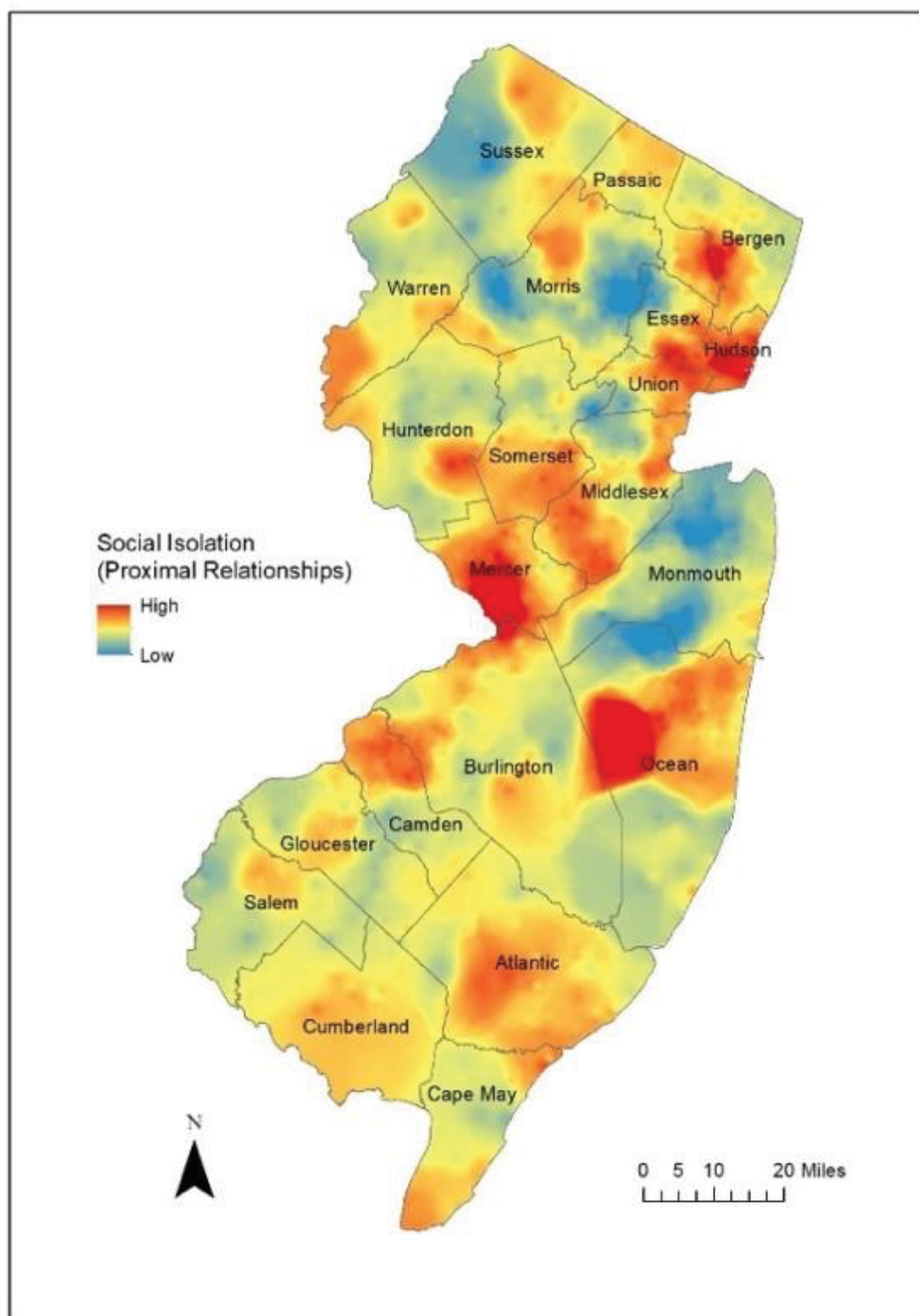


Figure 3 Hotspots of proximal social isolation among older adults in NJ

The clustering of loneliness, distal, and proximal social isolation

In the loneliness map (Figure 1), the hot spots representing clustering of loneliness among older adults were located primarily near New York urban areas in the northeast and near the Philadelphia and Camden areas in the southwest. The maps of neighborhood indicators indicates that the northeastern region was more deprived, had a higher percentage of African Americans, and was more densely populated (Appendix 1, 4, and 5). The southwestern hot spot shared some characteristics with the northeastern but also had some distinctive characteristics. It was not located in an urban region as was the case for the northeast hot spot but located in a suburban area in the middle of Camden and Gloucester Counties. It was also in a region with a high percentage of African Americans and a relatively high population density (Appendix 4 and 5).

The concentration of older adults who were isolated from distal relationships had a more scattered distribution (Figure 2). Hot spots were identified in Hudson and nearby regions, Warren County, Camden County, and parts of Ocean County. The Hudson hot spot, with a smaller scope than the hot spot of loneliness, includes the urban region near New York City. The Warren County hot spot, however, is in a quite different environment mainly comprised of White rural neighborhoods with low diversity but relatively higher deprivation compared to nearby neighborhoods (Appendix 1 and 2). The Camden region, similar to the Hudson region, is an urban region with surrounding suburban townships. The hot spot in the Ocean region had a concentration of older adults (Appendix 6); specifically, many retirement communities were located there. The hot spots of distal social isolation were located in very different environments across the state. However, some hot spots were near county boundaries. One typical example is the

Camden County hot spot of isolated older adults stands in contrast to the adjacent Gloucester County cold spot of isolated older adults, which suggests that county administration or policies may affect distal social connections among older adults.

Proximal social isolation was less likely to be affected by neighborhood characteristics, but those with an absence of proximal relationships may prefer certain kinds of neighborhoods and are therefore concentrated in certain areas. The map of proximal social isolation identified hot spots in the northeast near New York City, Trenton in Mercer County, and the west part of Ocean County (Figure 3). The first two regions were typical urban environments with high population density, high deprivation and diversity, a concentration of African Americans, and low stability (Appendix 1, 2, 4, 5, and 7). The Ocean County hot spot, consistent with the hot spot of distal social isolation, had a very high concentration of older adults (Appendix 6) living in retirement communities. The distribution of older adults who tended to live alone or/and without a partner suggests that they may choose to stay in either urban regions or aging-friendly places with easy access to services and facilities.

Comparisons across the clustering of loneliness, distal, and proximal social isolation

In general, older adults reporting high levels of loneliness and/or isolation were more likely to live in urban or surrounding suburban regions with high population density, greater neighborhood deprivation, and a concentration of African Americans. Disparities regarding loneliness relative to the two types of social isolation are also discernible. For example, older adults in Warren County were isolated from distal social connections but remained connected within their proximal or intimate relationships (Figures 2 and 3). These respondents also reported relatively high loneliness (Figure 1).

In contrast, respondents near Trenton in Mercer county reported proximal social isolation but being connected in distal relations and engaged in social activities (Figures 2 and 3). Even though they also reported relatively high loneliness, the sources of loneliness are apparently different. Those living in western Ocean County where there was a concentration of older adults reported being isolated from both distal and proximal relationships; however, they reported relatively low loneliness (Figures 1, 2, and 3).

The concentrations of loneliness and social isolation can be aligned with the boundaries of counties or with smaller geographic units. The examples of distal social isolation in Gloucester and Camden Counties exemplified how distal social isolation can be correlated to administrative boundaries (Figure 2). In contrast, the hot spot of loneliness across the Camden-Gloucester region and across the northeast Bergen-Essex-Hudson-Union region exemplified how across-county similarities and within-county differences must be accounted for (Figure 1). In that case, looking at the average level of loneliness and social isolation at the county level can miss important information for research or practice. Smaller geographic units should be used to examine the patterns of loneliness and social isolation as well as to explore related risk or protective factors.

Bivariate analyses loneliness, social isolation, and neighborhood characteristics

Correlation matrices for loneliness/social isolation measures and neighborhood-level indicators are examined in Table 6 for the full sample. Deprivation and population density were positively associated with all three measures: loneliness, distal social isolation, and proximal social isolation. Diversity was only positively associated with loneliness and was isolated from distal relationships. Percentage of African Americans was positively associated with loneliness and proximal social isolation. Gini, increase in

percentage of Whites, and percentage of older adults was only associated with proximal social isolation. Income increase was negatively associated with all three outcome indicators. Stability was negatively associated with only social isolation regarding both distal and proximal relationships.

Table 6 Correlation matrix for all sampled older adults

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Loneliness	-												
2. Distal social isolation	.26***	-											
3. Proximal social isolation	.16***	-	-										
4. Deprivation	.07***	.11***	.12***	-									
5. Diversity	.05**	.05**	.03+	.24***	-								
6. % of African Americans	.07***	.03+	.04*	.54***	.34***	-							
7. Gini	-.01	-.03+	.05**	.11***	-.02	.09***	-						
8. Δ Income	-.04*	-.06***	-.07***	-.50***	-.11***	-.24***	-.10***	-					
9. Δ Diversity	-.00	.00	-.00	-.16***	.33***	-.14***	-.08***	.02	-				
10. Δ % of Whites	-.02	-.03+	-.03*	-.03+	-.64***	-.02	.10***	.05**	-.67***	-			
11. Population density	.06***	.05**	.07***	.37***	.27***	.28***	.14***	-.07***	-.15***	-.08***	-		
12. Stability	-.02	-.04*	-.06***	-.36***	-.22***	-.24***	-.24***	.20***	.06***	.03	-.22***	-	
13. % of 60+	-.01	.02	.07***	-.00	-.38***	-.20***	.16***	-.15***	-.00	.16***	-.20***	.09***	-

Notes: *** p<.001, ** p<.01, * p<.05, + p<.10

Separated correlation matrices were developed for Whites and African Americans. For Whites, the pattern was similar to the full sample, as the majority of the sample were Whites (87.6%) (see in Table 7). Deprivation and population density were consistently associated with higher levels of loneliness and social isolation. Diversity and percentage of African Americans were only positively associated with loneliness and distal social isolation. Higher Gini was associated with lower distal social isolation, meaning more social engagement, which suggested the opposite direction of relationship to the hypothesis. Higher Gini was also associated with higher proximal social isolation. Increase in income in a neighborhood was associated with lower distal and proximal social isolation among older adults. Older adults who lacked proximal relationships were more likely to live in neighborhoods with a smaller increase in percentage of Whites but also with low stability. They were also more likely to live in neighborhoods with a higher proportion of older adults.

Table 7 Correlation matrix for White older adults

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Loneliness	-												
2. Distal social isolation	.28***	-											
3. Proximal social isolation	.16***	-.01	-										
4. Deprivation	.06**	.12***	.10***	-									
5. Diversity	.04*	.05**	.03+	.26***	-								
6. % of African Americans	.06**	.04*	.02	.42***	.54***	-							
7. Gini	-.01	-.05**	.04*	-.03	-.03	.03	-						
8. Δ Income	-.02	-.06***	-.06***	-.51***	-.09***	-.19***	-.08***	-					
9. Δ Diversity	-.00	-.00	-.00	-.09***	.36***	-.04*	-.07***	-.01	-				
10. Δ % of Whites	-.02	-.01	-.04*	-.11***	-.67***	-.19***	.08***	.07***	-.73***	-			
11. Population density	.07***	.04*	.05**	.29***	.31***	.15***	.10***	-.03+	-.09***	-.13***	-		
12. Stability	-.02	-.03+	-.07***	-.31***	-.23***	-.22***	-.21***	.19***	.04*	.05**	-.17***	-	
13. % of 60+	-.00	.02	.07***	.09***	-.39***	-.19***	.20***	-.18***	-.04*	.19***	-.17***	.06***	-

Notes: *** p<.001, ** p<.01, * p<.05, + p<.10

Correlations for African Americans are shown in Table 8. Due to the smaller sample size ($n=217$), correlations at a significance level of .10 were also interpreted. Neighborhood deprivation remained positively correlated with all three indicators: loneliness, distal social isolation, and proximal social isolation. However, no additional neighborhood indicator was significantly associated with loneliness among African American older adults. Increase in neighborhood income, increase in percentage of Whites, and high stability were all associated with lower distal social isolation for African American older adults. Gini and population density were only associated with proximal social isolation. African American older adults who lived alone or did not have a partner tended to live in areas with high Gini and population density, mostly urban environments.

Table 8 Correlation matrix for African American older adults

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Loneliness	-												
2. Distal social isolation	.28***	-											
3. Proximal social isolation	.19**	-.01	-										
4. Deprivation	.15*	.17*	.16*	-									
5. Diversity	-.02	.08	-.07	-.09	-								
6. % of African Americans	.07	.03	.07	.48***	-.38***	-							
7. Gini	.02	.07	.20**	.60***	-.07	.28***	-						
8. Δ Income	-.05	-.16*	-.03	-.59***	.06	-.38***	-.40***	-					
9. Δ Diversity	.02	-.04	.01	-.27***	.26***	-.26***	-.10	.25***	-				
10. Δ % of Whites	.00	-.13+	.03	.32***	-.36***	.41***	.33***	-.19**	-.16*	-			
11. Population density	.06	-.05	.13*	.47***	-.17*	.34***	.21**	-.21**	-.32***	.11	-		
12. Stability	-.01	-.12+	-.06	-.44***	-.03	-.13+	-.33***	.24***	.05	-.05	-.28***	-	
13. % of 60+	-.02	.01	.02	-.38***	-.15*	-.20**	-.11	.10	.11	-.07	-.31***	.28***	-

Notes: *** p<.001, ** p<.01, * p<.05, + p<.10

The bivariate analysis confirmed some of the relationships observed in the hot spot analysis. High neighborhood deprivation and population density, both characteristics of urban settings, were positively associated with loneliness and social isolation among older adults. Concentration of minorities and high diversity were associated with loneliness and social isolation for White older adults, and in addition, White older adults who lived alone or did not have a partner also tended to live in neighborhoods with a higher proportion of older adults, as in the Ocean County retirement community region.

Although one of the primary neighborhood indicators, neighborhood deprivation, showed consistent positive association with loneliness and social isolation, most indicators showed different patterns between Whites and African Americans. For example, White older adults were more sensitive to racial/ethnic composition and the proportion of minorities in their neighborhoods, while only isolation from distal relations among African American older adults, and then only marginally, was associated with an increase in the percentage of Whites. White older adults' feelings of loneliness and their isolation status were all associated with population density, while for African American older adults, population density was only associated with proximal social isolation. The difference suggests racial differences between Whites and African Americans for the neighborhood effects on loneliness and social isolation among older adults.

The results from the bivariate analysis can be biased due to missing both individual and neighborhood-level covariates, and there is a major concern regarding the association observed for neighborhood deprivation. Neighborhood deprivation both represents disadvantage at the neighborhood level and the concentration of older adults who are not affluent. The association may be due to the aggregation of population and

thereby function as a compositional effect rather than a contextual effect. In addition, the correlations across neighborhood indicators in the three correlation matrices were mostly small to moderate, suggesting that many of the observed correlations can be due to other neighborhood-level confounding indicators. Therefore, both individual and neighborhood-level covariates were controlled in the following multivariate analyses to reduce the aforementioned bias.

Multivariate regressions

The results predicting loneliness, distal social isolation, and proximal social isolation are shown from Table 9 to Table 14. Different predictors were observed across loneliness, distal social isolation and proximal social isolation, and different patterns were identified between White older adults and African American older adults.

Table 9 Multilevel linear regression models predicting loneliness among White older adults (n=3,076)

	Model 1		Model 2		Model 3		Model 4		Model 5		Robust Test	
	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE
Independent variables												
Deprivation			.07	.05	.04	.05	-.06	.05	-.06	.05	-.04	.05
Diversity			-4.21**	1.50	-4.37**	1.52	-4.06**	1.49	-4.87***	1.42	-4.59**	1.43
Diversity^2			11.46**	4.27	11.67**	4.30	10.71*	4.19	12.80**	4.01	12.27**	4.01
Diversity^3			-8.46*	3.64	-8.73*	3.65	-7.98*	3.56	-9.78**	3.41	-9.41**	3.41
% African Americans*1000			2.78	2.99	4.12	3.02	4.08	2.95	5.51+	2.82	-	-
Gini			-.42	.45	-.62	.47	-.16	.47	.01	.45	-.01	.45
Δ Income*10^6			.41	12.80	-.01	1.29	-.02	1.26	.19	1.20	.17	1.21
Δ Diversity			.41	.40	.57	.41	.27	.40	.55	.38	.13	.27
Δ Percentage of Whites*1000			7.08	5.48	7.51	5.49	3.66	5.36	5.38	5.14	-	-
Neighborhood-level covariates												
Population density*10^5					1.21**	.44	1.09*	.44	1.07*	.42	.98*	.41
Stability*1000					-.69	5.20	-.84	5.09	.81	4.87	1.29	4.87
% 60+					.07	.21	.12	.24	.03	.23	.02	.23
Individual-level covariates												
Income							-.09***	.01	-.03+	.01	-.03+	.01
Male							-.05	.04	-.03	.04	-.03	.04
Age*10							-.09***	.03	-.15***	.03	-.15***	.03
Self-rated health							-.19***	.02	-.16***	.02	-.16***	.02
Education							.04+	.02	.03+	.02	.04+	.02
Housing type												
Active adult retirement housing							-.08	.10	-.08	.09	-.09	.09
Affordable and subsidized senior housing							-.31+	.18	-.42*	.17	-.42*	.17
(Other residential communities)												
Unit type												
Apartment or townhouse							.11	.08	-.03	.08	-.03	.08

<i>(Independent house)</i>													
Urban/rural													
<i>(Urbanized area)</i>													
<i>Urban cluster</i>													
<i>Rural area</i>													
Distal social isolation													
Proximal social isolation													
Intercept	.64***	.02	1.20***	.25	1.29*	.58	2.80***	.61	2.67***	.59	2.72***	.59	
σ_u	.289		.261		.257		.245		.232		.240		
σ_e	1.157		1.157		1.157		1.123		1.079		1.079		
ICC	.059		.048		.047		.046		.044		.047		

Notes: *** p<.001, ** p<.01, * p<.05, + p<.10

Robust models based on model 5 (covariates were omitted)

Table 10 Multilevel linear regression models predicting loneliness among African American older adults (n=211)

	Model 1		Model 2		Model 3		Model 4		Model 5	
	B	SE	B	SE	B	SE	B	SE	B	SE
<i>Independent variables</i>										
Deprivation			.29*	.14	.36*	.16	.12	.17	.09	.16
Diversity			16.89+	1.09	17.71+	1.29	-.01	.67	-.05	.65
Diversity^2			-44.61+	25.55	-46.09+	25.90	-	-		
Diversity^3			34.30+	19.67	35.27+	19.89	-	-		
% African Americans*1000			2.25	4.21	2.66	4.32	4.18	4.24	3.63	4.06
Gini			-1.53	2.01	-1.67	2.05	-1.55	2.05	-1.92	1.97
Δ Income*10^6			1.75	7.69	3.06	7.90	2.55	7.70	2.39	7.43
Δ Diversity			.89	1.00	.85	1.03	1.11	1.03	1.09	.99
Δ Percentage of Whites*1000			-1.31	13.81	-1.82	13.97	-4.77	13.75	3.45	13.30
<i>Neighborhood-level covariates</i>										
Population density*10^5					-.11	1.16	-.16	1.22	-.21	1.17
Stability*1000					8.67	17.45	12.23	17.19	1.57	16.43
% 60+					1.10	1.42	.30	1.49	.19	1.43
<i>Individual-level covariates</i>										
Income							-.15*	.06	-.10	.07
Male							-.21	.19	-.21	.18
Age*10							-.34**	.12	-.45***	.12
Self-rated health							.02	.09	.02	.09
Education							-.03	.08	-.02	.08
Housing type										
Active adult retirement housing							-.38	.61	-.26	.59
Affordable and subsidized senior housing							.64+	.34	.51	.33

<i>(Other residential communities)</i>										
Unit type										
<i>Apartment or townhouse</i>							.03	.25	-.09	.25
<i>(Independent house)</i>										
Urban/rural										
<i>(Urbanized area)</i>										
<i>Urban cluster</i>							.20	1.01	.16	.98
<i>Rural area</i>							-	-	-	-
Distal social isolation									.32***	.09
Proximal social isolation									.27**	.10
Intercept	.79***	.10	-.65	1.51	-1.78	2.19	3.23	2.13	4.21*	2.08
σ_u^a	1.003		.999		1.010		1.020		.956	
σ_e	.976		.976		.976		.889		.888	
ICC	.514		.512		.517		.568		.537	

Notes: *** p<.001, ** p<.01, * p<.05, + p<.10

a. σ_u can be overestimated in the models

Table 11 Multilevel linear regression models predicting distal social isolation among White older adults ($n=3,133$)

	Model 1		Model 2		Model 3		Model 4		Model 5		Robust test	
	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE
Independent variables												
Deprivation			.18***	.04	.16***	.04	-.03	.04	-.02	.04	-.04	.04
Diversity			.38*	.17	.45*	.18	.41*	.17	.40*	.17	.27*	.12
% African Americans*1000			-3.46	2.44	-3.27	2.47	-2.77	2.26	-3.02	2.25	-	-
Gini			-		-		-		-		-	
			1.06**	.37	-1.23**	.40	-.61+	.37	-.60+	.36	-.59	.36
Δ Income*10 ⁶			-.71	1.06	-.55	1.07	-.52	.97	-.56	.97	-.55	.97
Δ Diversity			.29	.31	.26	.32	-.28	.29	-.37	.29	-.38+	.21
Δ Percentage of Whites*1000			8.35+	4.39	8.35+	4.41	2.97	4.04	1.65	4.03	-	-
Neighborhood-level covariates												
Population density*10 ⁵					.15	.36	.04	.34	.04	.34	.01	.33
Stability*1000					-1.19	4.37	-2.76	3.99	-3.19	3.97	-3.22	3.96
% 60+					.28	.17	.28	.18	.28	.18	.28	.18
Individual-level covariates												
Income							-.11***	.01	-.13***	.01	-.13***	.01
Male							.11**	.03	.09**	.03	.09**	.03
Age*10							-		-		-	
							1.58***	.34	1.48***	.34	1.46***	.34
Age*Age*100							.11***	.02	.10***	.02	.10***	.02
Self-rated health							-.15***	.02	-.14***	.02	-.14***	.02
Education							-.08***	.02	-.06***	.02	-.07***	.02
Housing type												
Active adult retirement housing							-.11	.08	-.09	.07	-.08	.07
Affordable and subsidized senior housing							.05	.14	.09	.14	.09	.14

<i>(Other residential communities)</i>													
Unit type													
<i>Apartment or townhouse</i>								.18**	.07	.22**	.07	.22**	.07
<i>(Independent house)</i>													
Urban/rural													
<i>(Urbanized area)</i>													
<i>Urban cluster</i>								.06	.10	.05	.10	.05	.10
<i>Rural area</i>								.15+	.08	.14+	.08	.14+	.08
Proximal social isolation										-.11***	.02	-.11***	.02
Intercept	-.01	.02	.45**	.17	.52	.47	7.42***	1.35	7.00***	1.34	6.94***	1.34	
σ_u	.285		.252		.251		.180		.177		.177		
σ_e	.963		.963		.963		.895		.889		.889		
ICC	.080		.064		.064		.039		.038		.038		

Notes: *** p<.001, ** p<.01, * p<.05, + p<.10

Table 12 Linear regression models predicting distal social isolation among African American older adults (n=217)

	Model 1		Model 2		Model 3		Model 4		Model 5		Robust test	
	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE
Independent variables												
Deprivation			.19+	.10	.24*	.11	.04	.12	.04	.12	.06	.12
Diversity			.19	.45	.33	.46	.45	.45	.41	.45	-	-
% African Americans*1000											-.35	2.66
Gini			-.40	2.84	.40	2.90	.31	2.82	.46	2.82		
			-.33	1.40	-.78	1.42	-.40	1.37	-.21	1.38	-.18	1.37
Δ Income*10 ⁶			-6.13	5.52	-4.33	5.66	-4.43	5.32	-3.83	5.33	-4.26	5.25
Δ Diversity			-.12	.68	-.27	.70	-.21	.91	-.17	.68	-	-
Δ Percentage of Whites			-.02*	.01	-.02*	.01	-.02*	.01	-.02*	.01	-.02**	.01
Neighborhood-level covariates												
Population density*10 ⁵					-.33	.82	.01	.83	.02	.83	.04	.80
Stability*1000					-8.75	11.92	-2.10	11.31	-1.57	11.31	-1.37	11.24
% 60+					1.41	1.00	1.03	1.02	.96	1.02	.85	1.01
Individual-level covariates												
Income							-.05	.04	-.06	.04	-.06	.04
Male							.00	.14	.00	.14	.00	.14
Age*10							.04	.09	.06	.09	.07	.09
Self-rated health							-.08	.06	-.07	.06	-.07	.06
Education							-.20**	.06	-.19**	.06	-.18**	.06
Housing type												
Active adult retirement housing							-.84*	.41	-.77+	.41	-.85*	.40
Affordable and subsidized senior housing (Other residential communities)							.18	.24	.23	.24	.23	.24
Unit type												
Apartment or townhouse (Independent house)							.05	.18	.09	.18	.08	.18

Urban/rural													
<i>(Urbanized area)</i>													
<i>Urban cluster</i>							1.13+	.58	1.07+	.58	1.05+	.58	
<i>Rural area</i>							-1.07	.98	-.96	.98	-1.00	.98	
Proximal social isolation									-.09	.07	-.09	.07	
Intercept													
σ_u^a													
σ_e													
ICC													

Notes: Model 1 was estimated as multilevel linear model. Model 2 to Model 5 were estimated as OLS models.

*** p<.001, ** p<.01, * p<.05, + p<.10

a. σ_u can be overestimated in the models

Table 13 Multilevel linear regression models predicting proximal social isolation among White older adults (n=3,133)

	Model 1		Model 2		Model 3		Model 4		Robust test 1		Robust test 2	
	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE
Independent variables												
Deprivation			.16***	.04	.11**	.04	.05	.04	.05	.04	.06+	.04
Diversity			-.03	.17	.06	.18	-.11	.16	.06	.12	-	-
% African Americans*1000			-3.48	2.44	-3.12	2.46	-2.26	2.24	-	-	-2.17	1.96
Gini			.76*	.37	.27	.39	.08	.36	.05	.36	.09	.36
Δ Income*10 ⁶			-.59	1.05	-.20	1.06	-.43	.95	-.44	.96	-.32	.96
Δ Diversity			-.50	.31	-.58+	.32	-.81**	.29	-.14	.21	-	-
Δ Percentage of Whites*1000			-9.32*	4.39	-9.86*	4.40	-12.16**	3.99	-	-	-3.56	2.18
Neighborhood-level covariates												
Population density*10 ⁵					.31	.36	-.36	.34	.12	.33	.12	.33
Stability*1000					-9.23*	4.34	-4.03	3.93	-2.91	3.96	-3.56	3.92
% 60+					.58***	.17	.01	.18	.02	.18	-.00	.17
Individual-level covariates												
Income							-.15***	.01	-.15***	.01	-.14***	.01
Male							-.21***	.03	-.21***	.03	-.21***	.03
Age*10							.96**	.34	.97**	.34	.97**	.34
Age ² *100							-.05*	.02	-.05*	.02	-.05*	.02
Self-rated health							.08***	.02	.08***	.02	.08***	.02
Education							.13***	.02	.13***	.02	.13***	.02
Housing type												
Active adult retirement housing							.18*	.07	.18*	.07	.18*	.07

<i>Affordable or subsidized senior housing (Other residential communities)</i>								.38**	.14	.39**	.14	.38**	.14
Unit type													
<i>Apartment or townhouse (Independent house)</i>								.38***	.07	.38***	.07	.38***	.07
Urban/rural													
<i>(Urbanized area)</i>													
<i>Urban cluster</i>								-.03	.10	-.03	.10	-.03	.10
<i>Rural area</i>								-.10	.08	-.10	.08	-.09	.08
Intercept	.00	.02	-.23	.16	.60	.46	-3.89**	1.34	-4.06**	1.35	-3.99**	1.34	
σ_u	.259		.224		.220		.140		.161		.153		
σ_e	.964		.964		.964		.890		.890		.890		
ICC	.067		.051		.050		.024		.032		.029		

Notes: *** p<.001, ** p<.01, * p<.05, + p<.10

Table 14 Multilevel linear regression models predicting proximal social isolation among African American older adults (n=217)

	Model 1		Model 2		Model 3		Model 4	
	B	SE	B	SE	B	SE	B	SE
<i>Independent variables</i>								
Deprivation			.15	.11	.17	.12	.09	.12
Diversity			-.61	.48	-.43	.50	-.48	.47
% African Americans*1000			-.02	3.12	-.31	3.17	1.47	2.95
Gini			3.11*	1.51	3.04*	1.54	2.12	1.42
Δ Income*10 ⁶			7.43	5.84	7.75	5.98	7.20	5.39
Δ Diversity			.653	.74	.72	.76	.01	.71
Δ Percentage of Whites*1000			-11.57	1.41	-1.07	1.49	-8.24	9.59
<i>Neighborhood-level covariates</i>								
Population density*10 ⁵					1.13	.88	.30	.84
Stability*1000					9.97	13.15	7.25	12.00
% 60+					1.10	1.05	-.56	1.04
<i>Individual-level covariates</i>								
Income							-.11*	.04
Male							-.02	.13
Age*10							.32***	.08
Self-rated health							.11+	.06
Education							.16**	.06
Housing type								
Active adult retirement housing							.73+	.41
Affordable and subsidized senior housing (Other residential communities)							.35	.23
Unit type								

<i>Apartment or townhouse</i>							.42*	.17
<i>(Independent house)</i>								
Urban/rural								
<i>(Urbanized area)</i>							-.66	.58
<i>Urban cluster</i>							1.20	.98
<i>Rural area</i>								
Intercept	.02	.08	-1.32+	.71	-2.58+	1.45	-4.44**	1.47
σ_u^a	.842		.822		.821		.714	
σ_e	.680		.680		.680		.602	
ICC	.605		.593		.593		.585	

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$, + $p < .10$

a. σ_u can be overestimated in the models

The ICCs represent the percentage of variance of the outcome variable that can be attributed to the differences at the census tract level. The ICCs for African American samples were not reported because, as stated by Clarke (2008), ICCs for samples with very small group size (<2) tend to be overestimated. In Table 9 Model 1, the ICC was 0.059, suggesting that 5.9% of variance of loneliness can be attributed to differences at the census tract level. From Table 9 Model 1 to Model 3, the included neighborhood indicators explained part of the variance at the census tract level, and the ICC was reduced from .059 to .048. The inclusion of individual indicators in Table 9 Model 5 reduced it further to .044, suggesting that older adults living in the same neighborhood reported similar sociodemographic status. Regarding distal social isolation among White older adults, the ICC was 0.080, suggesting that around 8.0% of variance of distal social isolation can be attributed to census tract-level differences (See Table 11). From Table 11 Model 1 to Model 3, the inclusion of neighborhood indicators reduced the ICC from .08 to .064. However, the inclusion of individual covariates dramatically reduced the ICC from .064 to .038, suggesting that the neighborhood effects on distal social isolation among White older adults are more compositional effects than contextual effects. In Table 12, the ICC for Model 1 was 0, and therefore the linear regression models were used to estimate distal social isolation for African American older adults. In Table 13 Model 1, the ICC was 0.067. The inclusion of neighborhood-level indicators in Model 3 reduced the value slightly to 0.050. In contrast, the inclusion of individual covariates in Model 4 reduced the ICC from .050 to .024, suggesting that the neighborhood effects on proximal social isolation among White older adults were also compositional effects.

Neighborhood deprivation

Results of the hot spot analyses and bivariate analyses suggest that the level of neighborhood deprivation was correlated with most of the loneliness and social isolation indicators. The multivariate analyses confirmed the observed relationships. However, after controlling for individual-level covariates, the effects of neighborhood deprivation became no longer significant for all models.

In models only including neighborhood-level indicators, neighborhood deprivation was positively associated with distal and proximal social isolation among White older adults (see Models 2 and 3 in Table 11 and Table 13) and loneliness and distal social isolation among African American older adults (see Models 2 and 3 in Tables 10 and 12). Based on models including key indicators and neighborhood-level covariates (Model 3), 1 unit increase in neighborhood deprivation was associated with 0.16 unit increase in distal social isolation and 0.11 unit increase in proximal social isolation among White older adults. For African American older adults, 1 unit increase in neighborhood deprivation was associated with 0.36 unit increase in loneliness and 0.24 unit increase in distal social isolation.

After controlling for individual-level covariates, however, none of the relationships remained significant. Consequently, it suggested that the association of neighborhood deprivation with loneliness and social isolation was due to the aggregation of older adults with similar individual characteristics within deprived neighborhoods. Their loneliness and social isolation status were therefore primarily attributed to their individual characteristics.

Racial/ethnic diversity

In this study, neighborhood diversity was a consistently significant predictor of loneliness and distal social isolation among White older adults (see Table 9 and Table 11). It was also a marginally significant predictor of loneliness among African American older adults before controlling for individual-level covariates (see Table 10).

In Table 9 Model 2 through Model 5, racial/ethnic diversity showed a cubic relationship with loneliness. The cubic relationship is demonstrated in Figure 4. Cutoffs were calculated based on the equation of Model 5. Loneliness decreased with the increase of diversity from 0 to 0.28. Then loneliness increased with the increase of diversity from 0.28 to 0.59. With a continuing increase in diversity from 0.59 to 0.8 (equal population in each racial/ethnic group), loneliness among White older adults decreased again. A closer look at the neighborhood racial/ethnic structure is reported beneath Figure 4. The statistics were calculated using 1,258 census tracts that any sampled White older adults resided in (see Table 15). About 47.3% ($n=595$) of census tracts were in the stage when racial/ethnic diversity was positively associated with loneliness. The majority were still White-dominated neighborhoods, but the average percentage of Whites was relatively low at 68.2%.

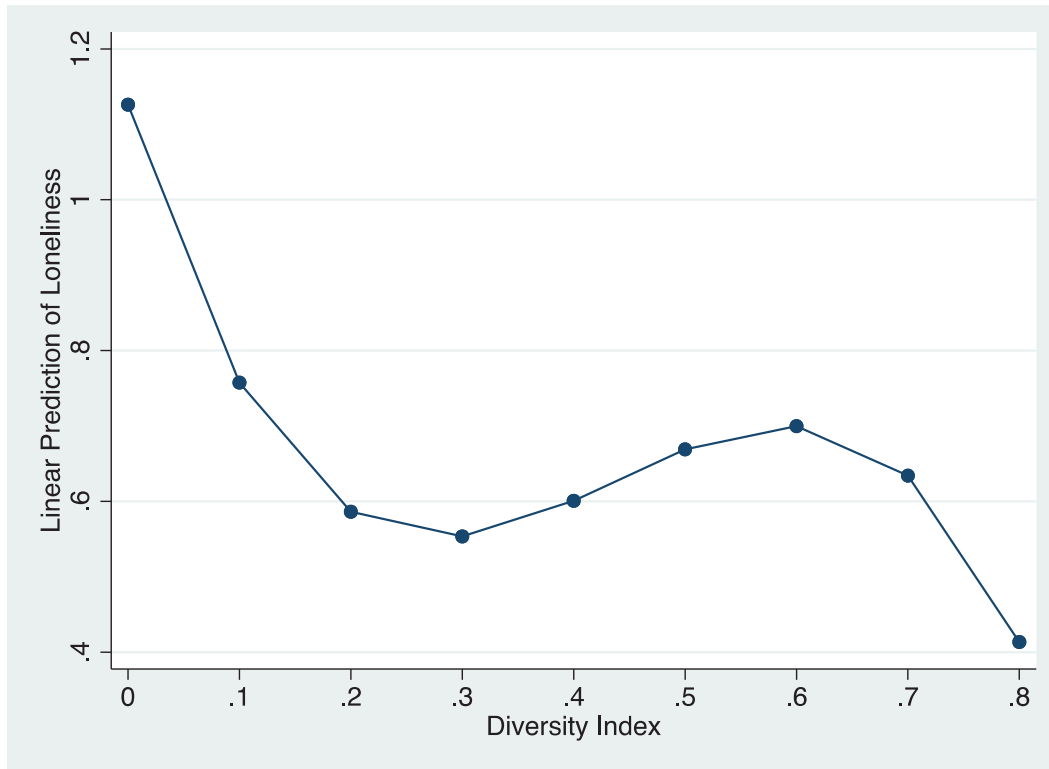


Figure 4 Linear prediction of loneliness by diversity among White older adults (based on Table 9 Model 5)

Table 15 Racial/ethnic composition by neighborhood diversity cutoffs from Figure 4

	0-.28 n=401	.28-.59 n=595	.59-.8 n=262
Average of % Whites	89.4	68.2	41.3
% of census tracts where Whites were the dominant group (>50%)	99.0	90.6	28.2
% mixed neighborhoods (no dominant racial/ethnic groups)	0	1.2	66.4

In contrast, for African American older adults, as indicated in Table 10 Model 2 and Model 3, diversity had a marginally cubic relationship with loneliness, and the relationship was in the opposite direction from the cubic relationships among White older adults. The relationship is demonstrated in Figure 5 based on Table 10 Model 3. With an

increase of diversity from 0 to 0.29, loneliness increased dramatically for African American older adults. From 0.29 to 0.59, loneliness decreased with the increase in diversity. From 0.59 to 0.8, the increase of diversity continued to be associated with the increase in loneliness. Even though the thresholds of diversity were similar, however, the meaning is different compared to White older adults. The racial/ethnic structure is shown in Table 16. For example, most homogenous neighborhoods where African American older adults resided were African American dominant regarding percentage (66.7%). This cubic relationship was only marginally significant and not robust when the individual-level covariates were included.

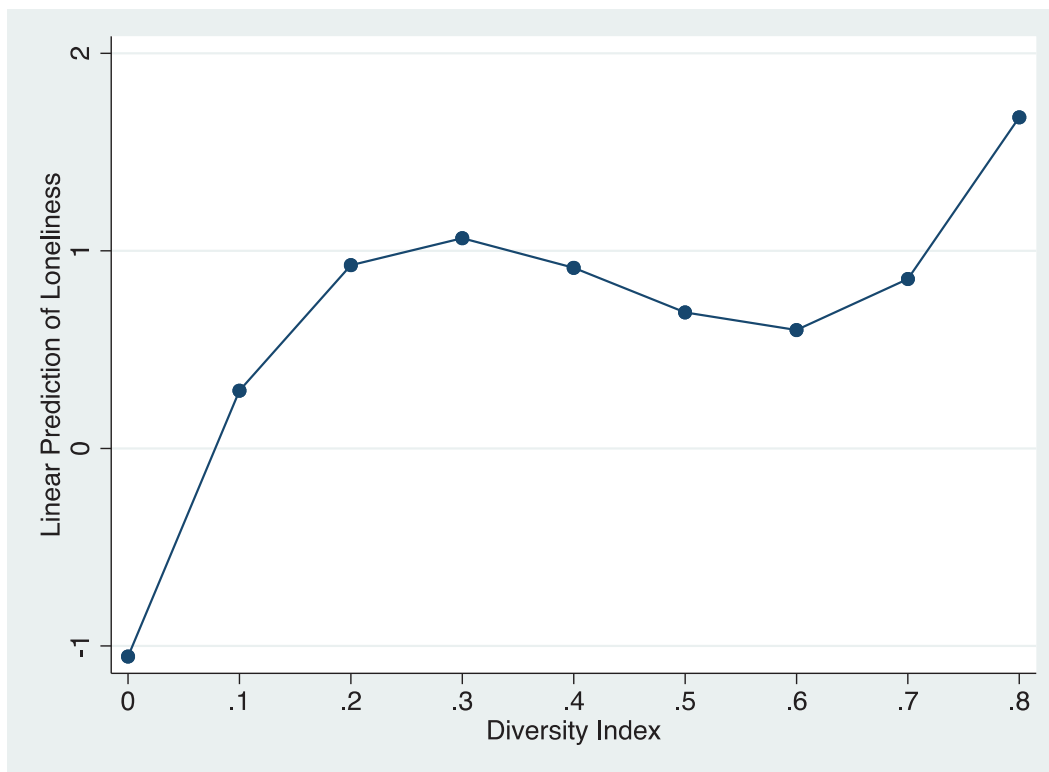


Figure 5 Linear prediction of loneliness by diversity among African American older adults (based on Table 10 Model 3)

Table 16 Racial/ethnic composition by neighborhood diversity cutoffs from Figure 5

	0-.29 N=39	.29-.59 n=97	.59-.80 n=54
% Whites	24.4	4.2	3.7
% blacks	6.8	31.4	29.2
% Whites dominate	25.6	49.5	11.1
% African American dominate	66.7	3.9	7.4
% mixed	0	2.1	77.8

In Table 11 Model 2 through Model 5, a positive relationship was observed between diversity and distal social isolation among White older adults. Based on Model 5, 1 unit increase in diversity was associated with .43 unit increase in distal social isolation. In the robust model, the coefficient was reduced to .27 but the relationships remained significant.

Percentage of African Americans

Although bivariate analyses found that the percentage of African Americans in their neighborhood was associated with loneliness and social isolation indicators among White older adults, the relationships were not supported in the multivariate models. The relationships were not supported even in Model 2 with only key neighborhood indicators included, suggesting the observed bivariate correlations can be because census tracts with a concentration of African Americans were also those with greater neighborhood deprivation and diversity.

The only marginal significance is indicated in Table 9 after controlling for all individual covariates and social isolation indicators, suggesting a trend: White older adults felt greater loneliness when they lived in census tracts with a higher percentage of African Americans regardless of their social isolation status.

Income inequality

In this study, income inequality was not associated with loneliness for both Whites and African American older adults. Its association with social isolation, although statistically significant, had an opposite direction relationship to the hypotheses. In Table 11, higher Gini was associated with lower distal social isolation for White older adults, suggesting that those living in census tracts with higher income inequality were less likely to be isolated from distal relationships. After including individual covariates in Table 11 Models 4 and 5, the relationship became marginally significant. In the robust model, the effect size was reduced further and became insignificant, but the trend of the negative relationship persisted.

Positive associations between income inequality and proximal social isolation are reported in Tables 13 and 14 for both Whites and African American older adults. Older adults who were lacking proximal relationships tended to live in census tracts with higher income inequality. For White older adults, the relationship was significant only when the key neighborhood predictors were included (Table 13 Model 2). After controlling for neighborhood-level covariates (especially the significant predictors: stability and percentage of older adults), the relationship became non-significant. For African American older adults, in Table 14 Model 2 and Model 3, 1 unit increase in Gini was associated with about 3 units increase in proximal social isolation. The positive relationship became insignificant after controlling for individual covariates (Table 14 Model 4).

Neighborhood dynamic indicators

Change in census tract-level median household income was not significantly associated with loneliness or social isolation indicators in this study. The only robust relationship observed was between change in percentage of Whites and distal social isolation among African American older adults (Table 12). The increase of 10% of Whites in neighborhoods from 2000 to 2015 was associated with 0.2 standard deviation decrease in the distal social isolation, or more social contacts and engagement, among African American older adults. In models controlling for covariates and in the robust model, the association persisted with little change in effect size. The relationship was also in the opposite direction to the original hypotheses.

Other neighborhood indicators

Neighborhood population density was positively associated with loneliness among White older adults (Table 9) in this study. Based on the full model (Table 9 Model 5), an increase of ten thousand people per square mile at the census tract level was associated with a 1.07 increase in loneliness. However, population density was not associated with loneliness among African American older adults nor social isolation indicators for older adults of African Americans or Whites.

Stability and percentage of older adults were significantly associated with proximal social isolation among White older adults (Table 13). Those who lacked proximal social relationships tended to live in census tracts with lower stability and a higher percentage of older adults. The association was fully explained by individual characteristics, housing type, and unit type. The association reflected the concentration of proximally socially isolated older adults in the Ocean County retirement community regions as shown in Figure 3.

There were significant associations for housing type and unit type. For White older adults, those with proximal relationships tended to live in areas of older adult concentration, like active adult retirement and senior housing (Table 13). For African American older adults, those isolated from proximal relationships were also more likely to reside in active adult retirement housing than in other residential communities (Table 14). However, the difference between choosing senior housing or regular residential housing was not associated with proximal isolation status. Living in senior housing was associated with significantly lower rates of loneliness for White older adults than for those living in other residential communities (Table 9). African American older adults living in active adult retirement housing reported the lowest distal social isolation (Table 12). Whites and African American older adults who were proximally isolated were also more likely to live in apartments or townhouses over independent housing (Tables 13 and 14). For White older adults, unfortunately, those in apartments or townhouses also tended to be more distally isolated even after controlling for their proximal social isolation status (Table 11 Model 5).

Urban/rural differences were only associated with distal social isolation, and the association was only marginally significant in both the Whites and the African American models. In Table 11, White older adults in rural areas tended to be more isolated from distal relationships than those in urbanized areas and urban clusters. For African American older adults, however, those in urban clusters were more likely to be isolated from distal relationships compared to those who lived in urbanized areas or in rural areas.

Individual covariates

For both Whites and African American older adults, as expected, social isolation indicators were significant predictors for their loneliness. For Whites, 1 standard deviation increase in distal social isolation was associated with .31 score increase in loneliness, and 1 standard deviation increase in proximal social isolation was associated with .22 score increase in loneliness (Table 9). For African Americans, 1 standard deviation increase in distal social isolation was associated with .32 increase in loneliness, and 1 standard deviation increase in proximal social isolation was associated with .27 increase in loneliness (Table 10). For White older adults, proximal social isolation was negatively associated with distal social isolation, suggesting those who were absent a partner or lived alone may be more likely to expand their distal social relationships (Table 11).

Age was negatively associated with loneliness for both Whites and African American older adults (Tables 9 and 10). There was a quadratic relationship between age and distal social isolation among White older adults (Table 11), and study participants around the age of 74 tended to report the lowest distal social isolation. Age was not associated with distal social isolation among African American older adults (Table 12). White and African American respondents on the older end of the scale (the quadratic relationship in Table 13 suggested that the highest proximal social isolation was reached around age 96) tended to be more likely to be isolated from proximal social relationships.

No gender difference was identified for loneliness in this study, and there was no gender difference in social isolation indicators among African American older adults. For Whites, however, being male was associated with higher distal social isolation but lower proximal social isolation (Tables 11 and 13). The direction of the relationships were

consistent with the findings in previous studies. The difference between Whites and African American also reflected how gender difference varied across different groups.

Better physical health was associated with lower loneliness and less distal social isolation among White older adults (Table 9 and Table 11) in this research. However, the relationships were not significant among African American older adults. In addition, better physical health was associated with higher proximal social isolation for both races (Tables 13 and 14). Since age was controlled for in both models, the findings suggested that older adults who reported better physical health tended to live alone and did not need their supporters to live with them.

The findings regarding income and education were consistent with the results found by previous researchers. Income was negatively associated with loneliness among both Whites and African American older adults (Tables 9 and 10). However, after controlling for social isolation status, the association with income attenuated to one third and became marginally significant for Whites (Table 9 Model 5) and insignificant for African Americans (Table 10 Model 5). This finding suggested that the association between income and loneliness was partly due to the association between income and social isolation status. Education, however, did not have a significant association with loneliness for both races. Both income and education were negatively associated with distal social isolation among White older adults (Table 11), and only education was negatively associated with distal social isolation among African American older adults (Table 12). Older adults of both races who were proximally socially isolated tended to have lower income, but since income was measured as household income, lower household income did not mean that these older adults were less self-sufficient than those

with more people in their household. In contrast to other results, education was positively associated with proximal social isolation for both races.

Discussions and Implications

This study examined how neighborhood characteristics are associated with social isolation and loneliness among older adults with a focus on neighborhood socioeconomic structure and racial/ethnic composition. The major contribution of the findings to the literature can be summarized into four themes: 1) neighborhood deprivation does not directly affect social isolation or loneliness among older adults. The observed associations can be fully explained by individual risk factors. 2) For White older adults, neighborhood racial/ethnic diversity has a curvilinear relationship with loneliness. In White-dominated homogenous and highly diverse neighborhoods, higher diversity is associated with lower loneliness among White older adults. In neighborhoods undergoing the transitional stage from homogenous to diverse, diversity is positively associated with loneliness among White older adults. In contrast, in the case of distal social isolation, greater racial/ethnic diversity has a consistently positive impact on distal social isolation among White older adults. 3) Regarding neighborhood dynamics, change in the percentage of Whites but not in neighborhood income affects distal social isolation among African American older adults. With the overall trend of a decrease in Whites across the state, the relationship suggests that abrupt decreases in the White population in neighborhoods can result in declining social connectedness among older African Americans. 4) Older adults living alone and/or without a partner tend to live in apartments that are easier to maintain and senior living communities that provide opportunities for developing new relationships. In addition, findings regarding other neighborhood-level and individual-level predictors are also briefly discussed.

The compositional effects of neighborhood deprivation

The findings of this study did not support previous studies on the detrimental effects of neighborhood deprivation on social participation (Bowling & Stafford, 2007; Franzini et al., 2005; Verhaeghe & Tampubolon, 2012). Even though the positive relationships between neighborhood deprivation and social isolation/loneliness were observed in the bivariate analysis and hotspot analysis, the relationships can be fully explained by individual-level risk factors. This type of effect, the compositional effect (Diez, 2002; Pickett & Pearl, 2001), suggests that the observed association between neighborhood indicators and individual-level outcomes are due to the aggregation of individuals with similar traits in the same neighborhoods. In this study, older adults with low SES and worse health conditions, and therefore at greater risk of loneliness and social isolation, were more likely to live in deprived neighborhoods. With limited resources and social connections, older adults living in deprived neighborhoods also lack the ability to move out and to travel and are therefore “stuck” in deprived neighborhoods (Lang, Llewellyn, Langa, Wallace, & Melzer, 2008; Strohschein, 2012). These study participants’ social isolation and loneliness status are not attributable to the contextual effect of neighborhood deprivation. But neighborhood deprivation still matters in terms of identifying geographic communities to invest resources.

The compositional effect emphasizes the importance of individual characteristics in determining social isolation and loneliness over neighborhood-level SES. Advantages in non-deprived and even affluent neighborhoods, like a safe environment and the availability of social services, may not significantly protect vulnerable older adults from social isolation and loneliness. Although the proximity to resources and amenities can

facilitate social activities for older adults (Levasseur et al., 2015; Richard et al., 2009) and affluent neighborhoods have more public facilities, like parks and community centers (Estabrooks et al., 2003), the proximity and the availability of these resources do not guarantee accessibility or utilization, which can be restricted by the mobility and financial capacity of older adults (Lehning et al., 2018). For example, if not driving, older adults are more likely to take public transit than to walk to social activities (Julien et al., 2015). The high costs of transportation services and limited public transportation in suburban and rural areas restrict the access of already vulnerable older adults to community resources. In addition to accessibility barriers, in affluent neighborhoods, the needs of marginalized older adults can also be neglected because their demands do not reflect the major concerns of their local community (Keating et al., 2013).

Neighborhood deprivation has been linked to a series of risk factors for social isolation and loneliness among older adults, including low social cohesion, high crime, dilapidated physical conditions, and limited social programs and transportation services (King, 2008; Levasseur et al., 2015; Portacolone et al., 2018). Although the overall effects of neighborhood deprivation on social isolation/loneliness were not supported in this study, that fact does not contradict the previous evidence that each detrimental social and physical factor can still impose risks to the social wellbeing of older adults.

One explanation is that neighborhood deprivation is also related to a series of stressors that drive older adults to actively seek additional social support to cope with stress. Wilson's (2012) study found that residents living in neighborhoods of concentrated poverty actively sought help from relatives and friends. Rankin and Quane (2000) reported the unexpected finding that African American families have higher

participation in social organizations in more deprived neighborhoods. Residents living in deprived neighborhoods are able to take proactive actions to protect themselves from neighborhood threats (Cattell, 2001; Rankin & Quane, 2000). Kim's study (2010) controlled for the detrimental effects of neighborhood disorder and found that neighborhood deprivation actually facilitates building and maintaining social relationships that can help to cope with environmental stress.

In the current study, the quality and the various functions of social connections were not included in the analyses. There are concerns that neighborhood deprivation can still impose risks to the wellbeing of older adults by affecting the socioeconomic structure of their social network, which can limit its benefits (Abramson, 2015). Besides fulfilling social needs, social connections serve various functions regarding the wellbeing of older adults, including social influence, informal social control, and access to information and instrumental support (Berkman et al., 2000). Previous studies have indicated that residents of deprived neighborhoods have more unemployed and working-class connections (Rankin & Quane, 2000; Verhaeghe & Tampubolon, 2012), and the socioeconomic composition of a social network developed in a deprived neighborhood can have little benefit and even bring risks to the physical health of residents (Verhaeghe & Tampubolon, 2012). Older adults seek support from their families and friends regarding health knowledge, resources, and medical decision making (Cornwell et al., 2009). Due to increased medical needs, older adults' need for instrumental and informational support continues to rise with age (Shaw et al., 2007). The composition and structure of social networks are important as regards the availability of resources, access to information, and social control that discourages unhealthy behaviors.

Although this study did not support the direct impacts of neighborhood deprivation on social isolation and loneliness among older adults, the compositional effect is important for targeting community-based programs and services to the most isolated and lonely older adults. Studies have suggested the effectiveness of utilizing communities as the natural social setting to facilitate social engagement among older adults (Cattan et al., 2005; Findlay, 2003). Considering the restricted availability of community-based programs geographically, allocating federal and state resources in more deprived neighborhoods would be more effective in reaching the most isolated and lonely older adults.

As the impacts of individual characteristics are more important compared to the risk and protective factors brought by neighborhood-level SES, it is essential for community-based programs and services to expand their accessibility to the most vulnerable or marginalized older adults. Individual-level risk factors for social isolation and loneliness identified in this study include low income, being male, being in worse physical health, and having less education. For older adults with limited mobility due to physical or financial challenges, convenient public transportation systems and alternative transportation services for older adults can increase accessibility to programs and encourage social participation (Levasseur et al., 2015; Rosenbloom, 2009). Information and Communication Technology (ICT) interventions utilizing computer/mobile device training and free computer lab access have shown the potential to improve social connections for older adults whose social contacts are restricted because of limited mobility (Armitage & Nellums, 2020; Chen & Schulz, 2016).

Practitioners should not assume older adults living in non-deprived or even affluent neighborhoods, especially those who are vulnerable to social isolation and loneliness, are able to make full use of neighborhood resources and activities. Instead, practitioners should be sensitive to the marginalized position of vulnerable older adults and take actions to match neighborhood resources to specific needs. Programs or activities that are delivered to the whole community tend to be used by various types of older adults, from those who are not isolated but participate to maintain or expand their existing social network to those who are lonely and in need of social connections (Greenfield & Mauldin, 2017). Some isolated older adults are likely to be left out, either voluntarily or involuntarily, and studies have also indicated that those who are isolated tend to benefit less from these program (Dickens et al., 2011). This inefficacy may be due to low motivation of individuals to engage in activities, the mismatch between programs and their needs, or a lack of social skills, which further discourages the engagement of isolated older adults. Therefore, developing the screening and assessment tools for isolated older adults and designing alternative programs that are more friendly to marginalized older adults can increase the accessibility of programs among the most vulnerable populations.

Programs and services should also assess if older adults' social network and social support system are sufficient to meet their various needs. Local communities can also provide programs directly addressing the unmet medical and informational needs due to restricted functions of social network. For example, community-based health literacy programs that provide education sessions on basic medical knowledge and how to access health information and services can be beneficial for older adults to gain access to

accurate health information and to develop healthy behaviors (Bennett et al., 2009; Manafo & Wong, 2012). Although these programs do not directly address social isolation or loneliness, they can help to reduce potential risks born of being isolated from broader social connections.

Neighborhood diversity is associated with social withdrawal and adaptation to social isolation: insights regarding the discrepancy between social isolation and loneliness

Many previous studies' authors have pointed out the negative relationships between racial/ethnic diversity and a series of social indicators, including social capital, trust, social participation, collective efficacy, and sense of community (Putnam, 2007; Portes & Vickstrom, 2011; Townley et al., 2011; Twigg et al., 2010; van der Meer & Tolsma, 2014). However, this study revealed inconsistent effects of neighborhood diversity on social isolation and loneliness. Neighborhood diversity had a positive association with distal social isolation among White older adults in the study, for example, which is consistent with Putnam's (2007) "hunker down" effect; but by contrast, a cubic relationship was identified between neighborhood diversity and loneliness. With increases in neighborhood diversity, loneliness first increased, then decreased, and ultimately increased again. In short, the hypothesized positive effect of diversity on loneliness only occurred in neighborhoods with a moderate level of racial/ethnic diversity. The negative effect of diversity on loneliness, however, was identified in both White-dominated homogenous and highly diverse neighborhoods. The study's revealed discrepancy between social isolation and loneliness sheds light on the inconsistent patterns between social isolation and loneliness.

In neighborhoods with a moderate level of diversity, increases in diversity are associated with both higher social isolation and loneliness among White older adults. The specific context is likely to figure a transitional stage that a neighborhood changes from White-dominated to race/ethnicity mixed. The importance of the relative size of neighborhood racial/ethnic groups on perceived threats is summarized by the Racial Threat Hypothesis (Blalock, 1967). Chiricos and colleagues (2001) have also reported that, where other racial/ethnic groups slightly outnumber their own racial/ethnic group, people report the highest perceived threats of violent crime in their neighborhood. It is likely that, when the relative size of minority groups is trending toward outnumber the majority group in a neighborhood, perceived threats and social distrust are likely to be intensified, causing both social withdrawal and loneliness among the majority group, in this case, older White adults.

In both relatively homogenous and highly diverse neighborhoods, the effects of diversity on social isolation vary from the effects on loneliness for White older adults. In this study, higher diversity was associated with higher social isolation but lower loneliness. The discrepancy between social isolation and loneliness can explain the different patterns of change between social isolation and loneliness. How people perceive social isolation as loneliness can be affected by their personalities and social ideologies (Cacioppo & Hawkley, 2009). Older adults who are resilient to social isolation tend to score high in agreeableness and openness and low in anxiousness (Long & Martin, 2000; Schutter et al., 2020). The ideology of individualism also contributes to lower expectations for social connections and a preference for independence and solitude (De Jong Gierveld & Van Tilburg, 1999). Increased neighborhood diversity may nurture

openness and agreeableness, which protects older people in this context from loneliness. Diversity erodes social cohesion and the process fosters the pro-individualism ideology, which in turn lowers social needs (Heu et al., 2019). That is, older adults can choose solitude over social activities (Birditt et al., 2019). The changes can be achieved through the self-selection process and reflected at the population-level. Those who are less able to tolerate diversity may choose to move out, so-called “White flight” (Laurence & Bentley, 2015), leaving those who are able to successfully adapt to diversity in the now diverse neighborhood.

Resilience and adaption to social isolation can be beneficial to older adults. For example, a longitudinal study found that older adults whose response to loneliness was robust reported the best cognitive performance of all study participants over time (McHugh et al., 2017). This means it is important to distinguish voluntary social withdrawal from involuntary social withdrawal. For older adults, social withdrawal can be by choice in order to pursue other important things, for example, and indeed, solitude has been associated with a variety of emotional and spiritual benefits and embracing solitude can be a choice by older adults (Finlay & Kobayashi, 2018; Lay et al., 2018; Long & Averill, 2003). Moreover, socioemotional selectivity theory suggests that, when people perceive their lifetime is limited, they tend to narrow their social network and focus only on relationships that are emotionally meaningful to them (Carstensen et al., 1999). This preference may drive them to focus on existing and long-term social relationships while hesitating to develop in-depth interracial relationships that may be new to their experience and therefore more emotionally costly.

Even though some older adults are able to emotionally adapt to diverse neighborhoods and a narrowed social circle, there are concerns related to the changes at both the individual and community level. At the individual level, the narrowed and selected social networks may not provide enough influence and control to prevent health risk behaviors, like being overly sedentary and substance use (House et al., 1988, Shankar et al., 2011). The same concern also applies at the community level. Social capital and community studies researchers are concerned with declining community functionality, like reciprocity, neighboring behaviors, informal social control, and collective efficacy in diverse neighborhoods. Organized and cohesive neighborhoods serve as sources of social support that can be especially beneficial to older adults who have limited social and financial resources (Cramm et al., 2012; Greenfield & Reyes, 2015). Neighbors, because of their proximity, are important for providing emergency assistance (Lau et al., 2012) and helping older adults with daily chores (Gardner, 2011); and therefore, declining interpersonal and community functionality are likely to harm the wellbeing of older adults.

Neighborhood diversity significantly influences social isolation and loneliness among White older adults, suggesting that, when designing and delivering programs, practitioners should be conscious of the neighborhood climate when delivering and managing social activities. Local communities are important settings for older adults to develop and maintain social connections and to engage in a broader range of activities. Even though community engagement brings many benefits, it can also bring physical and emotional costs, however, especially where there are conflicts among older adults whose physical functioning is declining (Attree et al., 2011; Ziersch & Baum, 2004). For older

adults who are willing to engage in broader social activities, strategies should be developed to minimize the potential costs. Strategies to reduce physical barriers, like promoting transportation systems and opportunities for online engagement, have been discussed in the previous section. In addition, neighborhood context may also impose emotional costs on social participation. Therefore, community practitioners should stay aware of sensitive issues in the neighborhoods they serve, like intergroup tension, and be culturally sensitive when designing and delivering programs.

Practitioners should also respect the choices of older adults regarding social participation. Gerontologists have raised concerns regarding active aging. By emphasizing social engagement and promoting productive and active lifestyles, practitioners inadvertently stigmatize older adults who are not capable, or are unwilling, to fulfill the expectations imposed on them (Dillaway & Byrnes, 2009; Martinson & Berridge, 2014; Stephens, 2017; Martinson & Minkler, 2006). Instead of over-emphasizing the importance of social engagement to older adults, practitioners should expand the types of activities in their neighborhoods and provide more choices for older adults. Program designers should also adopt an evidence-based approach to design personalized activities for older adults. For example, for older men, solitary activities, like reading and gardening, can be more beneficial than activities that involve more social interactions (Adams et al., 2011).

Considering the discrepancy between social isolation and loneliness as well as the proposed adaptation to narrowed social connections among older adults, practices regarding social participation should emphasize social satisfaction rather than mere social participation. Programs that facilitate adaptation and the development of resilience can

also assist older adults to better adjust to declining social connections, no matter if the changes are driven by neighborhood changes or an individual's declining functionality. In fact, programs that address maladaptive social cognition, like cognitive behavior therapy (CBT), can be more efficient in reducing loneliness than programs that focus on expanding social connections (Masi et al., 2011).

To offset the risks from declining informal control and support accompanied by social withdrawal, policies and public service agencies should aim at identifying the unmet needs of older adults and supplementing reduced informal support with formal support. Practitioners can function to bridge social ties in order to connect public service resources to older adults and their families. To regain the social cohesion of communities, scholars have suggested that, through interactions and gained mutual understanding, new identities can be developed to override original racial/ethnic identities that resulted in social fragmentation in diverse neighborhoods (Laurence, 2011; Lancee & Dronkers, 2011; Schmid et al., 2014). Intergenerational programs linking students with older adults in the community have achieved success in gaining intergenerational understanding and in reducing ageism (Drury et al., 2016; Thompson & Weaver, 2015). The same strategies have the potential to enhance contacts across different racial/ethnic groups and to regain social cohesion in diverse neighborhoods.

Neighborhood racial/ethnic dynamics and social engagement among African American older adults

Recent studies of the influence of neighborhood dynamics on the social wellbeing of older adults have primarily focused on the gentrification process and its impacts on social wellbeing. Most previous studies, using in-depth observations and interviews

within gentrifying inner-city neighborhoods, reported that gentrification has restricted existing social connections among minority older adults (Buffel & Phillipson, 2019; Burns et al., 2011; Versey, 2018). This study, by adopting the quantitative approach and expanding the scope to the whole state, identified the negative relationship between changes in the White population and social isolation among African American older adults. In other words, African American older adults living in neighborhoods with greater increases in White population tend to be more socially connected. This finding runs counter to the findings of previous studies, but that does not mean the findings contradict previous studies. The neighborhoods used in this study were diverse regarding patterns of change, and gentrification was only happening in a small subset of the sampled neighborhoods. Even if we do not restrict the measure for gentrification, only 12% of census tracts in NJ experienced both increases in percentage of Whites and median household income from 2000 to 2015. In the current sample, only 9% of older adults resided in those census tracts. Though inconsistent with previous studies, the current findings provide more insight into the whole picture of neighborhood dynamics and the impacts on social isolation and loneliness among older adults.

Although gentrification is an eye-catching phenomenon in many minority-concentrated inner-city neighborhoods, the typical trend in NJ neighborhoods is a decrease in White population and an increase in diversity. The majority of census tracts in New Jersey experienced modest decreases in the percentage of Whites, and counties in middle New Jersey experienced the fastest decrease in the percentage of Whites over the previous 15 years (Appendix 10), changes driven by increases in Hispanic and Asian immigrants to the state. With the increase in diversity in the US, White flight persists and

extends to middle-class suburban neighborhoods as they become more diverse (Farrell & Lee, 2011; Kye, 2018). These changes have resulted in decreases in White population in most of the census tracts in the state, with the fastest drop in highly diverse regions. In contrast, the few census tracts that have experienced quick increases in White residents are scattered across the state with some concentrated in the northeastern urban areas. A comparison of maps in Appendix 10 and Appendix 9 indicates that most census tracts with increases in Whites also have decreases in diversity. It further suggests that most census tracts with increases in Whites are actually undergoing a re-concentration of Whites in predominantly White neighborhoods rather than gentrification.

With most of the neighborhoods experiencing decreases in percentage of Whites, the findings can be largely driven by that the correlation between decreases in Whites and increased distal social isolation among African American older adults. The finding can be partially explained by the concentrated poverty studies (Massey & Denton, 1993; Wilson, 2012). With decreases in percentage of Whites, inner-city neighborhoods were isolated from the mainstream society and left with limited resources due to isolation from the mainstream society. African Americans living in inner-city neighborhoods experienced declining social capital and restricted social contacts (Rankin & Quane, 2000; Small, 2007; Tigges et al., 1998).

While most concentrated poverty studies have focused on the static economic status and racial composition, this study emphasized neighborhood dynamics. The findings suggest that, regardless of current neighborhood economic status and racial composition, the declining percentage of White triggers reduced social contacts among African Americans. One possible explanation is that the change in racial composition

affects expectations regarding the future development of such neighborhoods. To some residents, decreases in percentage of Whites can act as a signal of neighborhood degradation, which results in declining social capital and reduced investments, even if no real negative changes have taken place. In addition, researchers have noted that there is a “tipping point” in dynamic racial/ethnic segregation, which suggests that once the proportion of minorities in neighborhoods exceeds that point all remaining Whites will move out (Card et al., 2008). In neighborhoods with fast decreases in Whites, the process is likely to reaffirm the initial expectation of greater racial segregation and neighborhood social isolation among African American residents, resulting in continuously declining social capital. The explanation regarding decreases in Whites as a signal of neighborhood degradation also explains why only changes in the percentage of Whites but not changes in income or diversity resulted in a significant difference in the findings, as changes in percentage of Whites is more visually identifiable.

That is, significant effects were only identified for changes in White population instead of changes in income, which suggests that changing racial proportions in neighborhoods have more influence on African American older adults than economic changes do. In racial segregation studies there is a central debate as to whether the motivation of White flight is race-based or class-based, that is to say whether White flight is primarily driven by racial prejudice or by the social problems in minority-dominated neighborhoods (racial proxy hypothesis) (Kye, 2018; Swaroop & Krysan, 2011). Recent studies have suggested that racial preference is the primary driver of the migration of Whites independent of neighborhood socioeconomic conditions (Kye, 2018; Swaroop & Krysan, 2011). Studies regarding the perception of African Americans regarding

decreases in Whites in their neighborhoods are scarce, but gentrification studies have suggested interracial tensions rather than interclass differences are a more important factor for African American older adults (Versey, 2018). The evidence suggests that, compared to the differences across socioeconomic classes, people can be more sensitive to divisions across racial/ethnic groups.

The discussion above only describes one possible explanation of the observed relationship between changes in the percentage of the White population and social isolation among African Americans, and there are other possible explanations. For example, with Whites moving out, African American older adults may lose their existing connections with their White neighbors. It is also possible that decreases in Whites actually increases neighborhood satisfaction among African Americans, and therefore they do not need to expand their social connections for additional support and comfort. Due to concerns regarding racial discrimination and hostility, some African Americans might prefer mixed-race neighborhoods with equal proportions of Whites and African Americans or neighborhoods with a higher percentage of African Americans over White-dominated neighborhoods (Krysan & Farley, 2002). This can also explain the stable loneliness among African American older adults regardless of increased social isolation and a decreased percentage of White neighbors. In addition, some qualitative studies have found evidence of enhanced social connections and social actions among African Americans to help them deal with the stress from increased numbers of Whites in gentrifying neighborhoods (Buffel & Phillipson, 2019). Therefore, this explanation can explain the changes in social isolation among African American older adults in both neighborhoods with increasing numbers of Whites and those with decreasing numbers of

Whites.

The preliminary findings reveal that neighborhood dynamics are diverse and complex. There are many possible mechanisms that can explain the relationships between changes in percentage of Whites and changes in social isolation among African American older adults, but none is sufficiently supported by existing evidence and some explanations are even contradictory. Additional studies need to be conducted in the future that examine these mechanisms in order to provide insightful implications. To test the proposed effects of gentrification on social wellbeing among African American older adults, quantitative studies should aim at collecting a larger sample of African Americans from neighborhoods that have experienced or are experiencing gentrification to supplement the qualitative findings on African American older adults' experience with gentrification.

Proximal living environment and proximal social isolation

Findings regarding proximal social isolation likely reflect the living preferences of those who are without proximal social connections. The assumption is that older adults who are proximally isolated, defined as without a partner and/or living alone, tend to seek living environments with alternative social connections. The findings suggest that, compared to the social and economic structure of neighborhoods measured at the census tract level, proximal living environments (i.e., housing type and unit type) are more important predictors of the location of proximally isolated older adults. To be specific, both White and African American older adults who are proximally isolated tend to live in an apartment or in retirement communities. The proximally isolated White older adults

also tend to live in subsidized and affordable senior housing rather than in general residential communities.

The relocation of older adults who are proximally isolated explains why they are more likely to live in apartments and senior living communities. Old age is accompanied by a series of life changes that potentially trigger relocation. One of the primary triggers is the loss of a partner (Bonnet et al., 2010; Chevan, 1995; Choi, 1996). Compared to moving in with their adult children or other caregivers, older adults are more likely to move to retirement communities and smaller houses/apartments to remain independent (Abramsson & Andersson, 2016; Bonnet et al., 2010). Older adults who are proximally isolated are more likely to relocate, as they live independently and are likely to have experienced the loss of a partner.

Push and pull theory, which highlights push factors within the current living environment and pull factors from potential relocation destinations (Wiseman, 1980), is widely used to explain how living environment characteristics are related to the relocation patterns of older adults. Loss of a partner, the accompanying decline in financial capacity, desire to move closer to children and kin, and the need for more social contact are major push factors for older adults to relocate (Choi, 1996; Smetcoren et al., 2017). Smaller household size and reduced financial capacity push older adults to downsize their housing (Abramsson & Andersson, 2016; Bonnet et al., 2010). Older adults who are not closely connected within their current communities, and therefore their social needs cannot be met within their communities, are more likely to move out of their neighborhood (Erickson et al., 2006; Roy et al., 2018; Wiseman, 1980). Meanwhile, neighborhoods and housing with desirable traits, especially recreation amenities, aging

services, and social activities, are pull factors that attract older adults to a given neighborhood (Ewen et al., 2014). Older adults who are proximally isolated likely experience both pull and push factors that facilitate their relocation decisions.

Relocation is accompanied by stress from transition and adjustment, but relocation can also act as a positive coping strategy to escape from stressful environments, seek opportunities for social connections, and move to a location more proximal to adult children and kin and with easy access to health services and aging-friendly amenities (Choi, 1996; Golant, 2011). The outcome of relocation can largely depend on whether the relocation is voluntarily or involuntarily. Involuntary relocations, triggered by health deterioration, like disability or falls, are generally associated with negative consequences. Voluntary relocation, by choice and with the purpose of seeking a better living environment, can bring positive changes. Smetcoren and colleague's study (2017) found that, for older adults who are not in a relationship, the need for social contacts is the primary reason for relocation while health difficulties are of less concern, and therefore relocation is more likely to be a positive adjustment. Previous studies have also identified positive changes among older adults who moved to senior living communities, including increased social contacts with friends and family and improvement in the overall quality of life (Buys, 2001; Rossen & Knafl, 2007). The findings from Table 9 and Table 12 predicting loneliness and distal social isolation also support this assertion that there are positive changes associated with moving into senior living communities. Living in senior housing is associated with lower rates of loneliness among White older adults and living in retirement housing is associated with lower distal social isolation among African American older adults.

Even though living in senior-concentrated neighborhoods facilitates the development of new connections, its capacity to reduce loneliness due to the absence of intimate relationships, especially among older adults who experience loss and grief, is not sufficient (Ayalon & Green, 2013). Evidence suggests that social relationships newly developed in senior living communities can be superficial and casual, with the purpose of fulfilling personal goals and not for developing in-depth emotional connections (Dupuis-Blanchard et al., 2009).

The relocation of older adults reveals that the settings of their old housing and neighborhoods are not able to satisfy their needs. It also raises the concerns that, compared to the relocated older adults, the truly vulnerable population should be the involuntary stayers, defined as those who desire to move out from their current residence but are not able to do so (Strohschein, 2012). Older adults with low SES, poor physical health, and higher risk of social isolation and depression do not have the capacity, either financially or socially, to move to a more desirable environment (Strohschein, 2012) and have to face the risks and difficulties that relocated older adults have managed to escape, including the lack of aging-friendly amenities and limited opportunities for social engagement and leisure activities.

The aggregation of proximally isolated older adults due to their preference for relocation makes it easier for service providers to deliver community-based services and programs to this population that are tailored to their specific needs. Considering their lack of proximal social connections, programs with the aim of relieving emotional loneliness should be developed. Ayalon and Green (2013) suggest building smaller and more private settings in senior living communities to facilitate the development of closer and

in-depth relationships among older adults. Community-based bereavement mutual support groups and education programs teaching coping strategies can also help older adults who are experiencing loss and grief.

Community agencies should be aware of the vulnerable position of those who must stay in their environment involuntarily and design programs to fulfill their needs, for example, providing information and easy access to a variety of services, including housekeeping, medical services, transportation, meals, and social and entertainment activities (Greenfield, 2012; Menec et al., 2011). As in mixed-age neighborhoods, local communities and schools can also develop intergenerational programs to increase social engagement and community cohesion among older adults.

Associations with other neighborhood-level and individual-level predictors

No significant relationship was identified between within-neighborhood income inequality and social isolation/loneliness among older adults after controlling for covariates. This finding is consistent with prior studies that suggest influences of income inequality on social cohesion and health are more likely to be salient at a larger geographic scope, such as at the state and country level, through the political structure and welfare system (Coburn, 2000; Muntaner & Lynch, 1999; Wilkinson & Pickett, 2006).

Additional neighborhood-level and individual-level covariates, including population density, urban/rural environments, income, gender, age, self-rated physical health, and education, were found to be associated with social isolation and/or loneliness. Population density and urban/rural environments are related variables, but their association with loneliness and social isolation are different. By including both measures,

this study is able to distinguish their unique effects. Population density is associated with higher rates of loneliness among White older adults; however, White older adults in rural areas report the highest rates of distal social isolation. Older adults living in rural areas have fewer opportunities to engage in broader social activities; however, previous studies suggest that close social relationships with friends and family members have more benefits on the subjective wellbeing among older adults in rural areas than those in urban areas (Mair & Thivierge-Rikard, 2010; Therrien & Desrosiers, 2010). For African American older adults, however, those who live in urban clusters report the highest distal social isolation. The racial disparity might be attributed to diminished access to public transportation system in suburban areas because accessibility to transportation is an important facilitator for social participation among older adults (Lehning et al., 2018; Levasseur et al., 2015). Lack of access to a public transportation system in suburban areas is more likely to impede social participation of African American older adults who do not own cars and find it not cost-effective to use alternative transportation services for non-essential social activities.

Consistent with previous studies, in general, older adults with lower household income, less education, and worse physical health are more likely to be lonely and isolated from distal social connections (Cudjoe et al., 2020; Iliffe et al., 2007; Nicholson, 2009; 2012; Pinquart & Sörensen, 2003); women tend to report more social connections than men (Cornwell et al., 2008; Cudjoe et al., 2020; Iliffe et al., 2007); and there is no gender difference regarding loneliness (Theeke, 2009; Victor et al., 2006). Previous evidence regarding the association between age and loneliness/social isolation are mixed. This study found that, when White older adults enter old age, they tend to expand their

social connections and contacts before their declining physical capacity starts to impose restrictions, which restates the U-shape relationship suggested by previous studies (Utz et al., 2002; Cornwell et al., 2008). Loneliness, however, decreases with age. Although this finding seems to be counterintuitive, some hypotheses have been developed to explain it, including that older adults tend to be more selective in choosing social relationships that can bring higher emotional satisfaction (socioemotional selectivity theory) and that older cohorts are less likely to disclose their loneliness status compared to younger cohorts (Heylen, 2010; Pinquart & Sörensen, 2003). This study also reveals some racial differences. For African American older adults, education, rather than income, is the key indicator for engaging in social activities. Gender difference regarding distal social isolation was only observed among White older adults.

The associations between individual covariates with proximal social isolation are different from associations with distal social isolation and loneliness. In some cases, the relationships are even in the opposite direction. These different patterns underline the necessity of distinguishing distal and proximal social isolation in conceptual and empirical studies. Older adults who are proximally isolated tend to be better-off as regards their socioeconomic and physical condition. Older women with higher education and in better physical condition are more likely to be proximally isolated because they have outlived their partners (Brown et al., 2012). Higher education and better physical condition also contribute to the willingness and capacity of older adults to live independently after losing their partners rather than moving in with their adult children (Keene & Batson, 2010; Reher & Requena, 2018).

Limitations and Future Studies

There are some limitations to this study regarding the sample, measurement, and research design. Future studies can expand on this study by including larger samples of minority groups, additional methods to construct social isolation and loneliness measures, and developing models that integrate both spatial and temporal effects. This study has proposed a series of underlying mechanisms for how neighborhood-level socioeconomics and demographics are related to social isolation and loneliness among older adults, and follow-up quantitative studies with additional measures and in-depth qualitative studies can help to achieve a better understanding of the underlying mechanisms.

The response rate of the survey is low (response rate :3.7%; cooperation rate: 37.5%), due to the use of RDD and phone interview for collecting data. The Pew Research Center (2017) reported that people who are active in civic and social engagement are over-represented in phone interview, that is important for this study as social engagement is one important indicator of social isolation. In addition, older adults with various risks for social isolation tended to be under-represented in the phone interview, including but not limited to, older adults who do not own a phone, live in rural areas with limited phone coverage, do not speak English, and have hearing loss.

The sample size for minority groups in this study is relatively small, which restricted the examination of the neighborhood effects among minority groups. The sample size of African American older adults is not large enough to detect associations with smaller effect size. Considering that the size of neighborhood effects tends to be small (Oakes et al., 2015), larger sample sizes are required to detect more potential relationships. The limited sample sizes of Hispanic and Asian older adults also restricted

the inclusion of other minority groups in analyses and comparisons, and the social engagement patterns of Hispanic and Asian older adults are different from their White and African American counterparts (Morrow-Howell & Wang, 2014). First generation Hispanic and Asian immigrants, many of them older adults, have encountered language barriers, an important risk factor for social isolation and loneliness (Schneider et al., 2014; Zhang & Ta, 2009). Hispanics and Asians also tend to aggregate in neighborhoods with a higher proportion of their own groups and even form subcultures within their neighborhoods (Lichter et al., 2010; Mui et al., 2007). The evidence suggest that neighborhood-level socioeconomic structures and racial/ethnic composition affect these groups' social isolation and loneliness differently than those observed in this study. Therefore, to better understand the whole picture of neighborhood effects and the differences across racial/ethnic groups, future studies should extend analyses to Hispanic and Asian populations. Moreover, as discussed previously, Hispanic and Asian groups are very diverse; therefore, future studies should use surveys and data that target specific racial/ethnic groups and/or use large sample sizes for each sub-ethnicity group. Recent studies have also highlighted different subgroups within the Black population (Taylor et al., 2019), and consequently, future studies should also explore the differences across subgroups, for example: African Americans, Black Caribbeans, and new African Immigrants.

In this study, we adopted principle component analysis to construct two types of social isolation—distal social isolation and emotional social isolation—which reflect social and emotional loneliness as proposed by Weiss (1973). However, the current survey does not include separate measures for social and emotional loneliness to examine

the concordance between the typology of social isolation with the typology of loneliness. Future studies can adopt the separate emotional and social loneliness scales developed by De Jong Gierveld and colleagues (1985; 2006) to examine the validity of the social isolation typology.

Constructing social isolation as continuous variables from the reverse-coded social connection and participation items, although widely used in previous studies, may suppress the findings regarding those who are truly isolated. The isolated older adults, especially those who are severely isolated, comprise a small proportion of the older adult population. For example, Cudjoe et al. (2020) categorized only 4% of community-dwelling older adults in the US as severely socially isolated. Because of that, it is likely that the findings regarding the variance between those who are isolated and those who are not isolated can be overshadowed by findings regarding the variance between those who are connected within smaller networks and those who are actively participating in social activities. The implications of these findings, therefore, do not directly target isolated older adults. For example, in this study, diversity was associated with increased distal social isolation among White older adults. It is possible that these older adults simply become more selective in social connections and activities but do not become socially isolated. One approach to generate findings and implications directly targeting vulnerable populations is to dichotomize or categorize the social isolation indices by distinguishing the severely isolated and isolated older adults from connected older adults. However, there are pros and cons. Although this approach is a common practice in medical studies to simplify interpretation and provide direct implications for policies and practices (Fedorov et al., 2009; Ragland, 1992), it has also been widely criticized for losing

statistical power and for incapacity to detect nonlinear relationships (Royston et al., 2006). Selecting different cut points is also likely to yield different results (Ragland, 1992). Researchers should be aware of the limitations and select cut points that are clinically meaningful.

This study used the simple approach to construct multilevel models to detect neighborhood effects. The geospatial effects were not included. Swaroop and Morenoff (2006) used spatial lag models to find that the participation of residents can be affected by the conditions in both their own neighborhoods and nearby neighborhoods. However, potential spatial effects can be more complicated than spill-over effects measured as spatial lag terms. The influence of nearby neighborhoods can be larger for those living near the boundary of their own census tracts than for those living at the center of their census tracts. The modifiable area unit problem (MAUP) raises the concern of selecting the optimal geographic unit (Wong, 2009), as different proxies for neighborhoods may yield inconsistent results. The choice could be significant for this study as it is designed to emphasize within-neighborhood division and between-neighborhood segregation. Future studies, especially qualitative studies, should be developed to identify the optimal geographic unit to examine social isolation and loneliness among older adults. Additional quantitative studies need to construct and compare models with different spatial specifications in order to select the model that best fits the data.

Although the three variables—changes in income, diversity, and percentage of Whites—were included to measure neighborhood dynamics, they were far from sufficient to fully capture the complexity of neighborhood dynamics. The identified significant effects of changes in percentage of Whites are difficult to interpret without a better

understanding of neighborhood dynamics and the underlying drivers of change. Future studies can distinguish and compare neighborhoods undergoing different change trends. For example, a latent trajectory model can be used to explore the typology of neighborhood dynamics.

A related limitation is causal inference. Although multilevel regression models have been widely used to examine neighborhood effects, these models are flawed for making causal inferences (Oakes, 2004; Oakes et al., 2015). More rigorous designs for causal inference, like experimental designs and natural experimental designs, although costly, can be achieved as parts of policy design, implementation, and evaluation, for example, the Moving To Opportunity (MTO) experiment. Alternative statistical methods, like the propensity score method and the longitudinal design of marginal structural models (MSM), have also been used to examine the causal effects of neighborhood characteristics (Oakes et al., 2015).

This study has proposed several underlying mechanisms for how older adults react to their environment; however, due to the cross-sectional design and limited neighborhood-level and individual-level indicators, this study is not able to empirically test these underlying mechanisms. Future studies with additional measures and longitudinal data can help to better understand the underlying mechanisms. Indicators of perceived neighborhood cohesion, connections with neighbors, and individual psychological indicators can help to understand how older adults process impacts from subjective neighborhood environments. Prior studies have identified the perceived neighborhood characteristics and individual psychological indicators that function as mediators when examining the relationships between subjective neighborhood

characteristics and health among older adults (Wen et al., 2006; Wilson-Genderson & Pruchno, 2013). In this study, for example, the development of adaption and resilience has been proposed to explain the discrepancy between social isolation and loneliness in response to neighborhood diversity. Studies with direct measures of adaptation and resilience can help to test these proposed mechanisms. In addition, longitudinal or retrospective data that track the mobility of older adults can help achieve an understanding of both the self-selection process of White older adults in diverse neighborhoods and the relocation of proximally isolated older adults. Qualitative studies with in-depth observations and interviews have great advantages in understanding underlying mechanisms.

Future studies should also distinguish different life stages of old age to best understand possible mechanisms. The associations between loneliness/social isolation and neighborhood characteristics can be various across different life stages. Joan Erickson expanded on Erick Erickson's psychosocial development stages by adding a ninth stage that includes older adults who are over 80 (Erikson & Erikson, 1998). In the ninth stage, older adults experience psychosocial developments characterized as withdrawing from social activities and engaging in more self-reflection (Brown & Lowis, 2003; Erikson & Erikson, 1998). The socioemotional selectivity theory also suggests that, as older adults approach the end of life, they increasingly focus on only the closest social relationships and their emotional value (Carstensen et al., 1999). It is likely that neighborhood environments carry more meaning for older adults in their 60s and 70s compared to those over 80. In addition, retirement age, which differs for individuals and by occupation, can divide older adults into pre-retirement and post-retirement categories,

with the peripheral social connections of the latter group more likely to be embedded within their local communities compared to the former group (Wrzus et al., 2013). Scholars have also adopted the developmental perspective and identified three unique phases of relocation throughout old age: the first phase driven by retirement, the second phase by declining physical capacity or widowhood, and the third phase by severe chronic disease and childlessness (Litwak & Longino, 1987). Distinguishing different life stages, therefore, can help to understand the relocation patterns of older adults and how their relocations are associated with key social relationships throughout their old age.

Additional neighborhood-level physical and social indicators can also enhance our understanding of neighborhood effects on the social wellbeing of older adults. The social indicators of neighborhoods in many previous studies have been constructed by aggregating individual survey data (Subramanian et al., 2003). Proxy variables have also been constructed to measure neighborhood-level social and civic engagement, like density of social organizations; voting rates; and density of arts, cultural, sports, and entertainment institutes (AARP, 2018). Indicators of physical environments, including accessibility and availability of aging-friendly amenities, transportation services, senior centers, community centers, and other spaces for social activities, have drawn great attention from neighborhood effect studies as they are considered as mutable via policies and interventions (Oakes, 2004). Integrating the additional indicators can also help us to learn how the sociodemographic structure of neighborhoods can facilitate or inhibit the development of aging-friendly social and physical environments. However, the caveat is that these types of data are largely from administrative data, and therefore availability can be limited and vary across regions, but some efforts regarding easier access to the data

have been made. For example, the AARP's Livability Index (AARP, 2018) integrates neighborhood indicators that are important to the wellbeing of older adults. However, because of the limited availability of data overall, most indicators are aggregated at larger geographic units, such as at the county and metropolitan statistical area (MSA). Future efforts should be made to expand the availability and accessibility of data for smaller geographic units, like census tracts and municipalities. In addition, measures of housing and unit types can be improved by linking the individual data to the property tax assessment records to identify detailed building types and characteristics to assess the living environments.

Last but not least, with the advent of modern technologies, there is a trend toward adopting new technologies to improve the social lives and wellbeing of older adults (Chen & Schulz, 2016; Khosravi et al., 2016). The current generation of older adults also shows less hesitation to embrace new technologies (Mitzner, 2010). Information technology extends the geospatial boundaries of social interactions, facilitating the transition of the concept of community from spatial-based to virtual-based. Because of COVID-19 and social distancing orders, there is increasing concern regarding social isolation and loneliness among older adults, and adopting technologies and virtual-based communities (for example, social media and forums) is a promising strategy to provide social support and connections to older adults (Armitage & Nellums, 2020). How virtual-based communities affect the social wellbeing of older adults, and how to develop virtual-based communities to meet the needs of older adults, will no doubt be prominent topics that for researchers as we explore how best to improve the social wellbeing of older adults.

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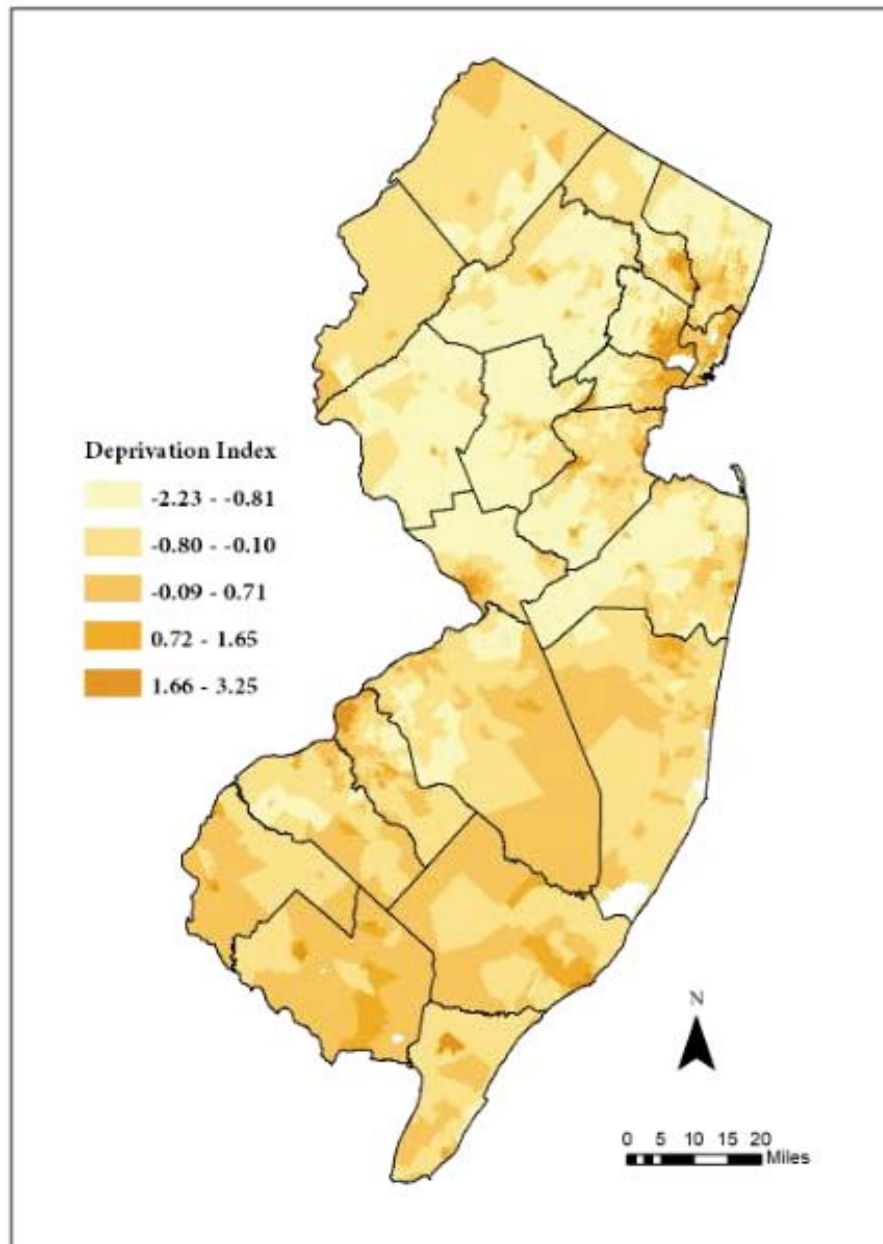
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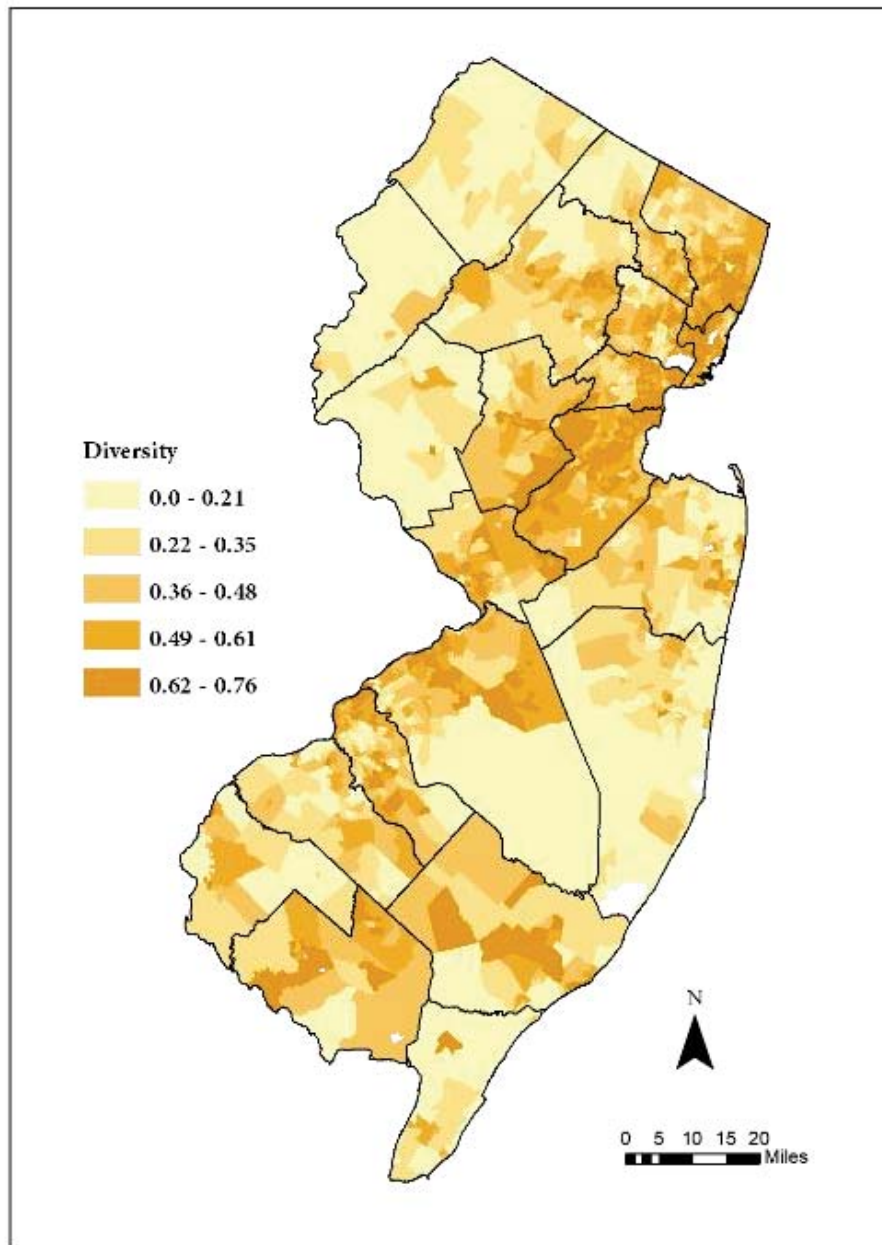
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Appendices:



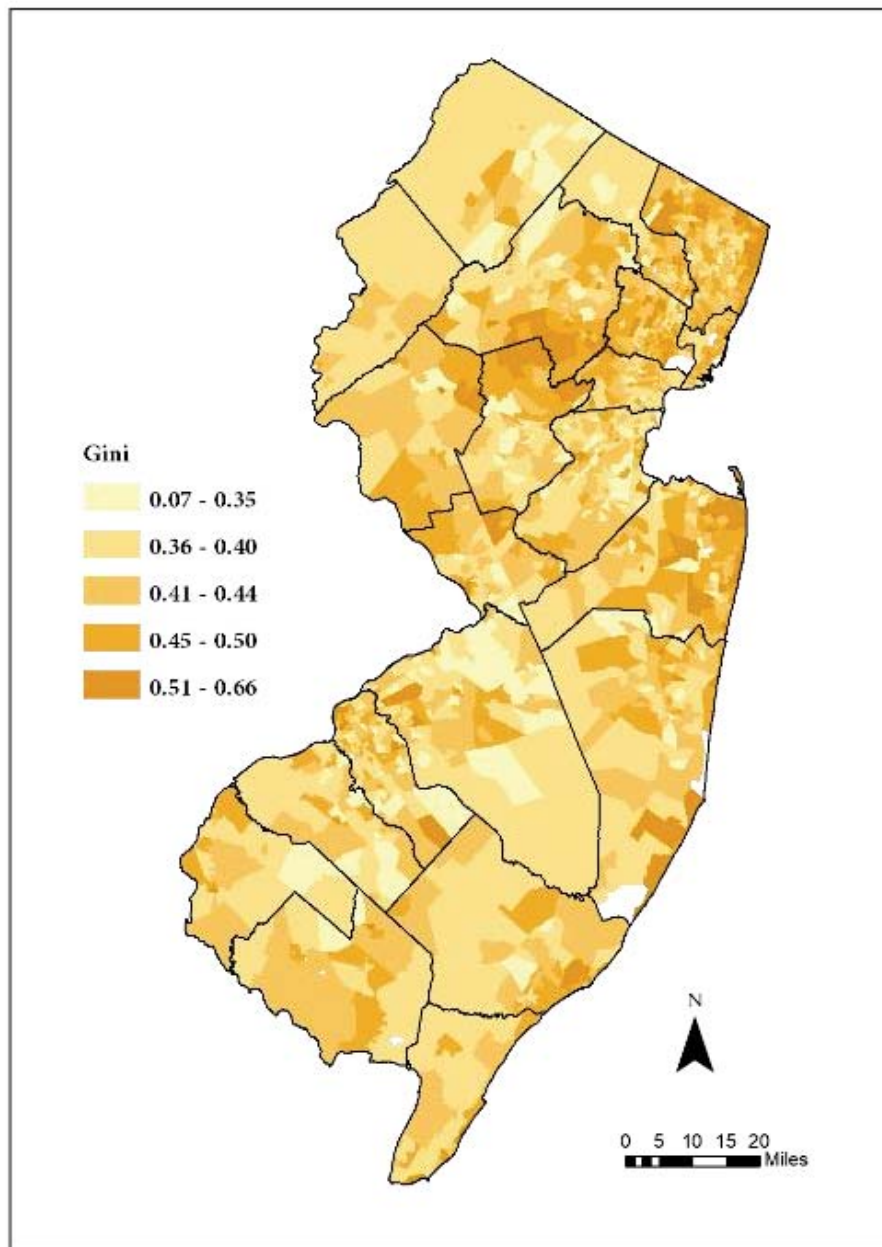
Appendix1 Neighborhood deprivation by census tract NJ

Appendix2 Neighborhood deprivation by census tract NJ



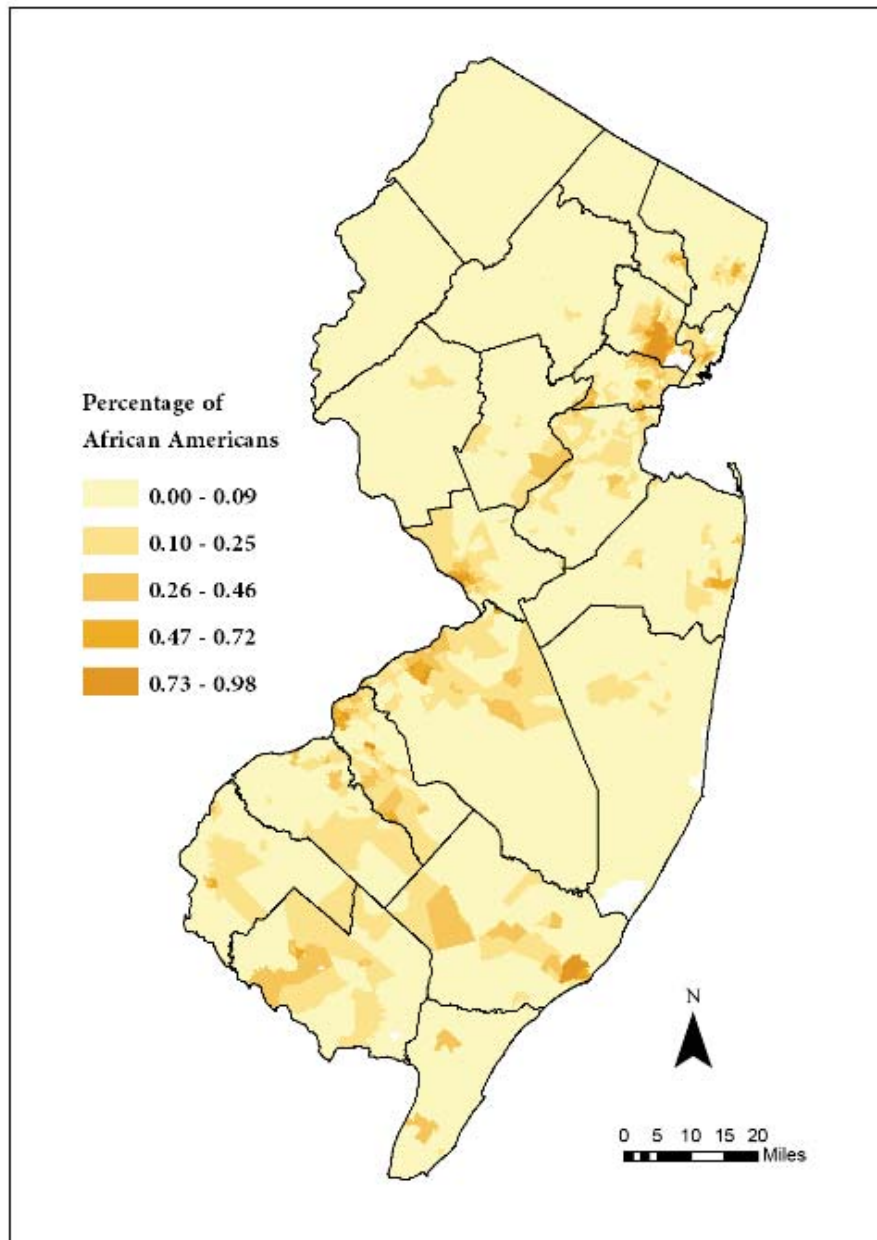
Appendix3 Racial/ethnic diversity by census tract NJ

Appendix4 Racial/ethnic diversity by census tract NJ



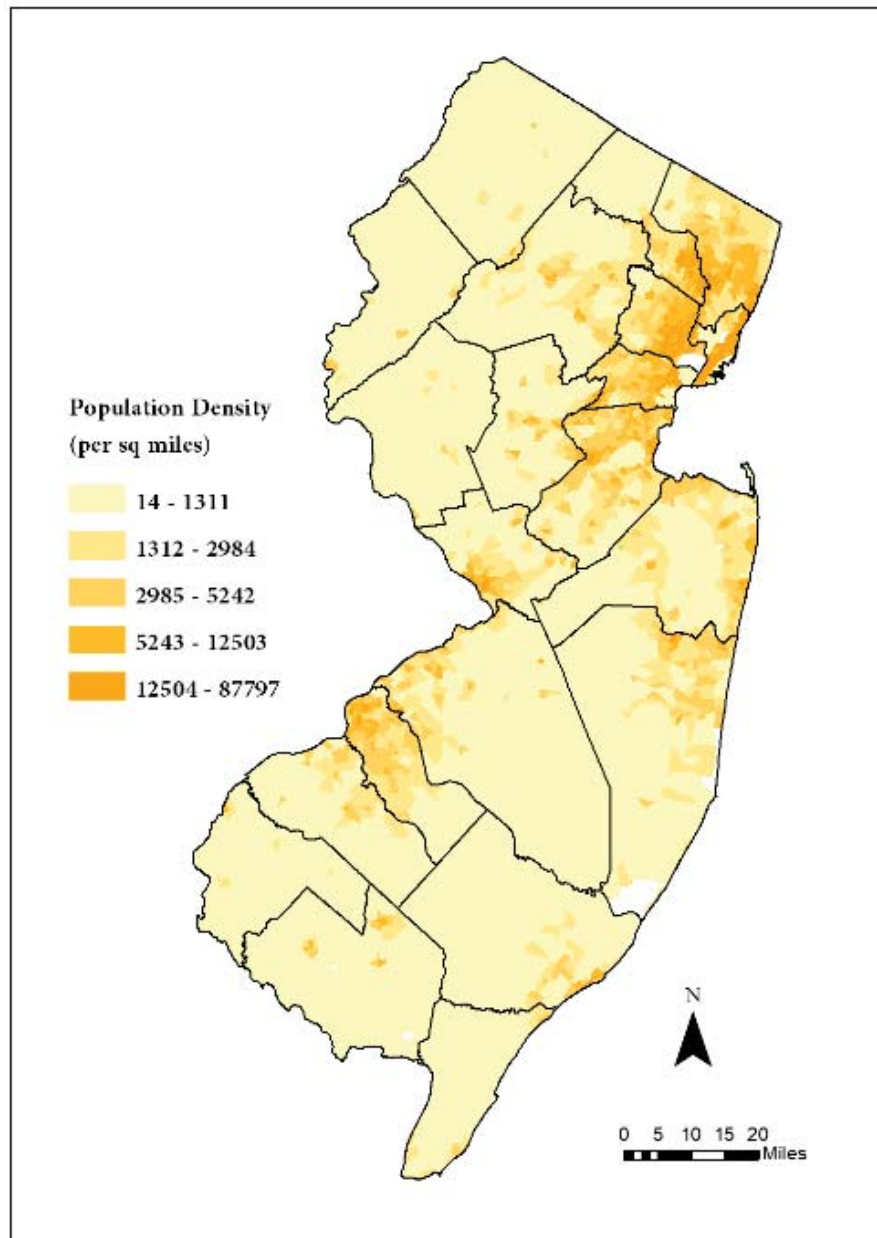
Appendix5 Gini Index by census tract NJ

Appendix6 Gini Index by census tract NJ



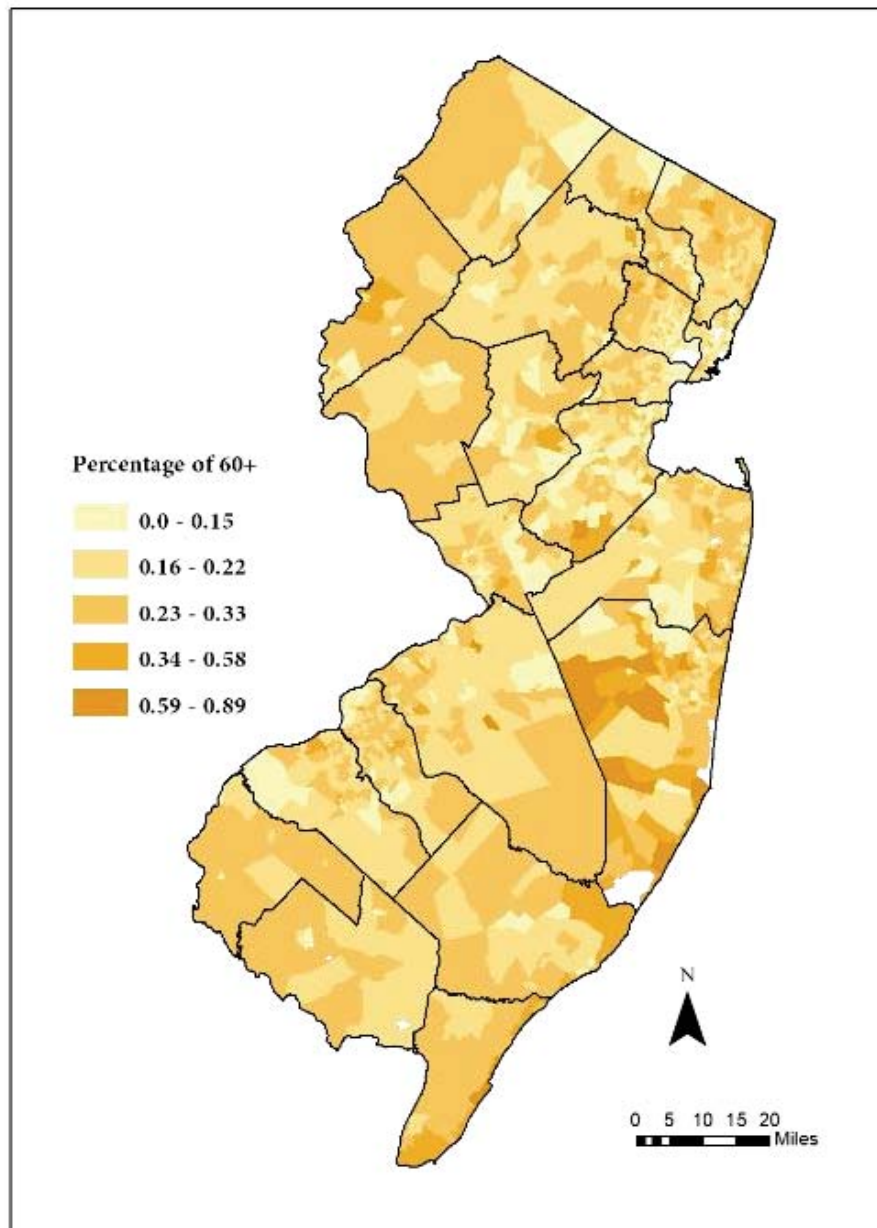
Appendix7 Percentage of African Americans by census tract in NJ

Appendix8 Percentage of African Americans by census tract in NJ



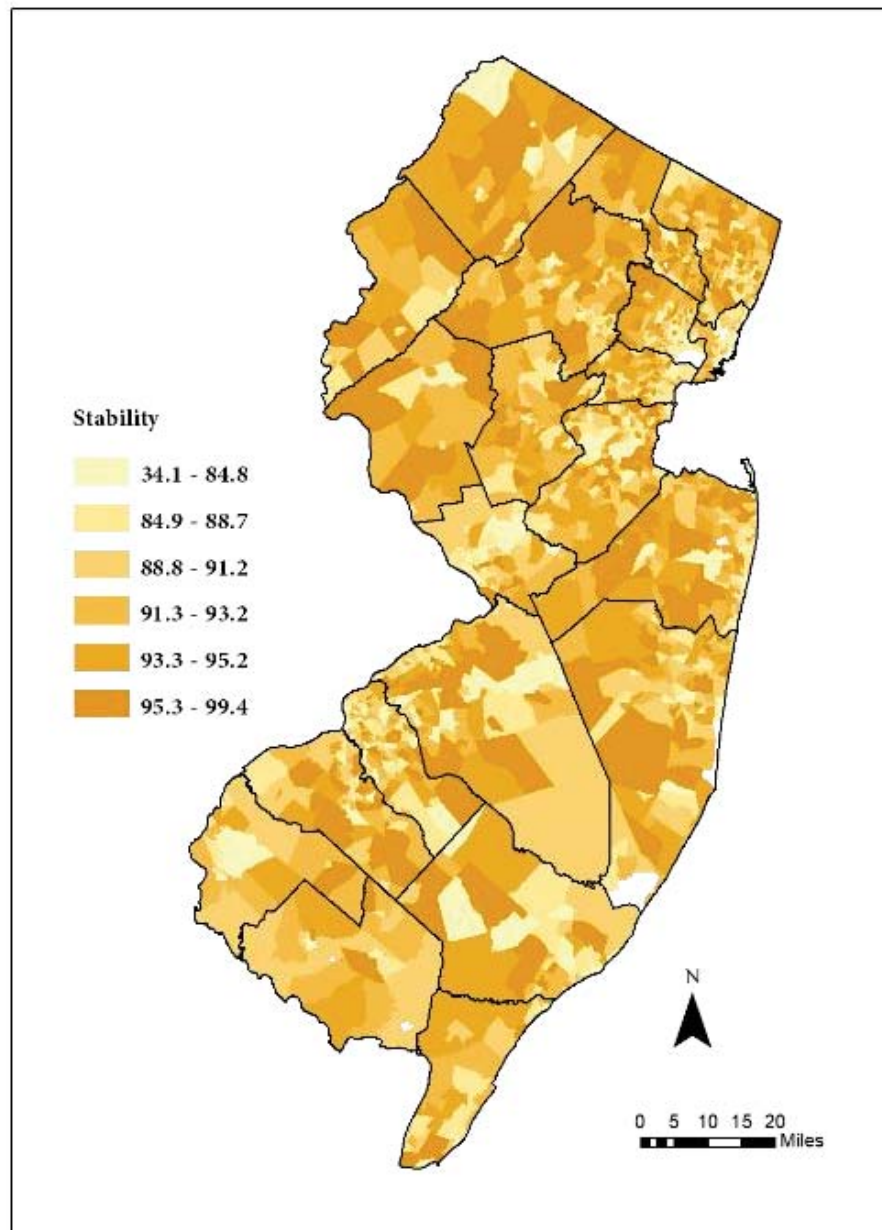
Appendix9 Population density by census tract in NJ

Appendix10 Population density by census tract in NJ



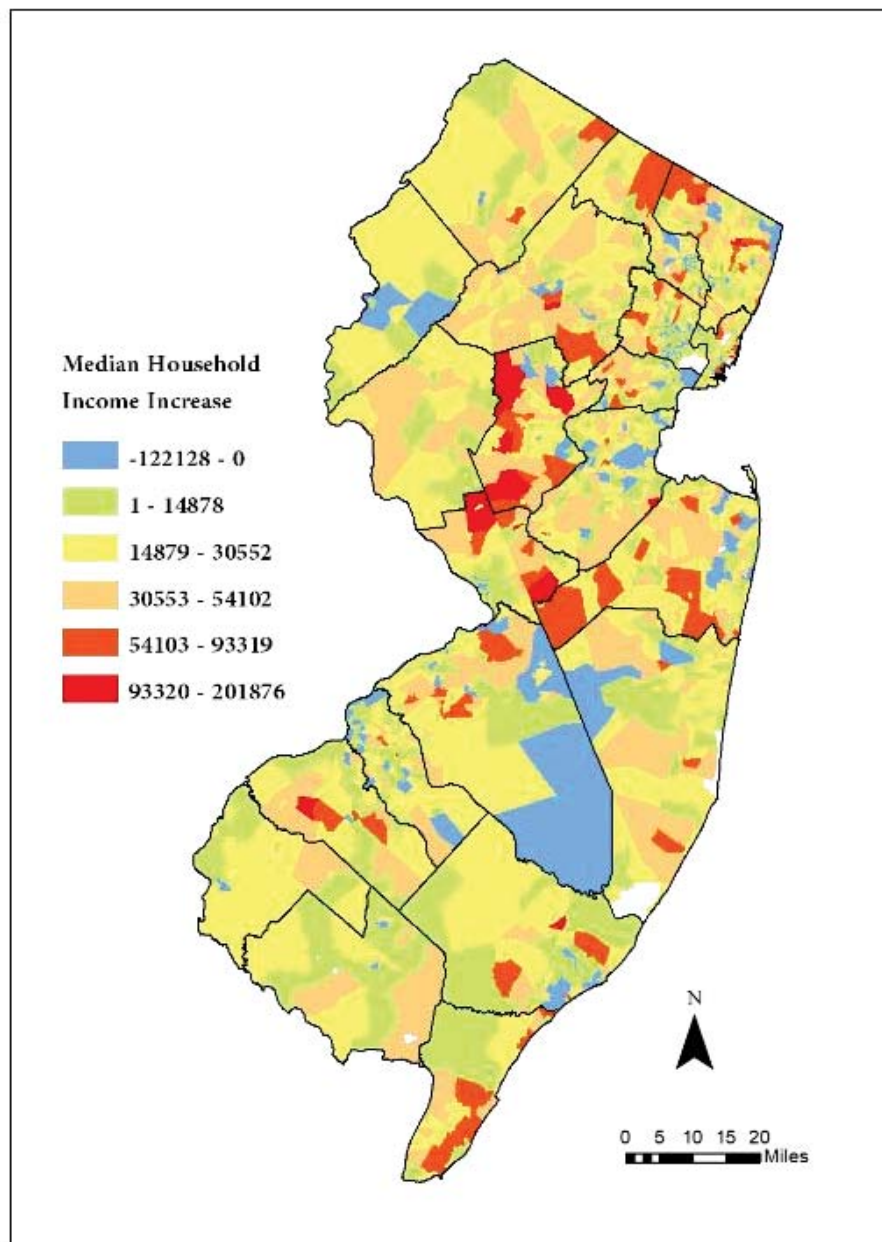
Appendix 1 Percentage of older adults (60+) by census tract in NJ

Appendix 2 Percentage of older adults (60+) by census tract in NJ



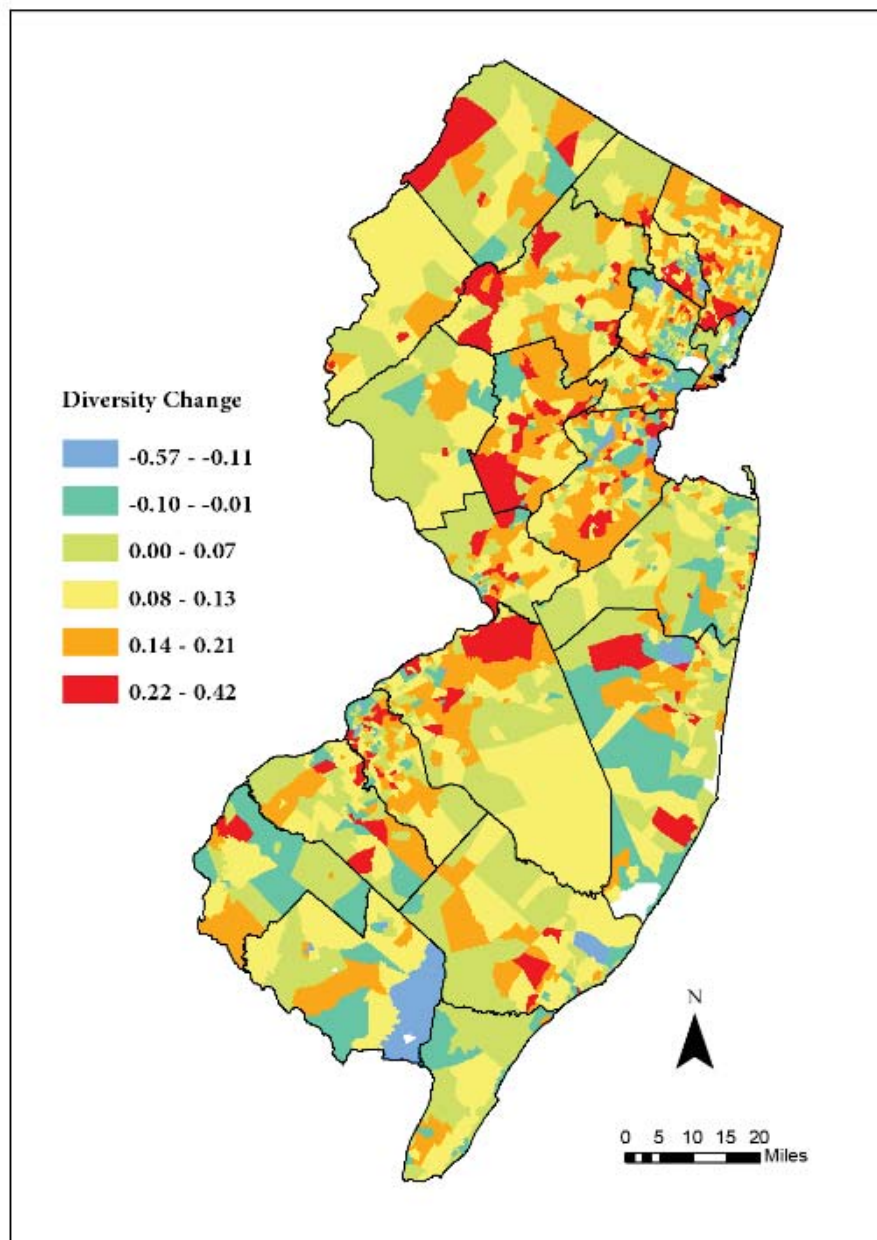
Appendix 13 Stability (% living in the same household last year) by census tract in

Appendix 14 Stability (% living in the same household last year) by census tract in



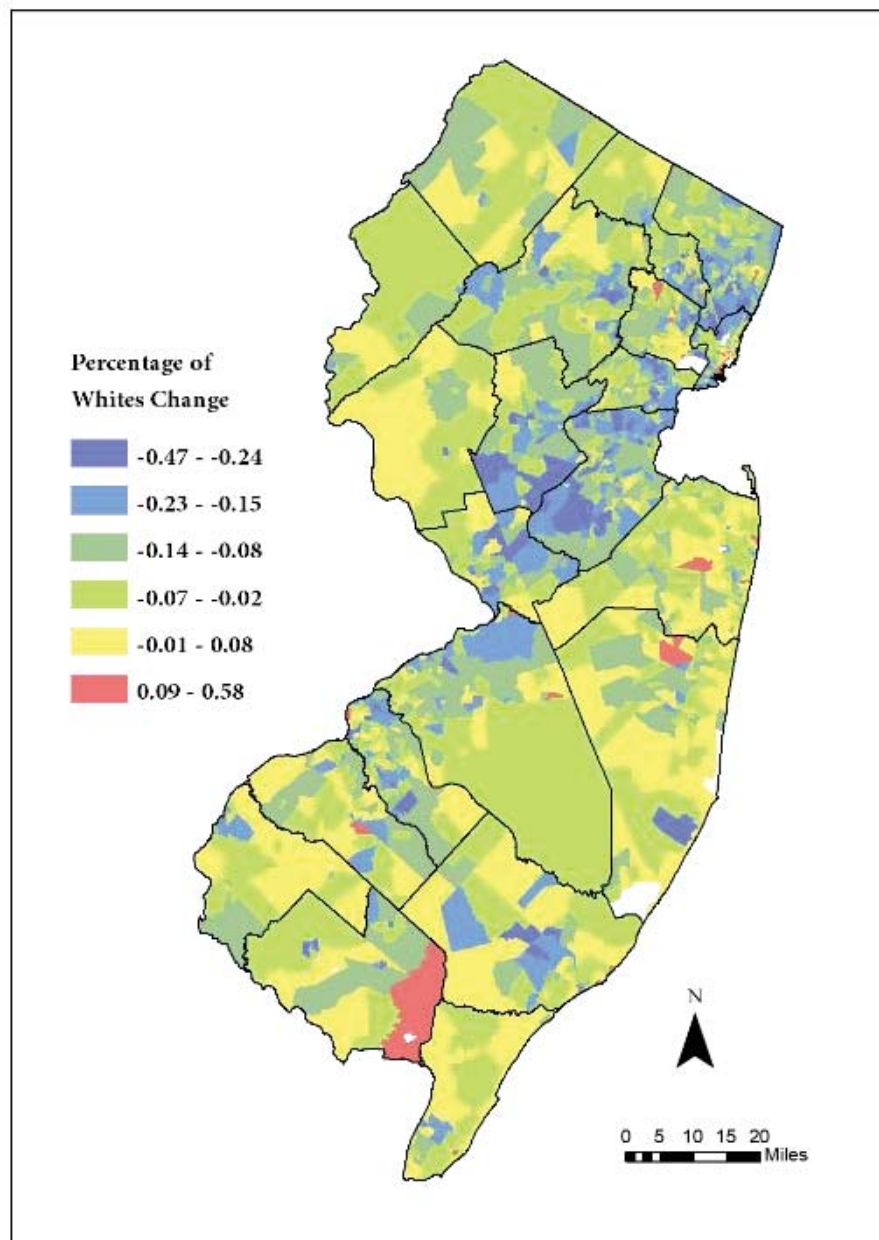
*Appendix 15 Changes in median household income from 2000 to .
by census tract in NJ*

*Appendix 16 Changes in median household income from 2000 to .
by census tract in NJ*



Appendix 17 Changes in racial/ethnic diversity from 2000 to 2010 by census tract in NJ

Appendix 18 Changes in racial/ethnic diversity from 2000 to 2010 by census tract in NJ



Appendix 19 Changes in percentage of Whites from 2000 to 2010 census tract in NJ

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